



AKTU

B.Tech 6th Semester
CS IT & Allied



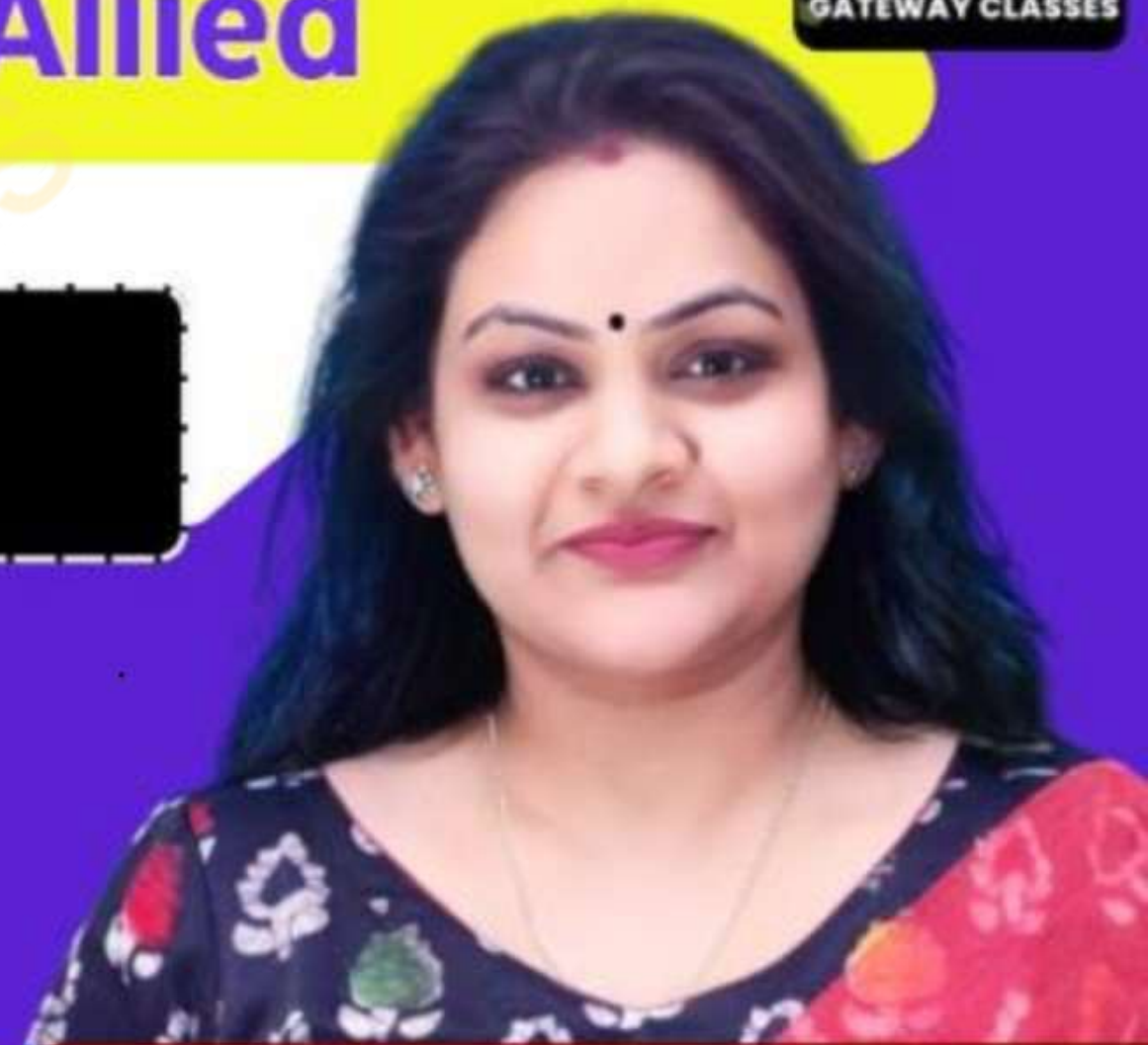
Computer Networks

Unit-3: Lecture 1

[Network layer]

Today's Target

- Responsibilities of Network Layer ✓
- IPV4 Datagram Format ✓
- AKTU PYQs



By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified

AKTU PYQs

1. Define IPv4 header Format in detail. (AKTU 2023-24) ✓
2. Define delivery in Network Layer. (AKTU 2023-24)
3. Discuss the role logical addressing. (AKTU 2021-22) ✓

Gateway Classes

Computer Networks

Unit-3

AKTU Syllabus

Network Layer: Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.



B.Tech 6th Semester

CS IT & CS Allied

COMBO PACK

of 3 Subjects

**Validity till
Sem Exam**



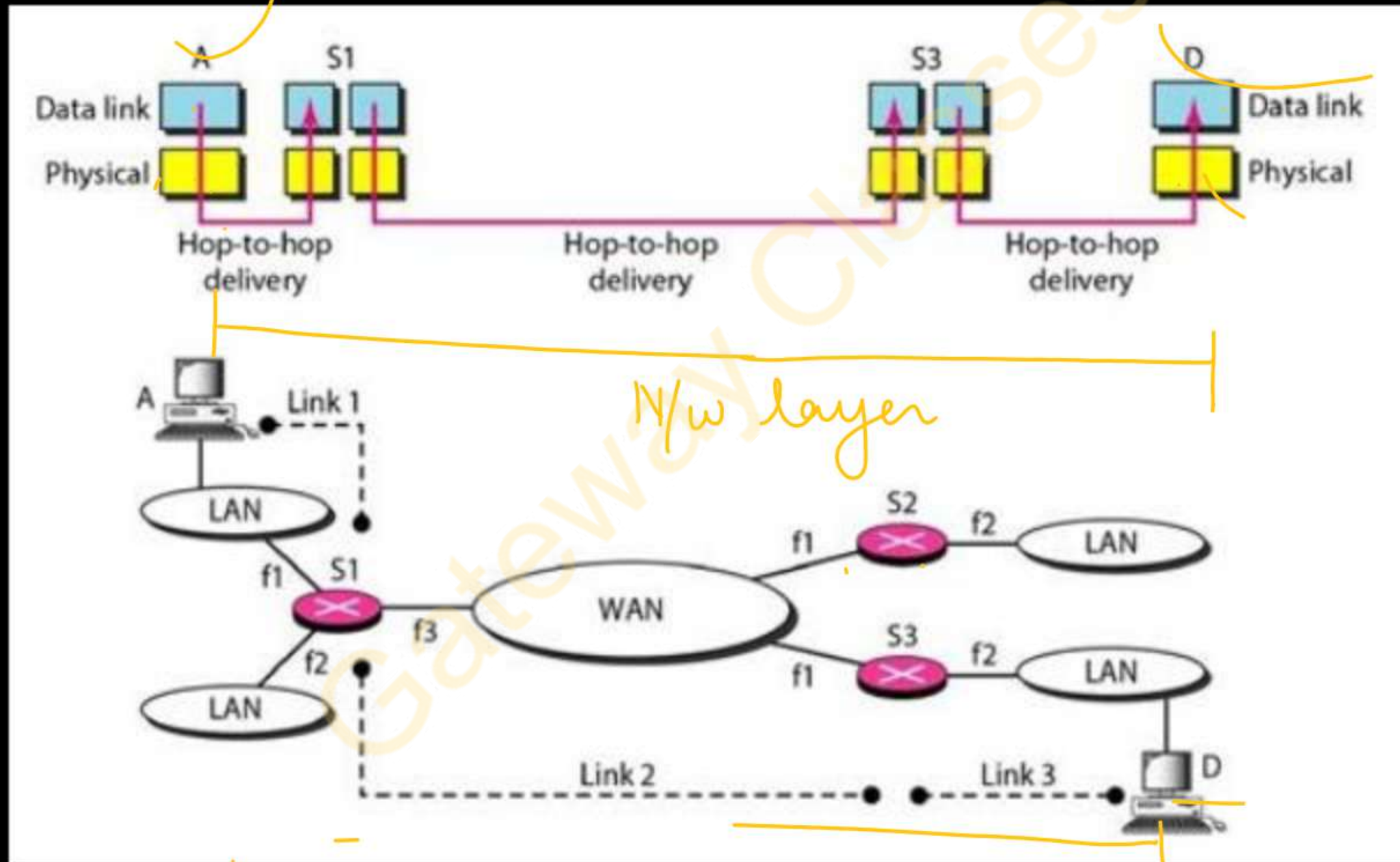
- **Software Engineering**
- **Compiler Design**
- **Computer Networks**

Helpline No- 7819 0058 53



Need for Network Layer

- The network layer is responsible for host-to-host delivery.
- It is also responsible for routing the packets through the routers or switches.



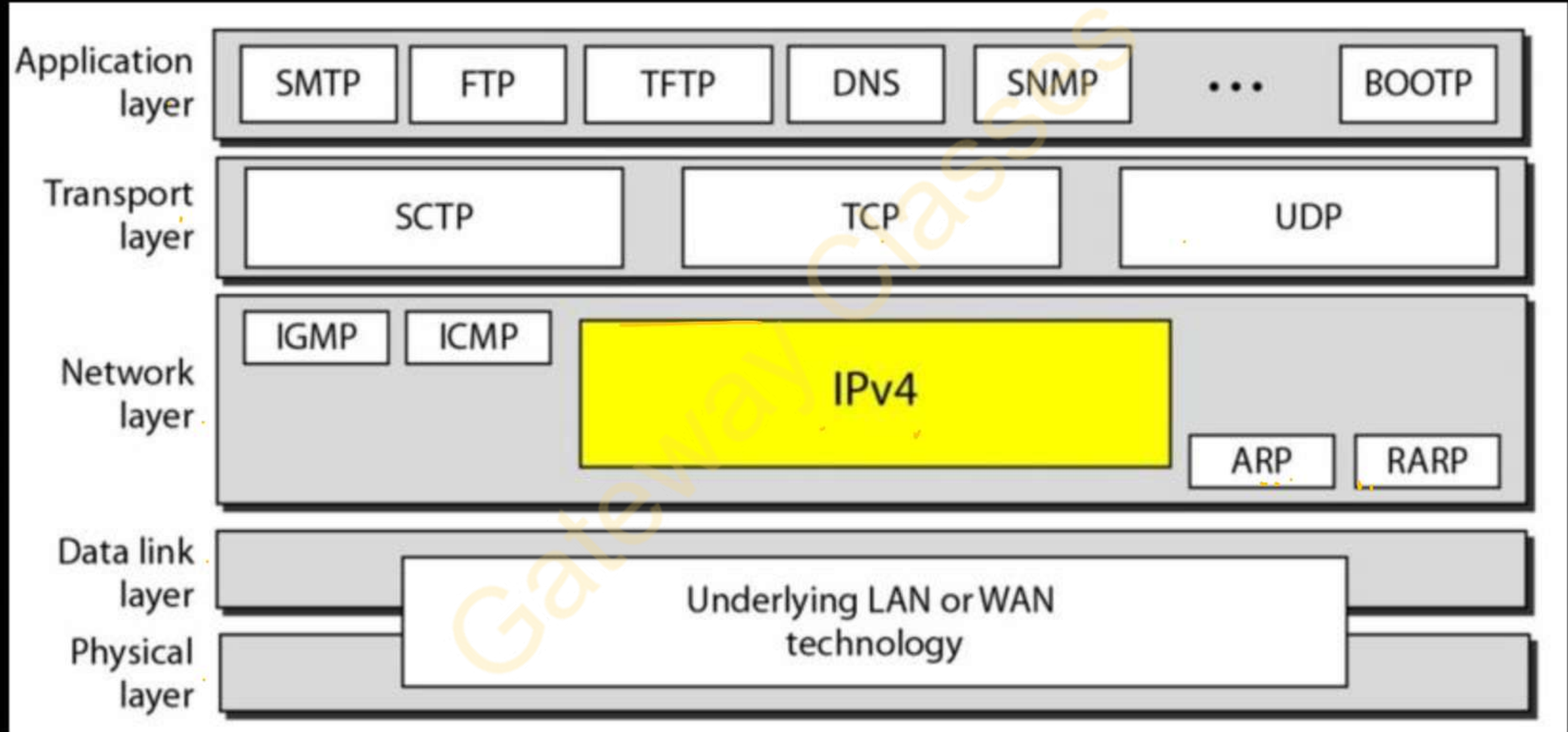
Responsibilities of Network Layer

- At source, it is responsible for creating a packet from the data coming from transport layer .
- The packet header contains logical addresses of the source and destination.
- The network layer is responsible for checking its routing table to find the routing information (such as the outgoing interface of the packet or the physical address of the next node).
- If the packet is too large, the packet is fragmented. (Ethernet frame: — 46 — 1500 bytes)
- The network layer at the switch or router is responsible for routing the packet by consulting its routing table and finds the interface from which the packet must be sent.
- The network layer at the destination is responsible for address verification; it makes sure, that the DA on the packet is the same as the address of the host. ✓
- If the packet is a fragment, the network layer waits until all fragments have arrived, and
- then reassembles them and delivers the reassembled packet to the transport layer.

IPv4

Internet Protocol

The Internet Protocol version 4 is the delivery mechanism used by the TCP/IP protocols.

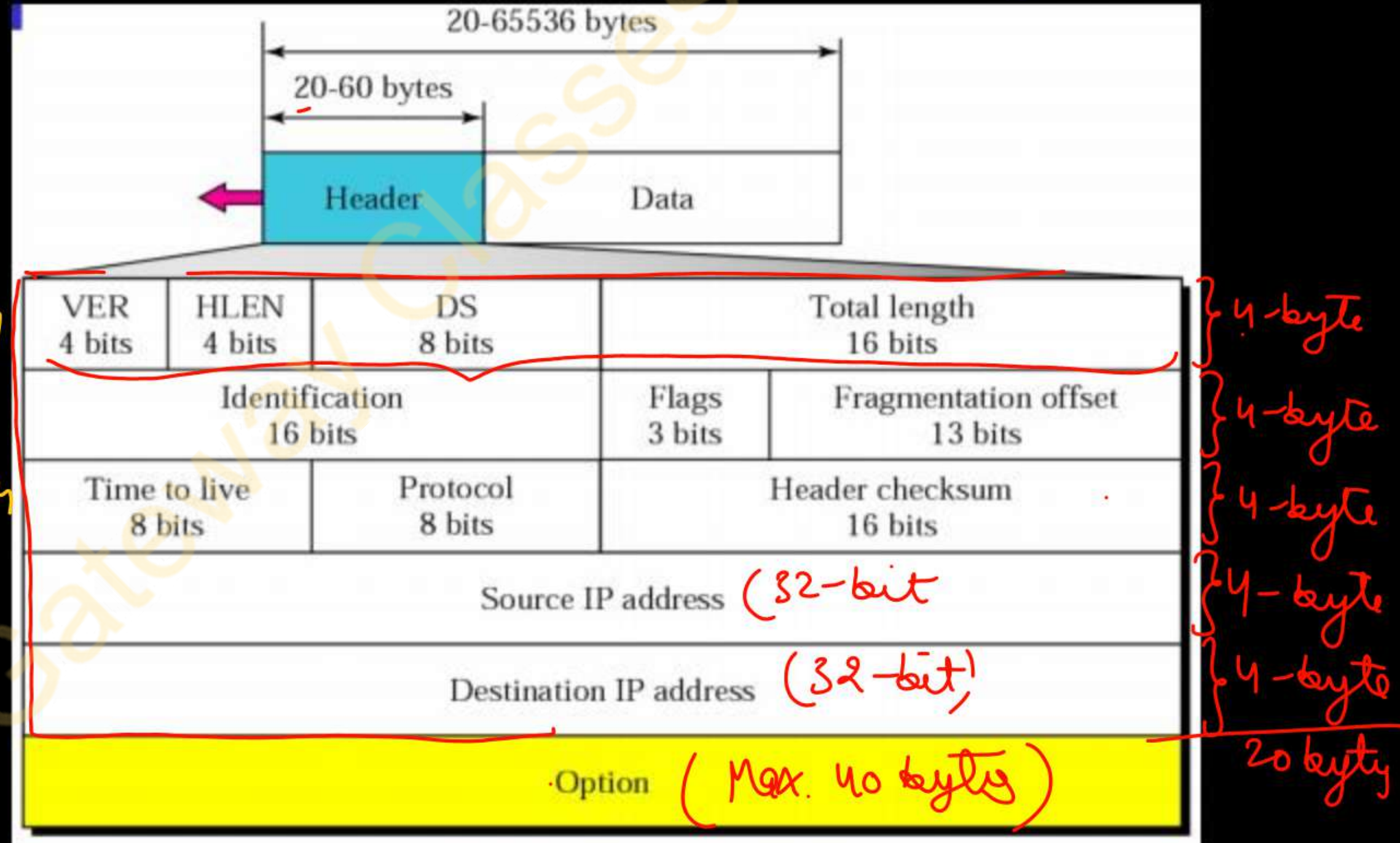
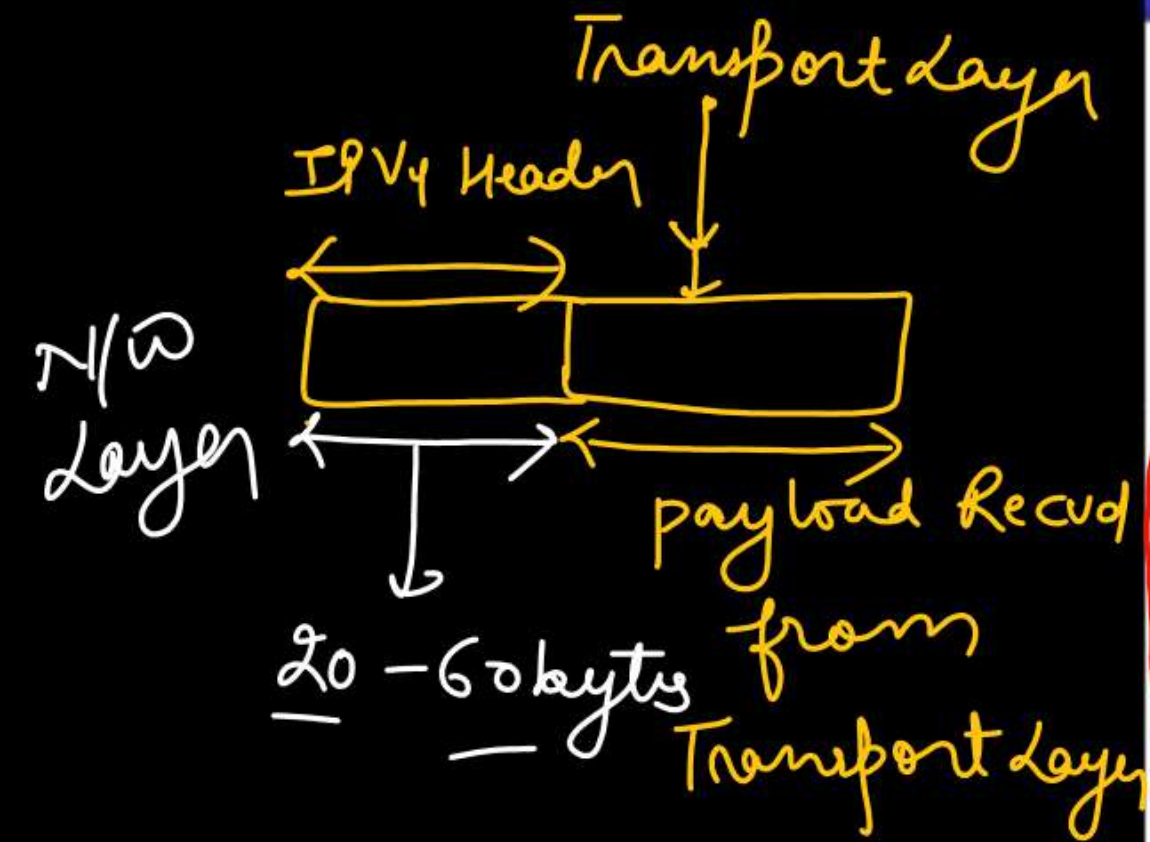


Important Points:

- Switching at the network layer in the Internet uses the datagram approach to packet switching.
- Communication at the network layer in the Internet is connectionless.
- IPv4 is an unreliable and connectionless datagram protocol.
- IPv4 provides no error control or flow control(except for error detection on the header)
- If reliability is important, IPv4 must be paired with a reliable protocol such as TCP. ✓ U.D.P. ✓
- Each datagram is handled independently and can follow a different route to the destination.
- Datagrams sent by the same source to the same destination could arrive out of order. Also, some could be lost or corrupted during transmission.
- IPv4 relies on a higher-level protocol to take care of all these problems.

Datagram: Packets in the IPv4 layer are called datagrams. A datagram is a variable-length packet consisting of two parts: header and data. The header is 20 to 60 bytes in length and contains information essential to routing and delivery.

IPv4 Datagram Header format

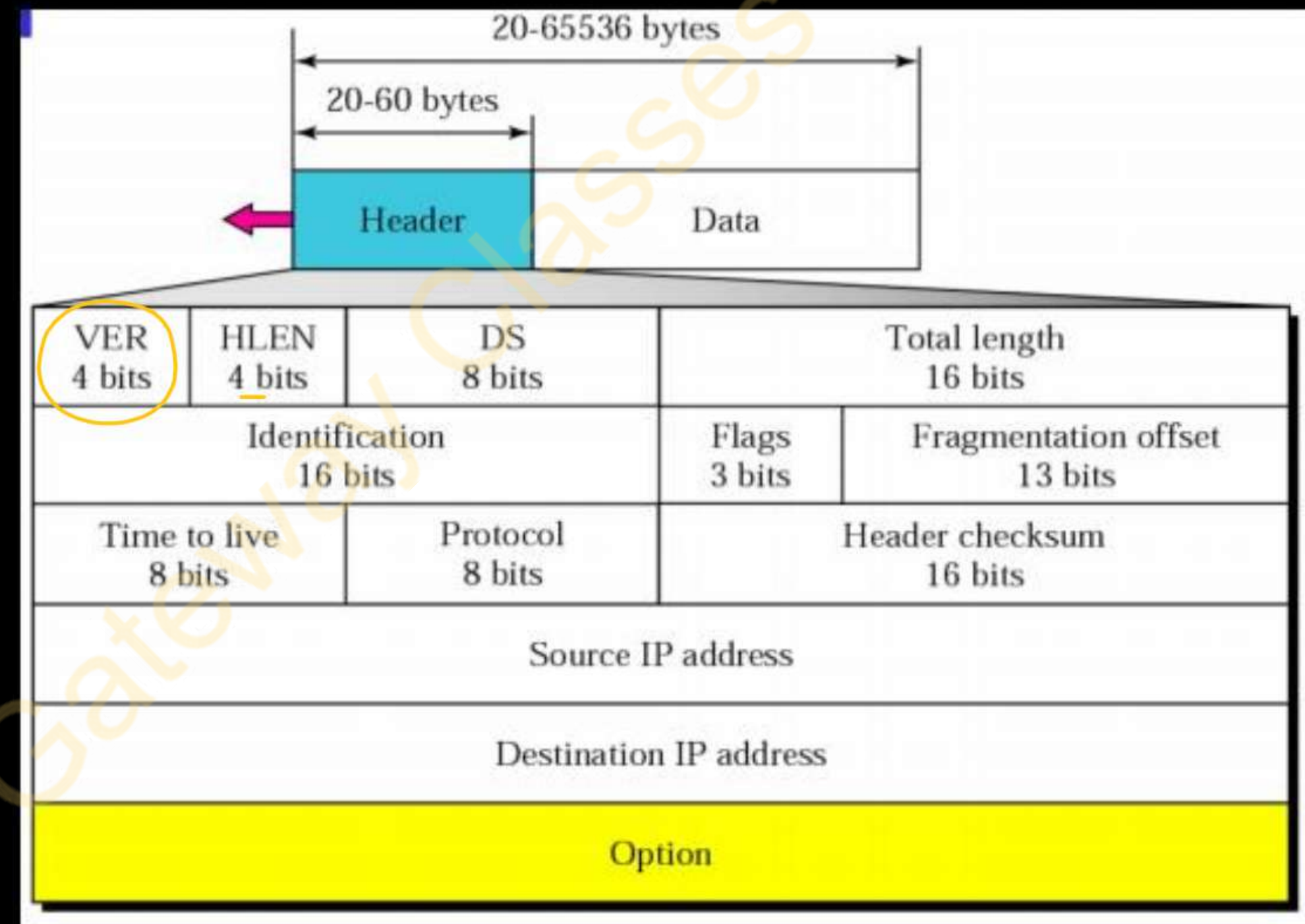


IPv4 Datagram Format

Datagram: Packets in the IPv4 layer are called datagrams. A datagram is a variable-length packet consisting of two parts: header and data. The header is 20 to 60 bytes in length and contains information essential to routing and delivery.

Version

0000	→ IPv4
0001	
0010	
0011	
0100	→ 4
0101	
0110	→ IPv6
0111	
1000	
1001	
1010	
1011	
1100	
1101	
1110	
1111	



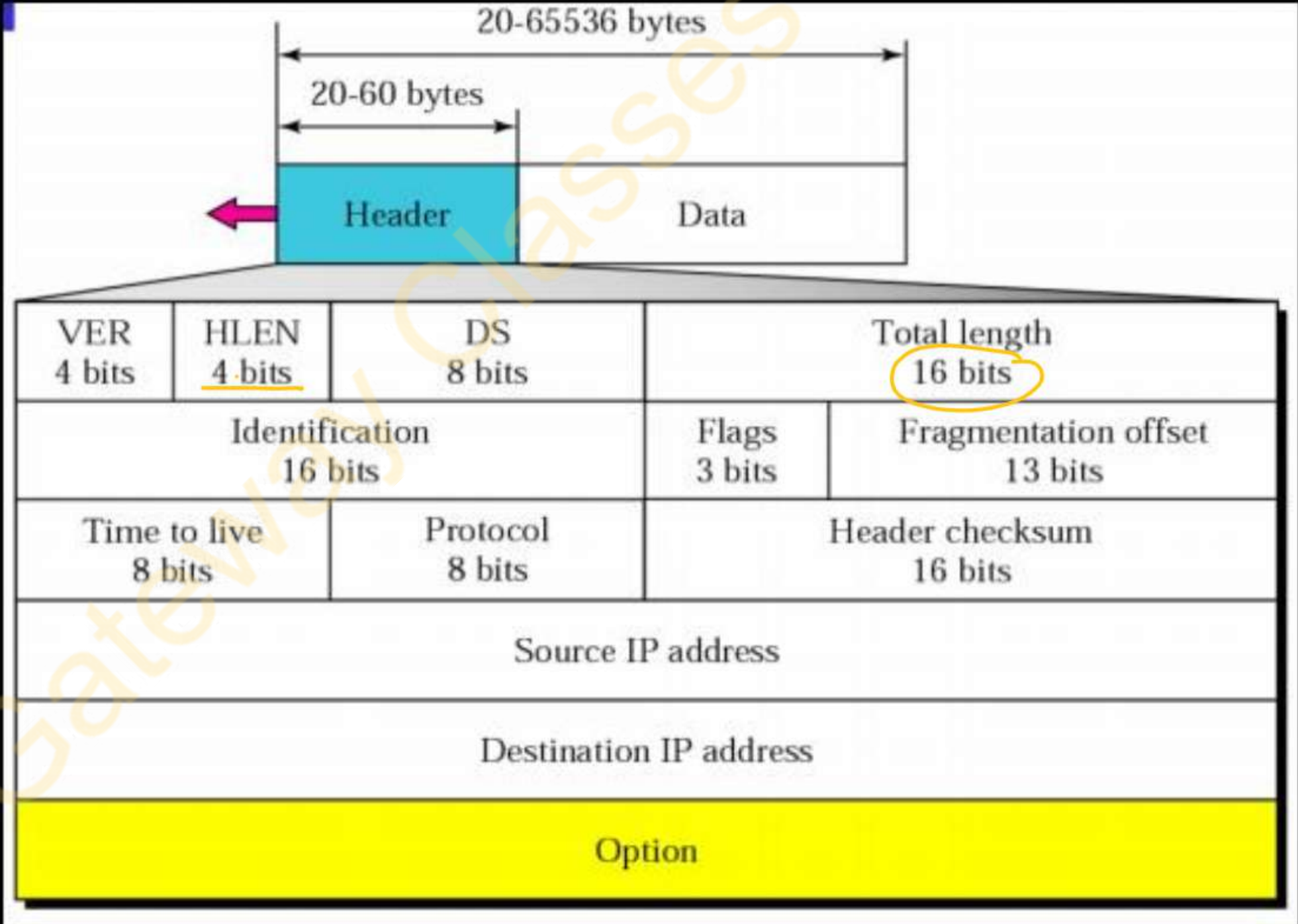
IPv4 Datagram Format

Datagram: Packets in the IPv4 layer are called datagrams. A datagram is a variable-length packet consisting of two parts: header and data. The header is 20 to 60 bytes in length and contains information essential to routing and delivery.

Header length

$2^4 = 16 \rightarrow (0 - 15)$

0	x 4 = 0	} unused
1	x 4 = 4	
2	x 4 = 8	
3	x 4 = 12	
4	x 4 = 16	
5	x 4 = 20	} 15 x 4 = 60 bytes
6	x 4 = 24	
7	x 4 = 28	
...	...	
...	...	



IPv4 Datagram Format

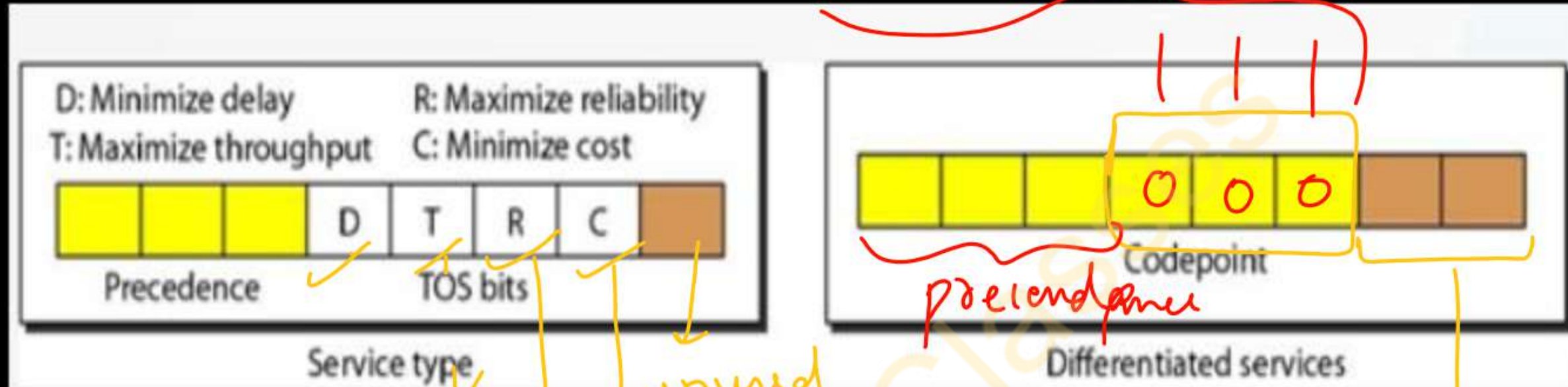
Version (VER):

- This 4-bit field defines the version of the IPv4 protocol.
- Currently the version is 4. However, version 6 may totally replace version 4 in the future.
- This field tells the IPv4 software running in the processing machine that the datagram has the format of version 4. (0100)
- If the machine is using some other version of IPv4, the datagram is discarded rather than interpreted incorrectly.

Header length (HLEN): ✓

- This 4-bit field defines the **total length of the datagram header** in 4-byte words.
- When there are no options, the header length is 20 bytes, and the value of this field is 5 ($5 \times 4 = 20$).
- When the option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$).

Services: This 8-bit field was previously called service type, is now called differentiated services.



1. Service Type

a. **Precedence:** Ranging from 000 to 111.

- Defines priority of datagram in issues such as congestion. If a router is congested and needs to discard some datagrams, datagrams with lowest precedence are discarded first. Some datagrams (datagram used for network management) are more important than others. The precedence subfield was part of version 4, but never used.

b. TOS bits: $D_s = 1$ for min delay $T_s = 1$ for Max Throughput $R_s = 1$ Max. Reliability $C = 1$ Min. cost

2. Differentiated Services:

First 6 bits make up the codepoint subfield, and the last 2 bits are not used.

- a. When 3 rightmost bits are 0s, the 3 leftmost bits are interpreted the same as precedence bits in the service type interpretation.
- b. When 3 rightmost bits are not all 0s, the 6 bits define 64 services based on the priority assignment by the Internet or local authorities

Total length:

16-bit $0 - 2^{16} - 1$

0 - 65535 bytes $\rightarrow$ Max data
65515 bytes

This is a 16-bit field that defines the total length (header plus data) of IPv4 datagram in bytes. To find

$$\text{Length of data} = \text{total length} - \text{header length}$$

Total length of IPv4 datagram is limited to 65,535 ($2^{16} - 1$) bytes, of which 20 to 60 bytes are the header and the rest is data from the upper layer.

Identification:

Flags:

Fragmentation offset:

Time to live: (TTL) 8-bit

} → Related to fragmentation



➤ This field is used mostly to control the maximum number of hops (routers) visited by the datagram.

When a source host sends the datagram, it stores a number in this field (approx. 2 times the maximum number of routers between any two hosts).

➤ Each router that processes the datagram decrements this number by 1.

➤ If this value, after being decremented, is zero, the router discards the datagram.

➤ This field is needed because routing tables in the Internet can become corrupted or to intentionally limit the journey of the packet.

Protocol: This 8-bit field defines the higher-level protocol that uses the services of the IPv4 layer such as TCP, UDP, ICMP, and IGMP.

Checksum: Used for error detection. (16-bit)

Source address:

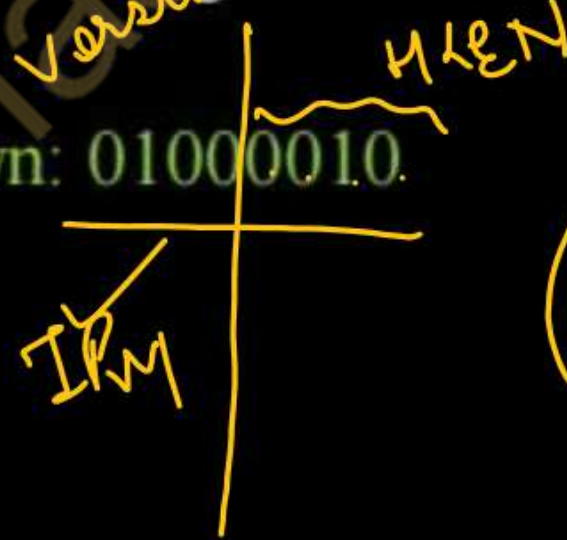
➤ 32-bit SA must remain unchanged during time the IPv4 datagram travels from source to destination.

Destination address:

➤ 32-bit DA must remain unchanged during time the IPv4 datagram travels from source to destination.

Q: An IPv4 packet has arrived with the first 8 bits as shown: 01000010.

The receiver discards the packet. Why?



$$(2) \times 4 = 8 \text{ bytes}$$

Q: In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

$$\text{Header length} = 1000$$

$$\Rightarrow 8 \times 4 = 32 \text{ bytes } (\underline{20} - 60 \text{ bytes})$$

$$\text{options} = 32 - 20 = 12 \text{ byte-long options are added.}$$

Q: In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is Ox0028. How many bytes of data are being carried by this packet?

$$\text{Hlen} = 5 \times 4 = 20 \text{ bytes}$$

$$\text{Total length} = (\underline{\underline{0x0028}})_{16}$$

$$(28)_{16} = (?)_{10}$$

$$2 \times 16^1 + 8 \times 16^0$$

$$= 32 + 8 = 40 \text{ bytes}$$

$$\underline{40 \text{ bytes}}$$

$$\text{Data} = \underline{40} - \underline{20} = \underline{20 \text{ bytes}}$$

Q: An IPv4 packet has arrived with the first few hexadecimal digits as shown.

0x45000028000100000102 ...

How many hops can this packet travel before being dropped? TTL = ?

Leave first 64 bit and read next 8-bit after that to find value of TTL.
16-hex digits 2 hex digits

$$\underline{TTL} \Rightarrow (01)_{16} = (?)_{10}$$

$$= 0 \times 16^1 + 1 \times 16^0$$

$$= 1$$

$$\text{TTL} = 1$$

The packet will travel to only 1 hop before being dropped.

Fragmentation:

- A datagram can travel through different networks.
- The format and size of the sent/received frame depend on the protocol used by the physical network through which the frame travel.
- Each data link layer protocol has its own frame format in most protocols in which maximum size of the data field is defined.
- We must divide the datagram to pass it through these networks. This is called fragmentation.
- When a datagram is fragmented, each fragment has its own header and required parts of the header must be copied by all fragments.

The host or router that fragments a datagram must change the values of three fields:

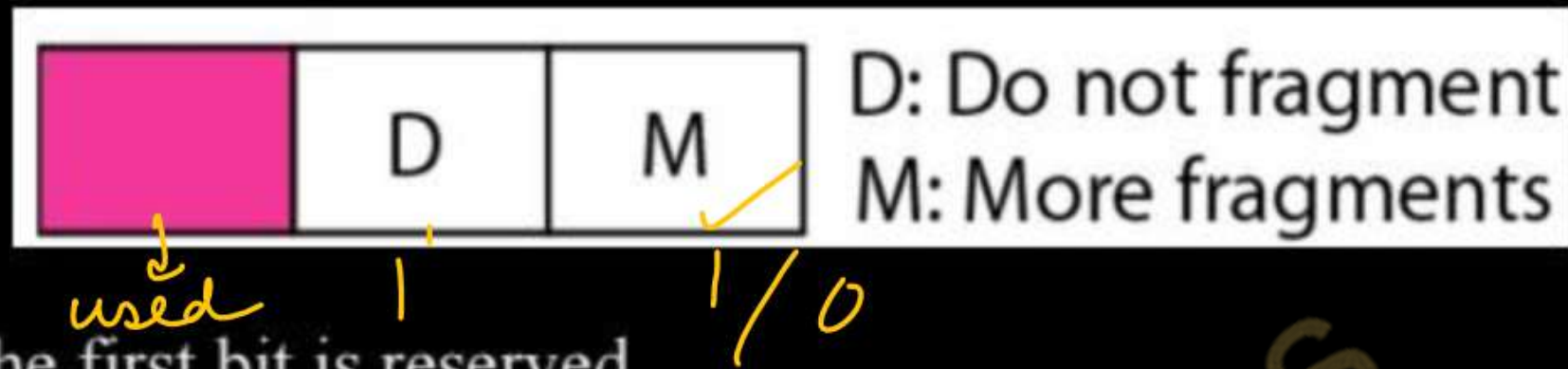
1. Identification *Total length*
2. Flags
3. fragmentation offset

The rest of the fields must be copied and checksum must be recalculated regardless of fragmentation.

1. Identification:

- This 16-bit field with source IPv4 address identifies a datagram from source uniquely.
- The identification number helps the destination in reassembling the datagram.
- IPv4 protocol label the datagrams.
- When a datagram is fragmented, value in the identification field is copied to all fragments.

2. Flags:

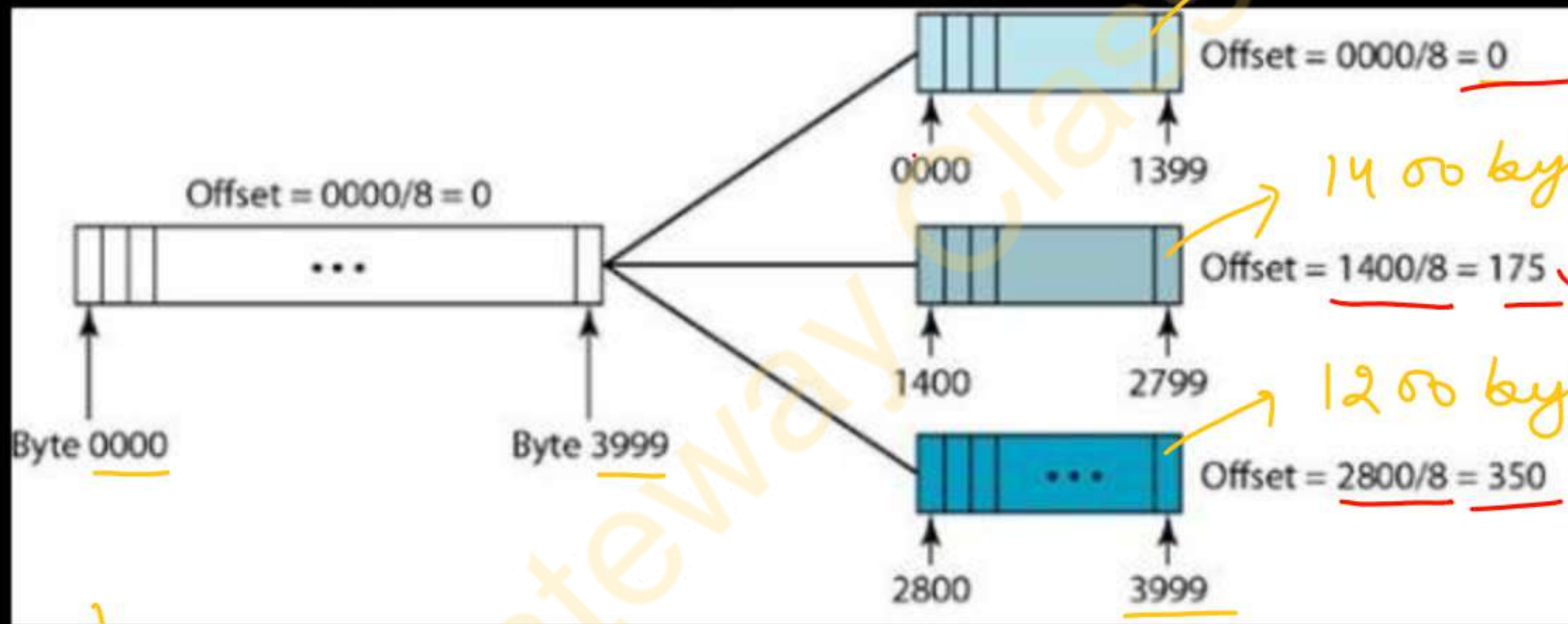


- This is a 3-bit field. The first bit is reserved.
- Second bit is called the **do not fragment** bit.
 - If its value is 1, the machine must not fragment the datagram. If it cannot pass the datagram through any available physical network, it discards the datagram.
 - If its value is 0, the datagram can be fragmented if necessary.
- The third bit is called the **more fragment** bit. If its value is 1, it means the datagram is not the last fragment; there are more fragments after this one. If its value is 0, it means this is the last or only fragment.

3. Fragmentation offset:

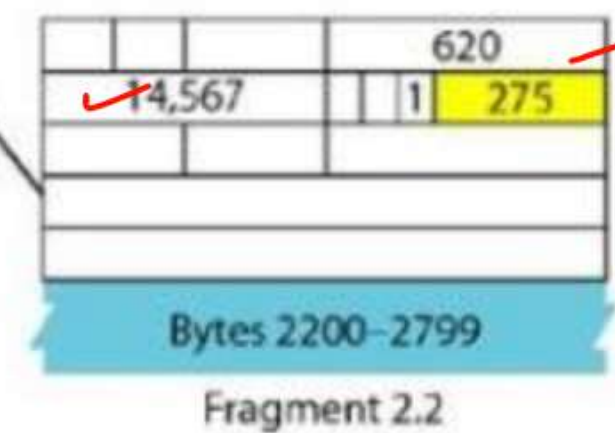
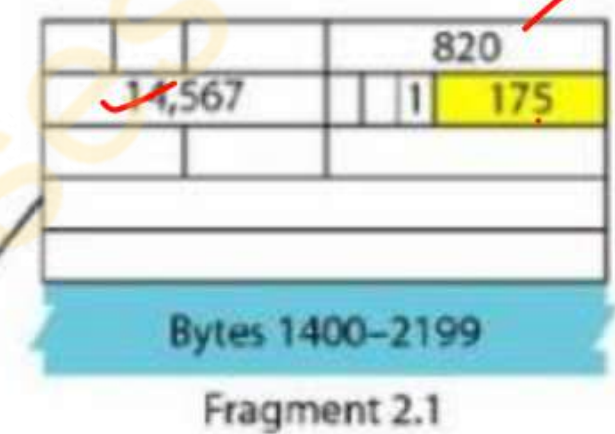
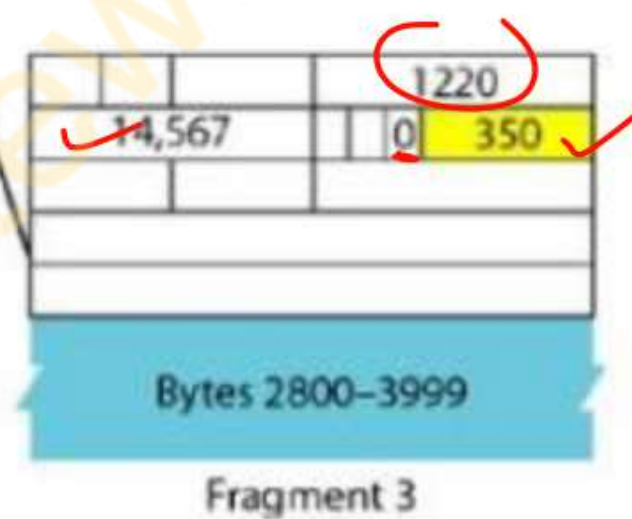
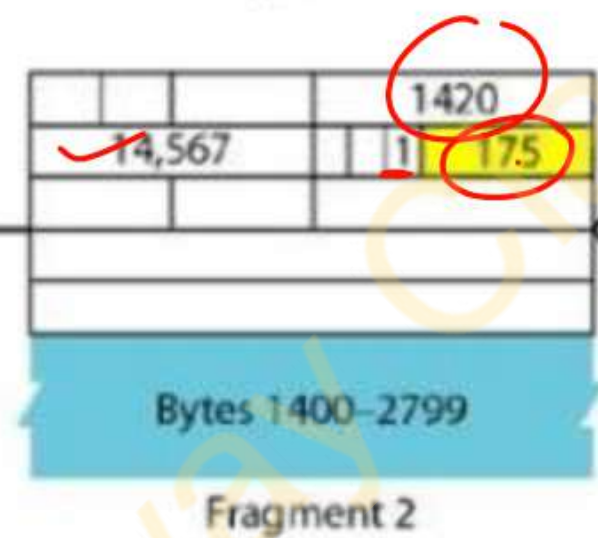
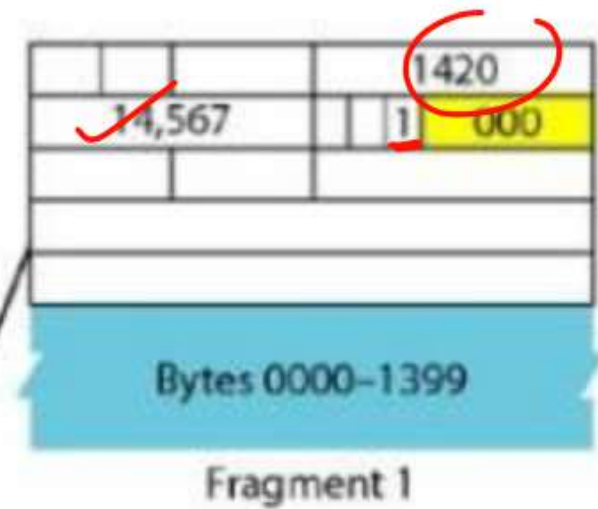
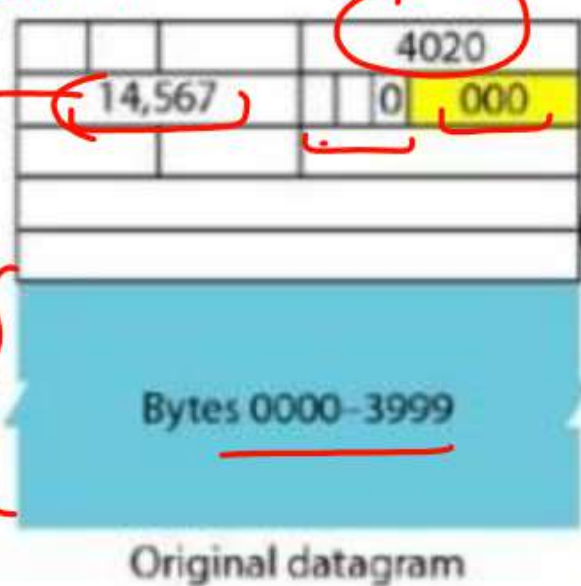
- This 13-bit field shows the relative position of this fragment with respect to the whole datagram.
- It is the offset of the data in the original datagram measured in units of 8 bytes.

pure data = 4000 bytes



f_4	f_3	f_2	f_1
40	60	60	60

identification
Header + data
data



800 byte data
600 byte

➤ Even if each fragment follows a different path and arrives out of order, the final destination host can reassemble the original datagram from the fragments received (if none of them is lost) by using the following strategy:

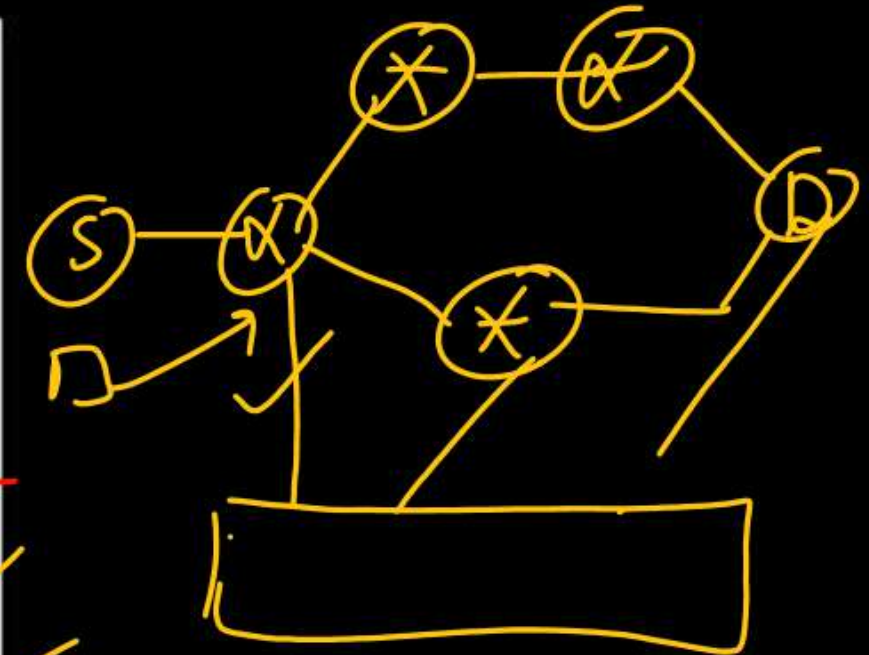
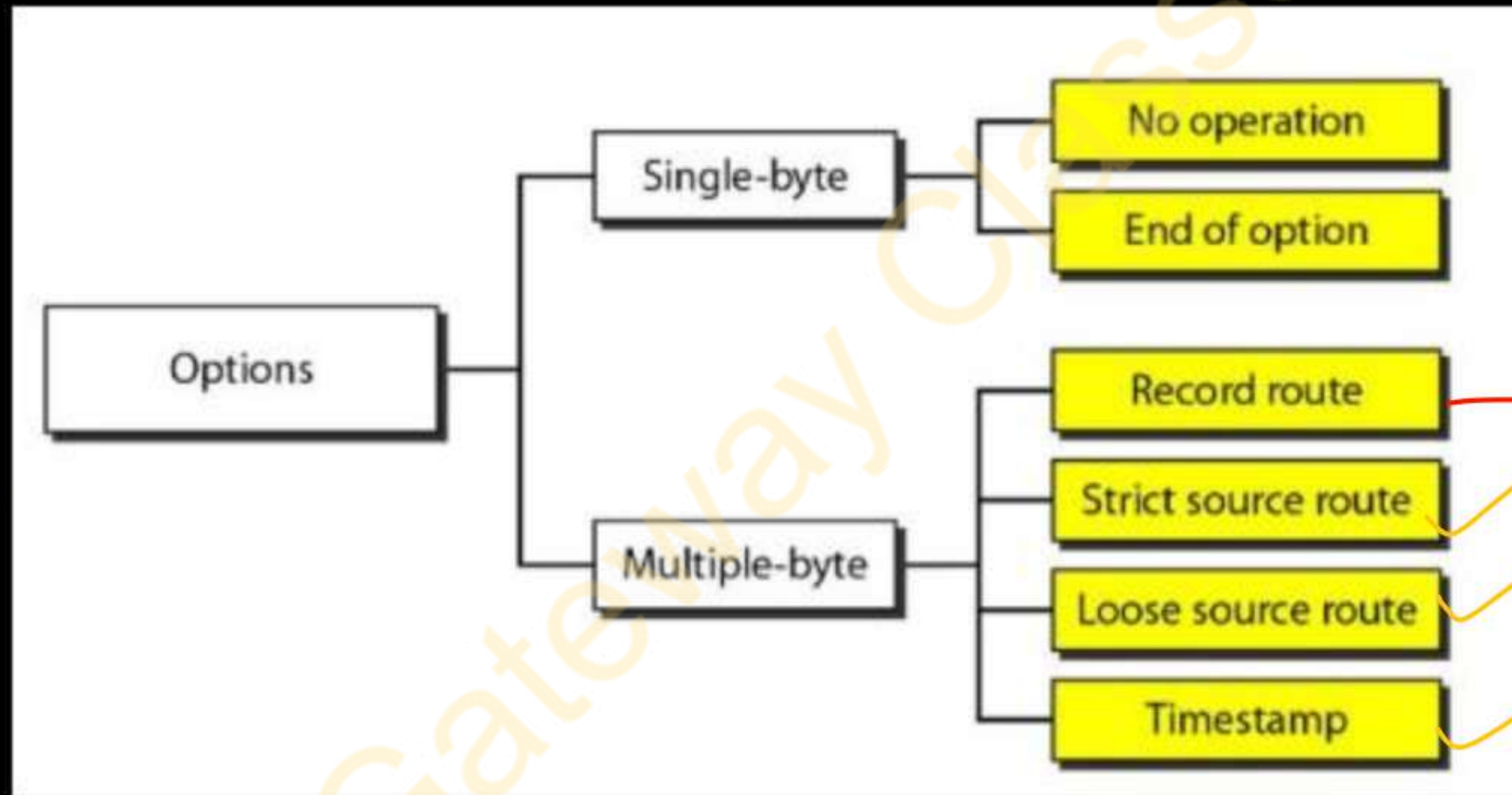
1. The first fragment has an offset field value of zero. ✓
2. Divide length of the first fragment by 8. Second fragment has an offset value equal to that result.
3. Divide total length of the first and second fragments by 8. The third fragment has an offset value equal to that result.
4. Continue the process. The last fragment has a *more* bit value of 0.

Options:

The fixed part of IP header is 20 bytes long.

The variable part comprises the options that can be a maximum of 40 bytes.

Options are not required for a datagram. They can be used for network testing and debugging.



No Operation: Used as a filler between options.

End of Option: Used for padding at the end of the option field. an only be used as the last option.

Record Route

- Used to record the Internet routers (up to nine router addresses) that handle the datagram.
- It can be used for debugging and management purposes.

Strict Source Route

- Used by the source to predetermine a route for the datagram as it travels through the Internet for minimum delay or maximum throughput or more reliable .

Loose Source Route

- Similar to strict source route, but less rigid. Each router in the list must be visited, but the datagram can visit other routers as well.

Timestamp

- Used to record the time of datagram processing by a router(in milliseconds).
- Can help users and managers track the behavior of the routers in the Internet.

AKTU PYQs

1. Define IPv4 header Format in detail. (AKTU 2023-24)
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3. Discuss the role logical addressing. (AKTU 2021-22)

Gateway Classes



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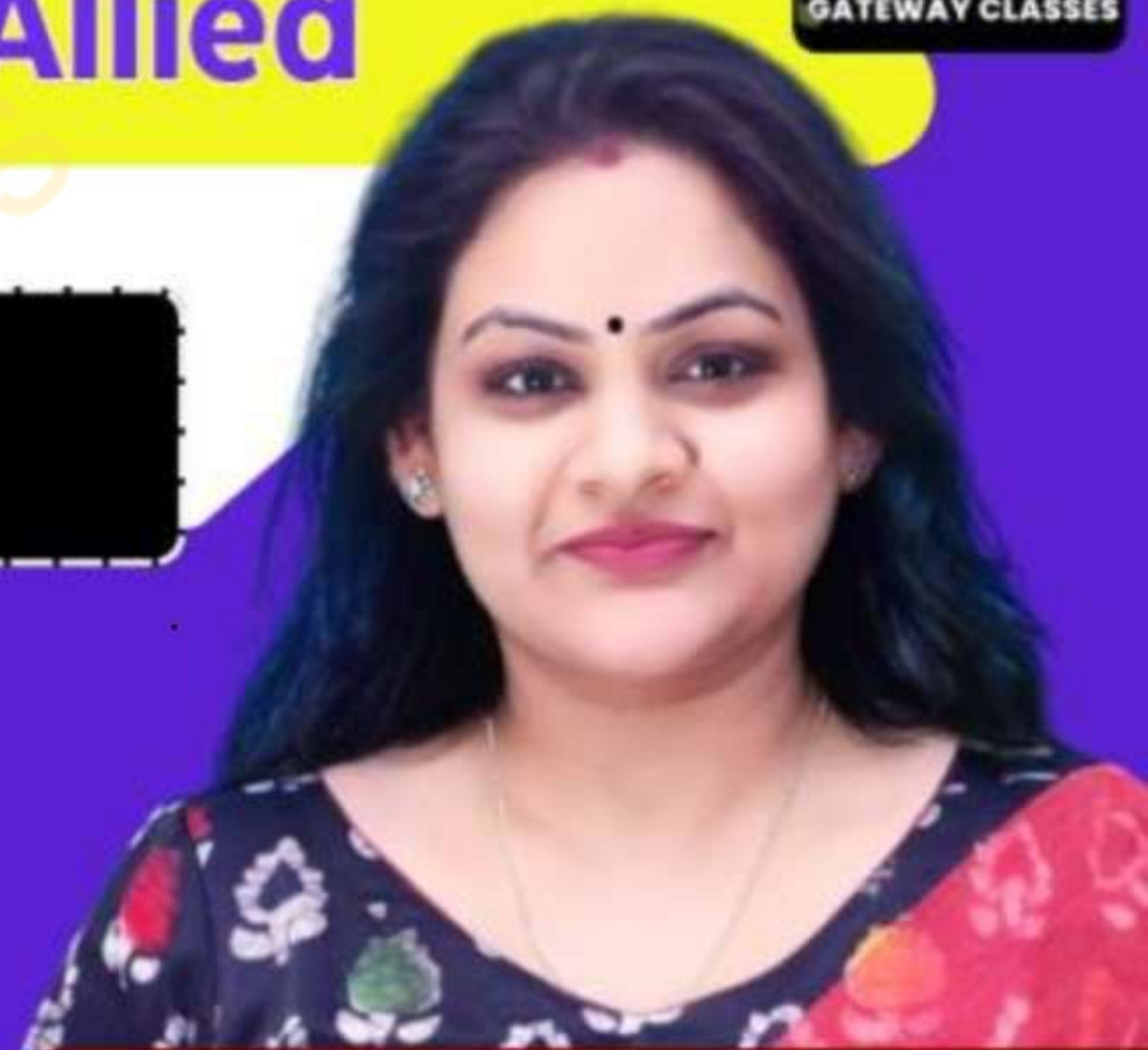
Unit-3: Lecture 2

Today's Target

- IP Addressing ✓
- Classful Addressing ✓
- AKTU PYQs

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- Net Qualified



IPv4 ADDRESSES

- An **IPv4** address is a 32-bit address that *uniquely* and *universally* defines the connection of a device to the Internet.
- An IPv4 address is 32 bits long.
- Two devices on the Internet can never have the same address at the same time.
- If a device operating at the network layer has m connections to the Internet, it needs to have m addresses. Example: router

1 IP address / connection

Address Space:

- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than 4 billion).

2^{32}

Notations

There are two prevalent notations to show an IPv4 address:

1. binary notation
2. Dotted decimal notation

Binary Notation:

In binary notation, the IPv4 address is displayed as 32 bits or a 4-byte address.

01110101 10010101 00011101 00000010
0-255 0-255 0-255 0-255

$$\begin{array}{r} 2^8 \\ \hline 256 \end{array}$$

Dotted-Decimal Notation:

To make the IPv4 address more compact and easier to read, IP addresses are usually written in decimal form with a decimal point (dot) separating the bytes.

117.149.29.2

As each byte (octet) is 8 bits, each number in dotted-decimal notation is a value ranging from 0 to 255.



Q: Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

a. 129.11.11.239

b. 193.131.27.255

Q: Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

Sol: 01101111 00111000 00101101 01001110

Q: Find the error, if any, in the following IPv4 addresses.

a. 111.56.045.78

b. 221.34.7.8.20

c. 75.45.301.14

d. 11100010.23.14.67

Sol: a) 045 is not a valid representation.

b) five bytes are incorrect.

c) The octet value 301 is invalid.

d) We can't mix binary & dotted decimal notation.

Classful Addressing

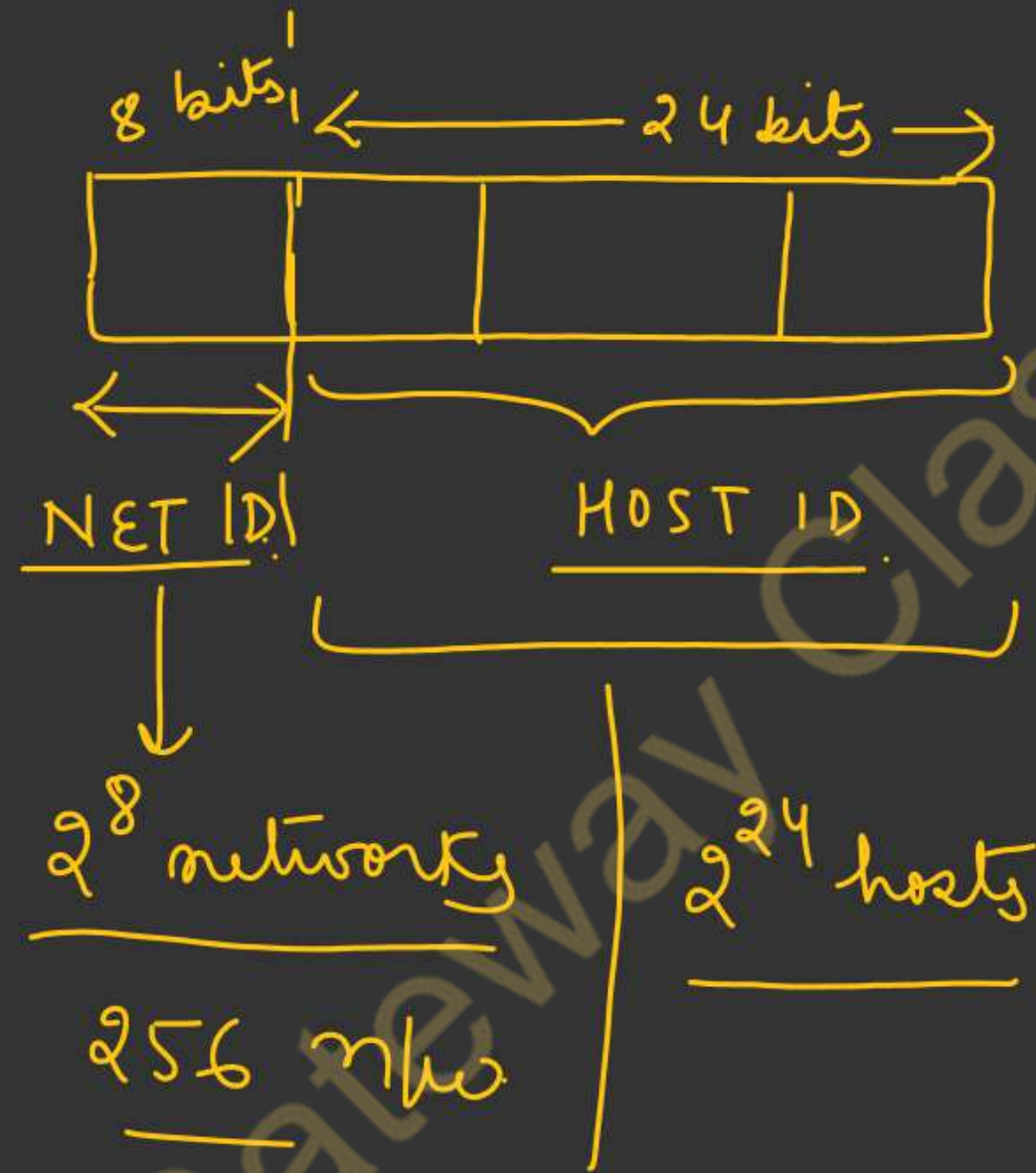
- IPv4 addressing, at its inception, used the concept of classes. This arch. is called classful addressing.
- In classful addressing, the address space is divided into five classes: A, B, C, D, and E.
- Each class occupies some part of the address space.

	First Byte	Second Byte	Third Bytes	Fourth Byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First Byte	Second Byte	Third Byte	Fourth Byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation



Q: Find the class of each address.

a. 00000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

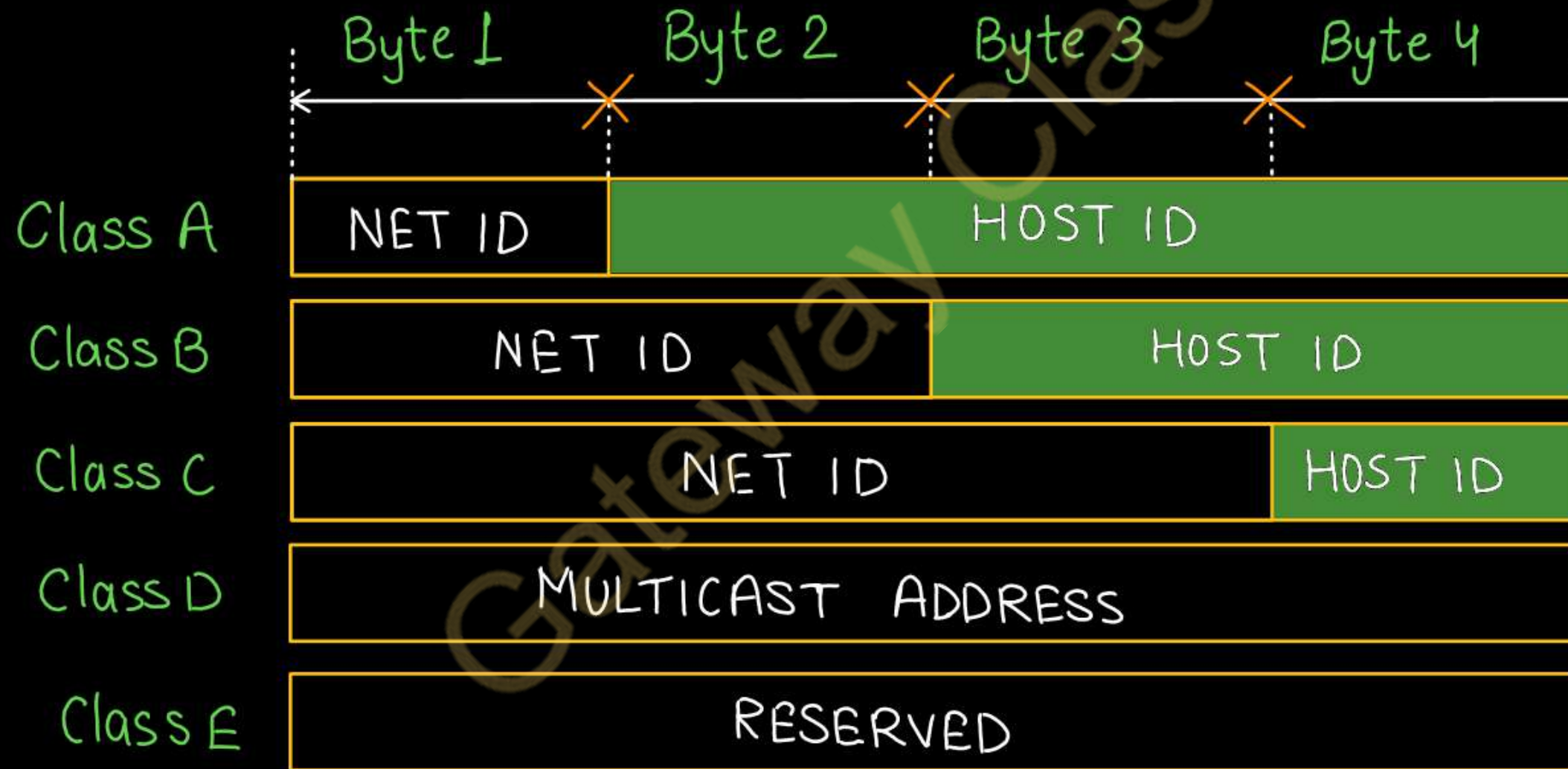
c. 14.23.120.8

d. 252.5.15.111

Sol:

Gateway Classes

- Each of these classes has a valid range of IP addresses.
- Classes D and E are reserved for multicast and experimental purposes respectively.
- The class of IP address is used to determine the bits used for network ID and host ID and the number of total networks and hosts possible in that particular class.



- IP addresses are globally managed by Internet Assigned Numbers Authority (IANA) and Regional Internet Registries (RIR).
- While finding the total number of host IP addresses, 2 IP addresses are not counted and are therefore, decreased from the total count because the first IP address of any network is the network number and whereas the last IP address is reserved for broadcast IP.

↓
address

↓
directed broadcast



Total No. of IP addresses = 2^31

Range → 0 - 127

0 . 0 . 0 . 0

Reserved

127 . 0 . 0 . 0

127.255.255.255

loop back address.
(used for diagnostic
purposes on a single m/c)

How many N/w are possible?

No. of N/ws using 7-bits = $2^7 = 128$

possible N/ws = $128 - 2 = 126$

(IP address)
↑
No of hosts possible in a class A n/w = 2^{24} .

Reason - ex. 32.1.2.3.

1) First IP address of any n/w is used as network ID.

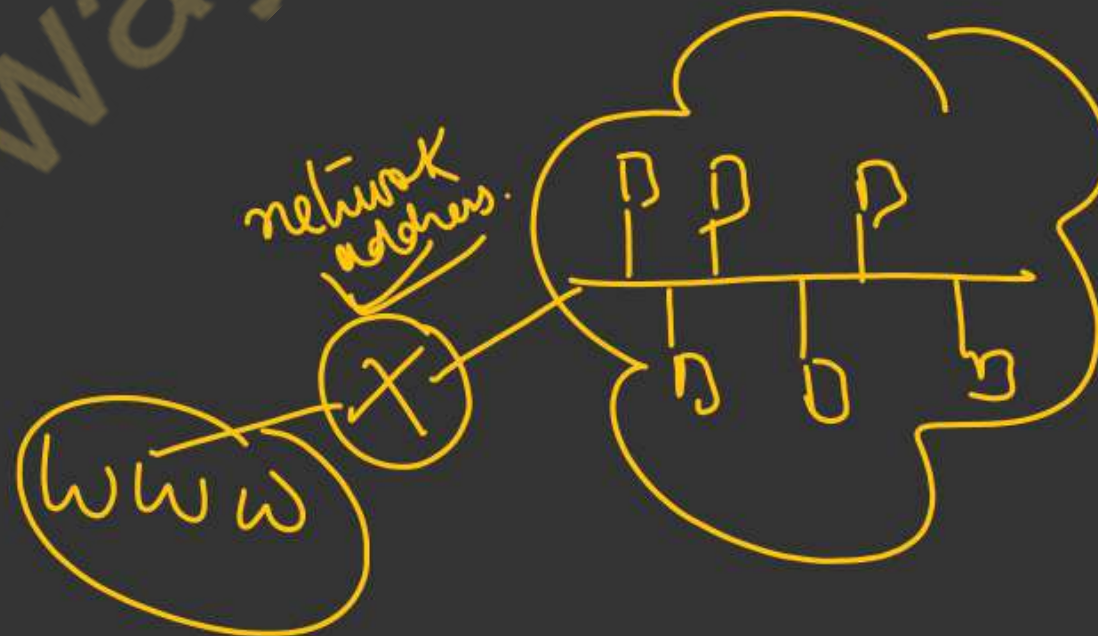
ex. 32.0.0.0.

2) Last IP address of the n/w is used as directed broadcast address.

ex. 32.255.255.255

out of these IP address, IP add.
available for valid hosts are

$$2^{24} - 2$$



Ex. one of the IP address of a class A n/w = 32.1.2.3

Mask for class A $\rightarrow$ 255.0.0.0

First address

AND

32.1.2.3

255.0.0.0

First address
used as Network
address to identify
this n/w over internet

32.0.0.0

last address
is reserved for
directed broadcast address

32.255.255.255

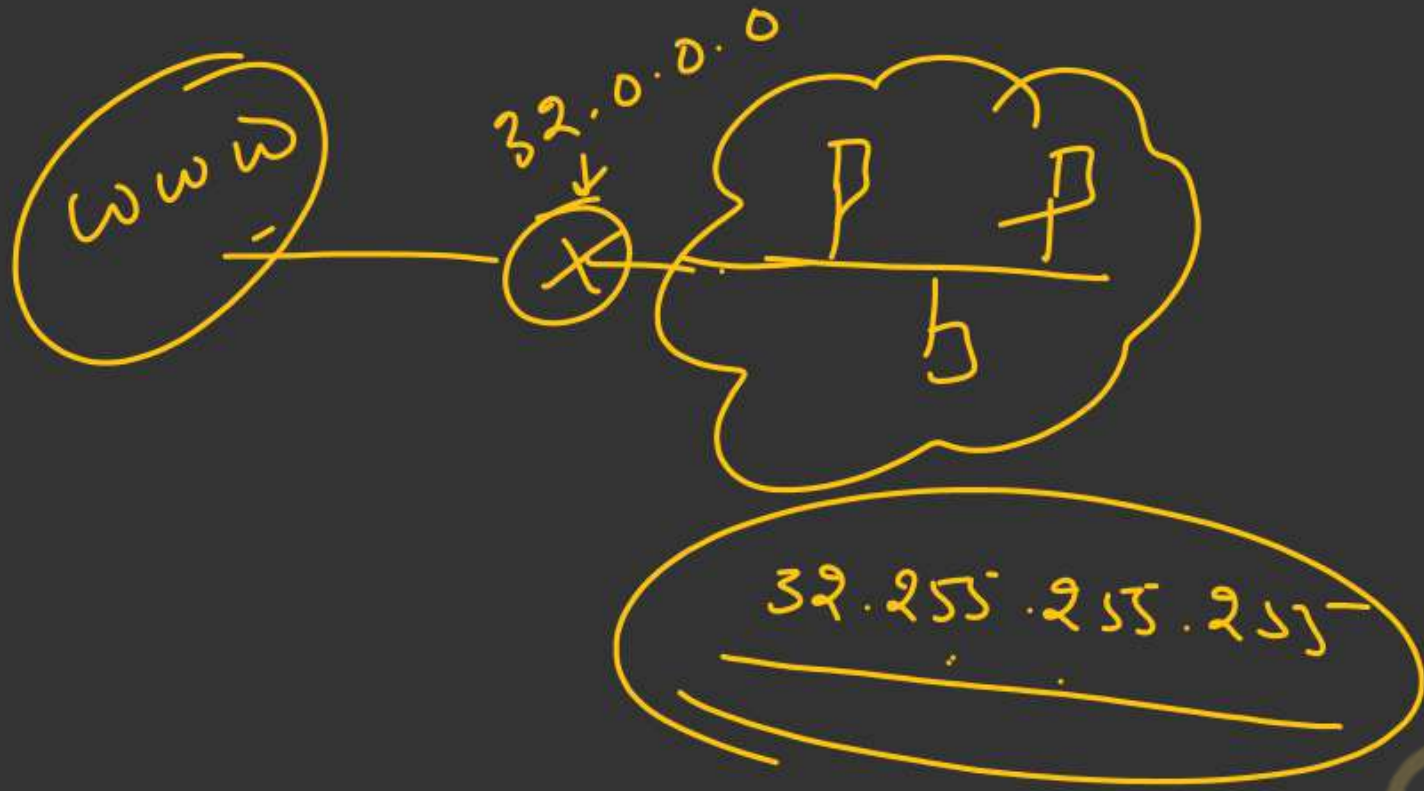
00100000.00000001.00000010.00000011

11111111.00000000.00000000.00000000

00100000.00000000.00000000.00000000

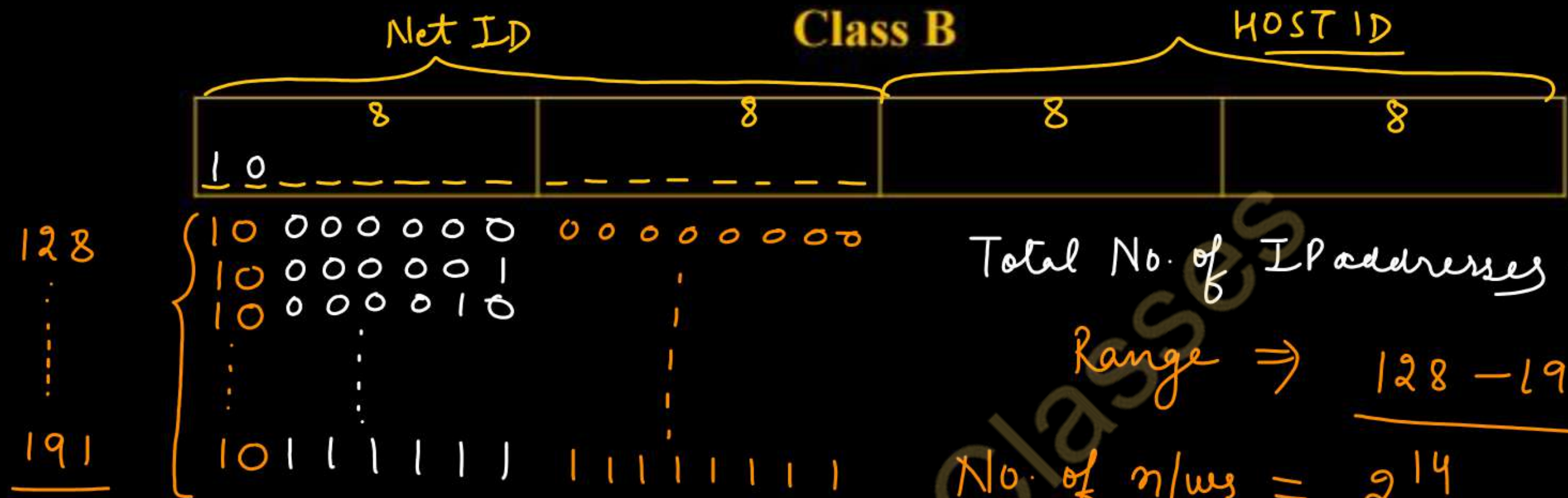
32 . 0 . 0 . 0

Directed Broadcast address



limited Broadcast

255.255.255.255



Total No. of IP addresses = 2^{30}

Range $\Rightarrow$ 128 - 191

No. of n/w = 2^{14}

No of valid host/nw = $2^{16} - 2$

Mask $\Rightarrow$ 255.255.0.0

$\downarrow$
net ID.

host ID.

Class D

1110----			
----------	--	--	--

11100000

11100000

11101111

224-239

Range = 224-239

Total available IP addresses = 2^{28}

Reserved for
multicast groups.

Class E

1111----			
----------	--	--	--

11110000

11110000

11111111

240-255

No. of IP addresses

= 2^{28}

Reserved

Range $\rightarrow$ 240-255

Structure of Classful Addressing

Class	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORKS	START ADDRESS	END ADDRESS
Class A	0	8	24	$2^7 (128)$	$2^{24} (16,777,216)$	<u>0.0.0.0</u>	<u>127.255.255.255</u>
Class B	10	16	16	$2^{14} (16,384)$	$2^{16} (65,536)$	<u>128.0.0.0</u>	<u>191.255.255.255</u>
Class C	110	24	8	$2^{21} (2,097,152)$	$2^8 (256)$	<u>192.0.0.0</u>	<u>223.255.255.255</u>
Class D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	<u>224.0.0.0</u>	<u>239.255.255.255</u>
Class E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	<u>240.0.0.0</u>	255.255.255.255

Mask

- Although the length of the netid and hostid (in bits) is predetermined in classful addressing,
- We can also use a mask (also called the default mask).
- It is a 32-bit number made of contiguous 1s followed by contiguous 0s.
- The masks for classes A, B, and C are given but the concept does not apply to classes D and E.

Class	Binary	Dotted-Decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

- Class A addresses were designed for large organizations with a large number of hosts or routers.
- Class B addresses were designed for midsize organizations with tens of thousands of hosts or routers.
- Class C addresses were designed for small organizations with a small number of hosts or routers.

Q: Suppose IP address 201.20.30.40 belong to any organization. Find network ID, 5th host ID, last host ID, and Broadcast address.

Solⁿ.

IP address $\rightarrow$ 201.20.30.40

1) Finding Net ID :- First address of n/w.

class of n/w = class 'C' n/w (Range of class C
n/w = 192-223)

Mask of class C = 255.255.255.0
AND 201.20.30.40

201.20.30.0 ✓✓

↓
First IP address used as network ID

②

First host address $\Rightarrow$

200.20.30.1

2nd host $\Rightarrow$

200.20.30.2

3rd host $\Rightarrow$

200.20.30.3

$\rightarrow$ will remain fixed as they represent the network part.

...

5th host $\Rightarrow$ 200.20.30.5

...

(254th host)

last host $\Rightarrow$ 200.20.30.254

Directed broadcast

address (last address) $\Rightarrow$

200.20.30.255

AKTU PYQs

Q: Discuss about the IP ranges of Class A, B, C and D. (AKTU 2021-22) 2 marks

Q: If a class B network on the internet has a subnet mask of 255.255.248.0. What is the maximum number of hosts per subnet? (AKTU 2018-19) 2 marks

Q: What is IP Addressing? How it is classified? How is subnet addressing performed?(AKTU 2018-19)

10 marks

Gateway Classes



AKTU

B.Tech 6th Semester
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Computer Networks

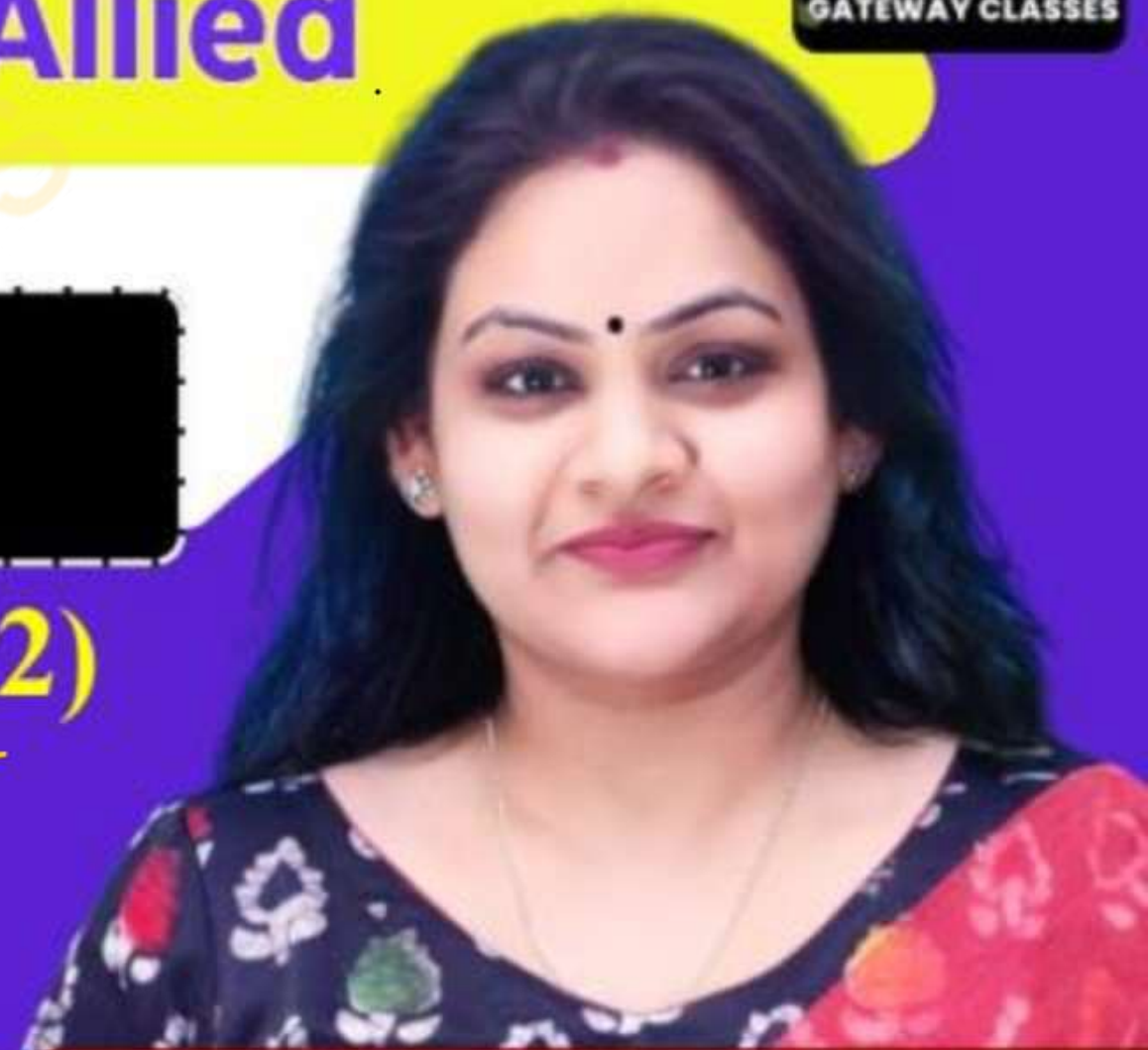
Unit-3: Lecture 3(Part1+part2)

Today's Target

- Subnetting in Classful Addressing ✓
- AKTU PYQs

By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Subnetting

- A subnet is like a smaller group within a large network.
- It is a way to split a large network into smaller networks so that devices present in one network can transmit data more easily.
- For example, in a company, different departments can each have their own subnet, keeping their data traffic separate from others.
- Subnet makes the network faster and easier to manage and also improves the security of the network.

The subnetting process

- Work with host portion of the original network (do not touch the original network portion).
- Convert some of the leftmost host bits into subnet bits.

How many host bits to convert to subnet bits?

$$2^{\text{subnet bits}} = \text{number of subnets created}$$

$$2^{\text{host bits}} - 2 = \text{number of hosts per subnet}$$

For example:



$$\text{No. of subnets} = 2^1 = \underline{2}$$

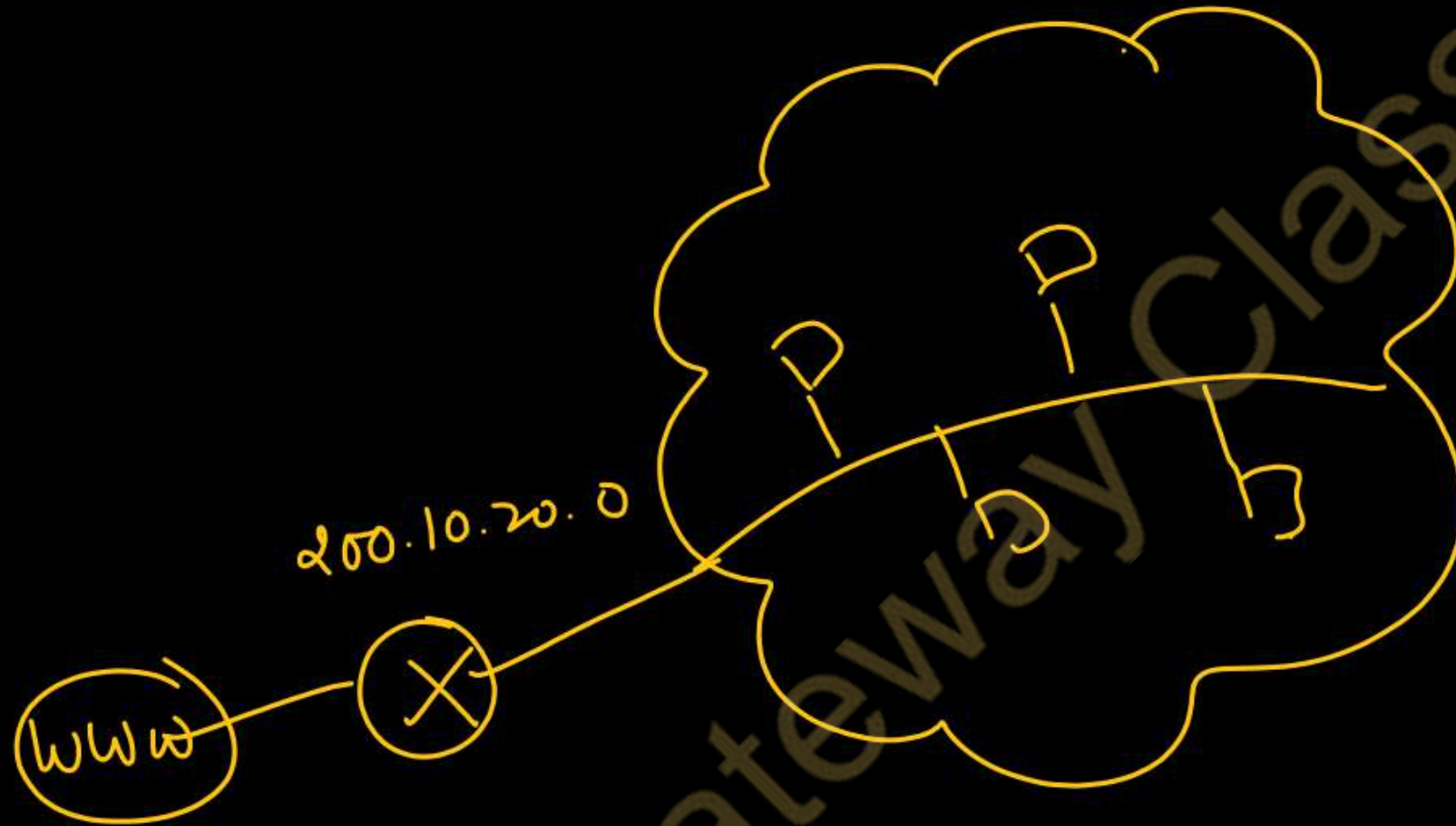
$$\text{No. of valid hosts/subnet} = 2^7 - 2 = 126 \text{ hosts/subnet}$$

- Number of subnet bits must be sufficient to create as many sub-networks as required. Remaining host bits must also be sufficient to ensure that we have as host IP addresses as needed.

Example

Suppose company owns IP addresses starting from 200.10.20.0 .Divide the network into two subnets.

Boundaries of this network are: Network ID: 200.10.20.0 , Broadcast address: 200.10.20.255

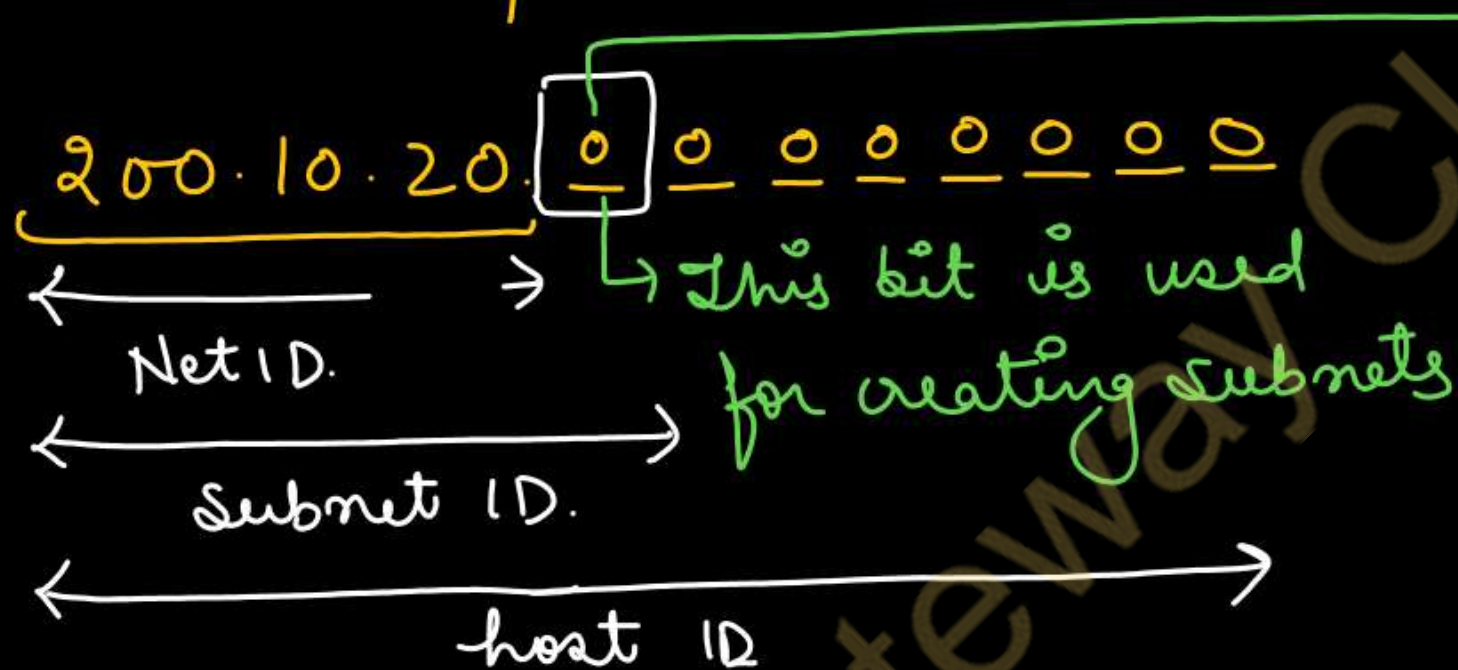


N/w ID = 200.10.20.0

↓
class C IP address

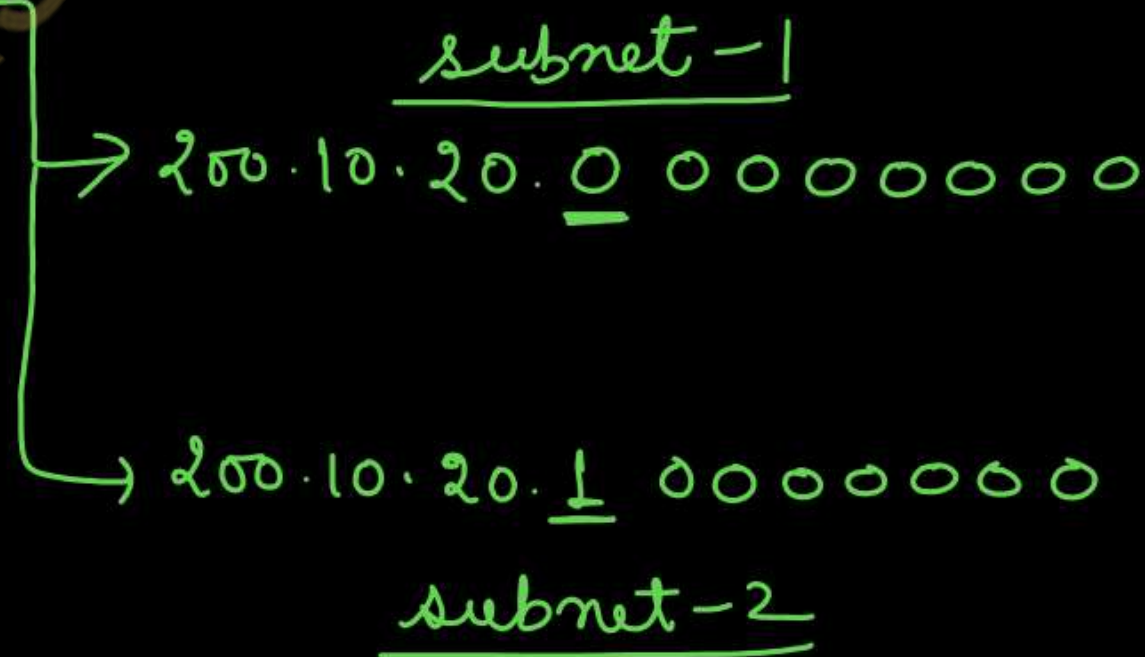
200.10.20.0

N/w part host part

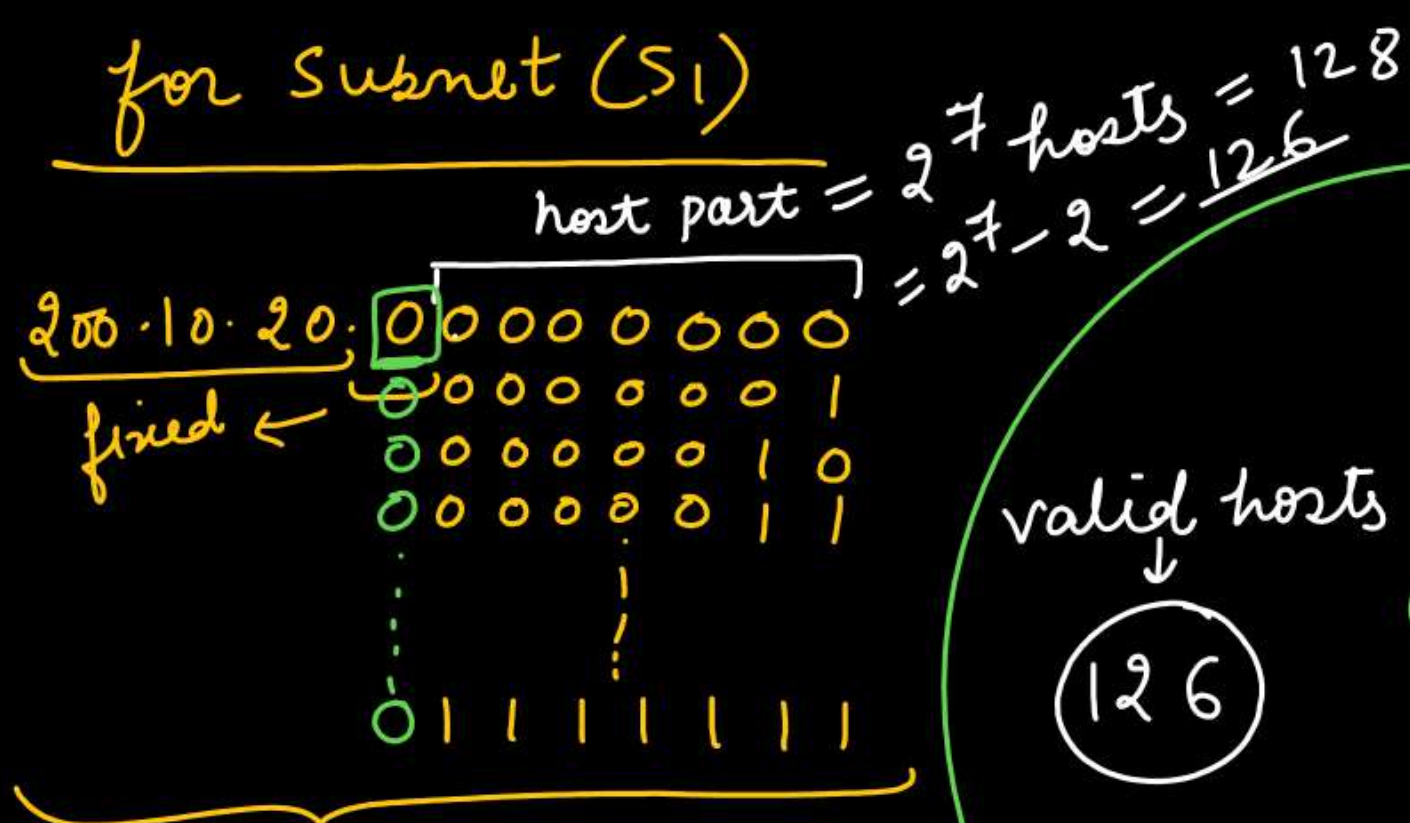


using n bits, no. of subnets = (2^n)

$n=1$, no. of subnets = $2^1 = 2$



for subnet (S₁)



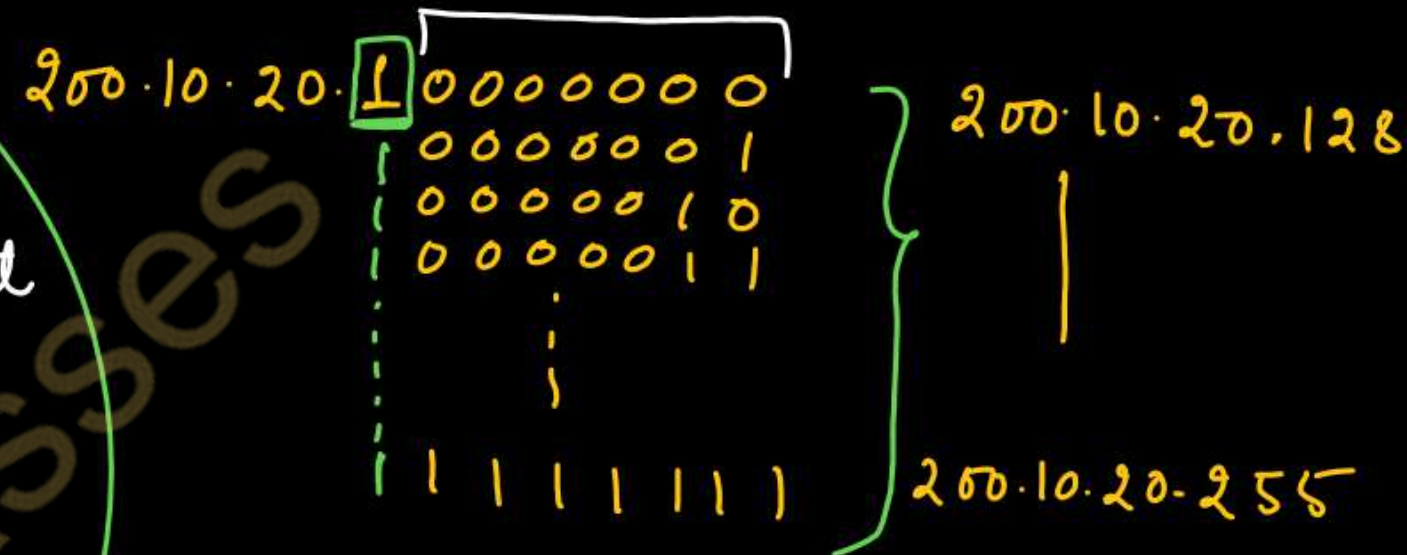
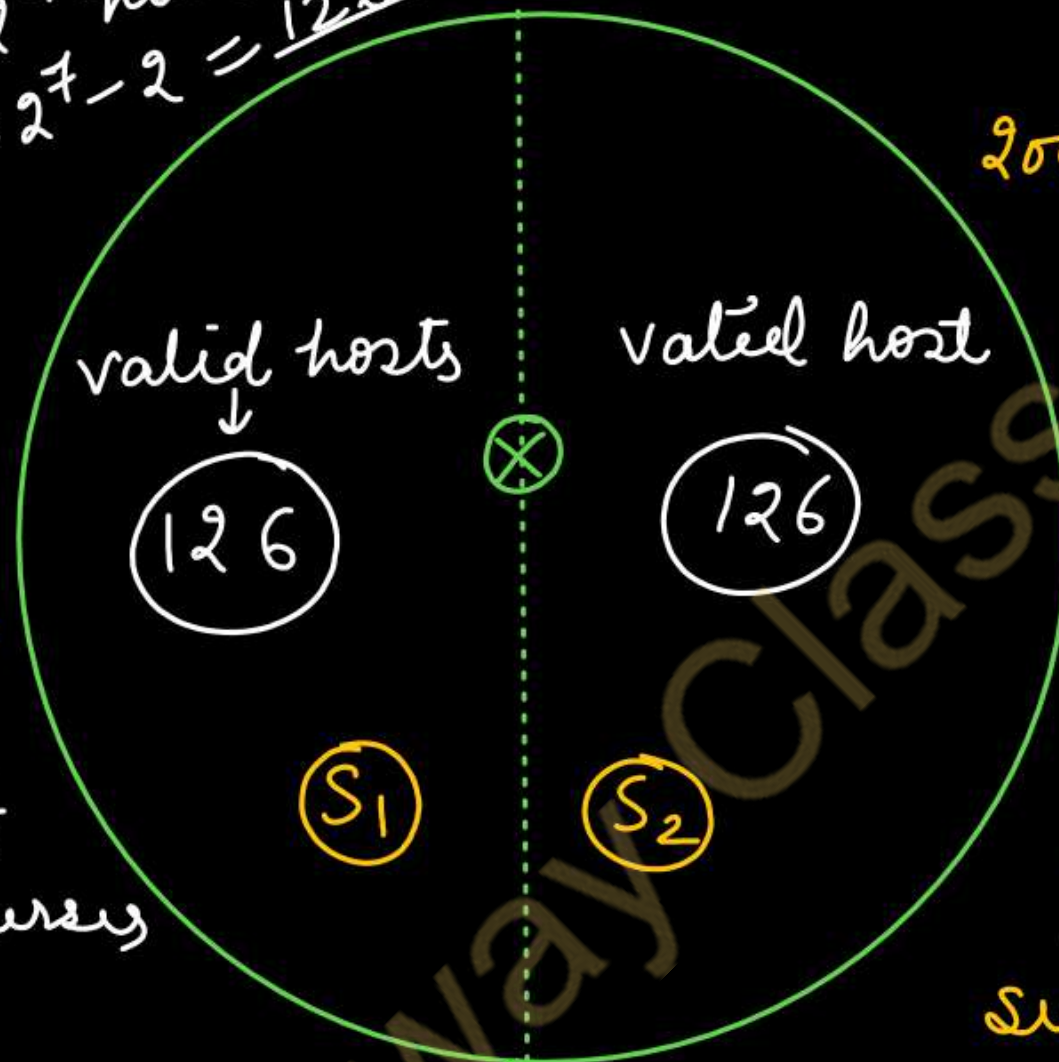
200.10.20.0
 |
 200.10.20.127
 ⇒ Range of IP addresses in S₁

Subnet ID ⇒ 200.10.20.0

Directed broadcast

address of S₁ ⇒ 200.10.20.127

valid host IP addresses =
 in S₁



200.10.20.128
 |
 200.10.20.255
 ⇒ Range of IP addresses in S₂

Subnet ID ⇒ 200.10.20.128

Directed broadcast address = 200.10.20.255

valid host in S₂ = $128 - 2 = 126$

Subnet Mask

No. of bits used for subnet ID. $\Rightarrow$ Mark all 1's.

$200.10.20. \boxed{0}0000000$
 $\underbrace{\hspace{1.5cm}}$

$255.255.255. \underline{1}0000000$



$255.255.255.128$

$\Rightarrow$ subnet mask for S_1, S_2

Example (I) Suppose a packet with destination address $\Rightarrow$ 200.10.20.64

To find the subnet

Subnet Mask $\Rightarrow$ 255.255.255.128

AND operation

200.10.20.01000000

255.255.255.10000000

200.10.20.00000000

200.10.20.0 $\Rightarrow$ subnet ID of S₁
so, packet will be delivered in S₁

(II) Suppose packet arrives with destⁿ address = 200.10.20.130

To find subnet

AND

200.10.20.10000010

255.255.255.10000000

200.10.20.10000000

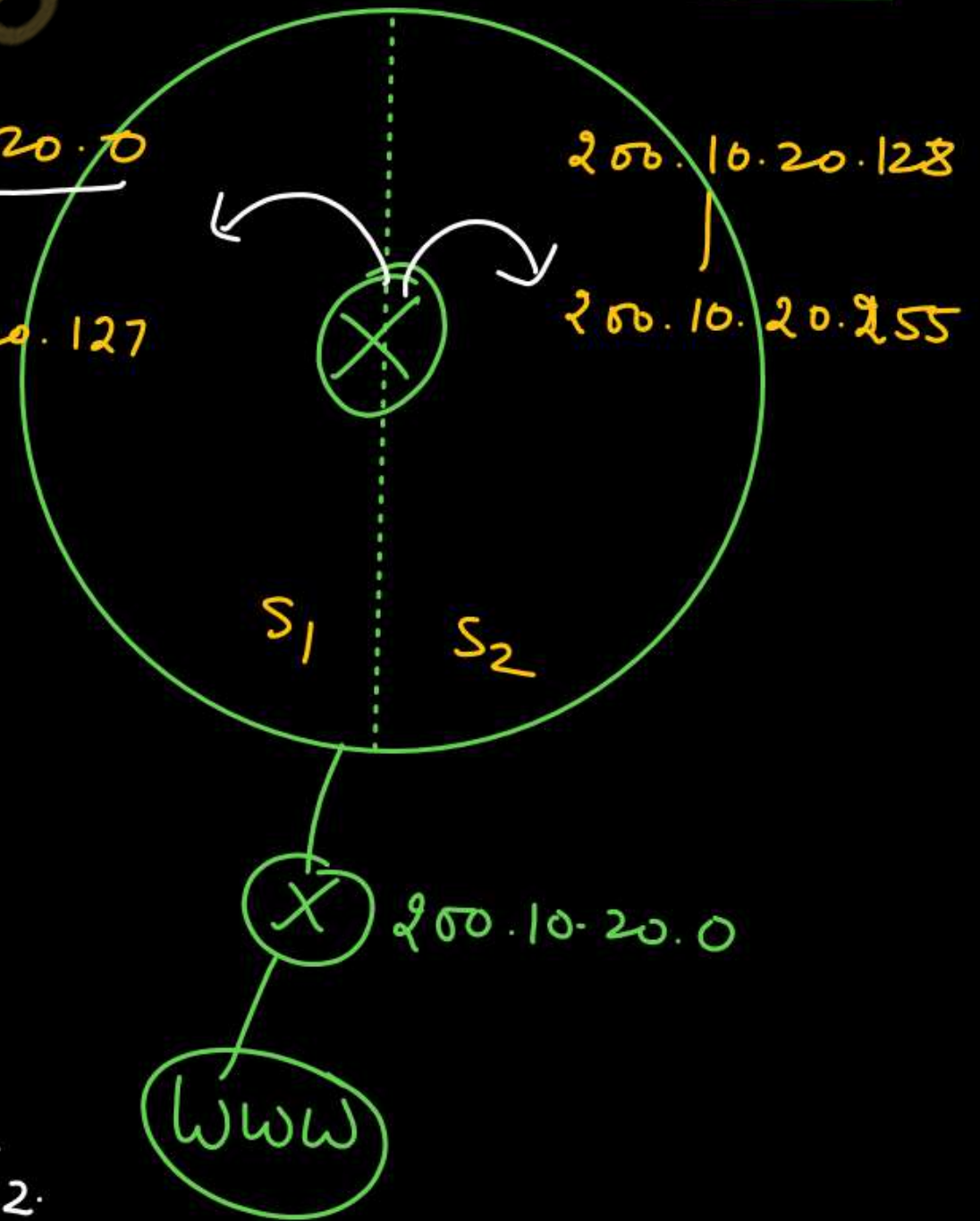
200.10.20.128 $\Rightarrow$ subnet ID for S₂
 $\downarrow$
packet is delivered in S₂

200.10.20.0

200.10.20.127

200.10.20.128

200.10.20.255



Limited Broadcast Address:

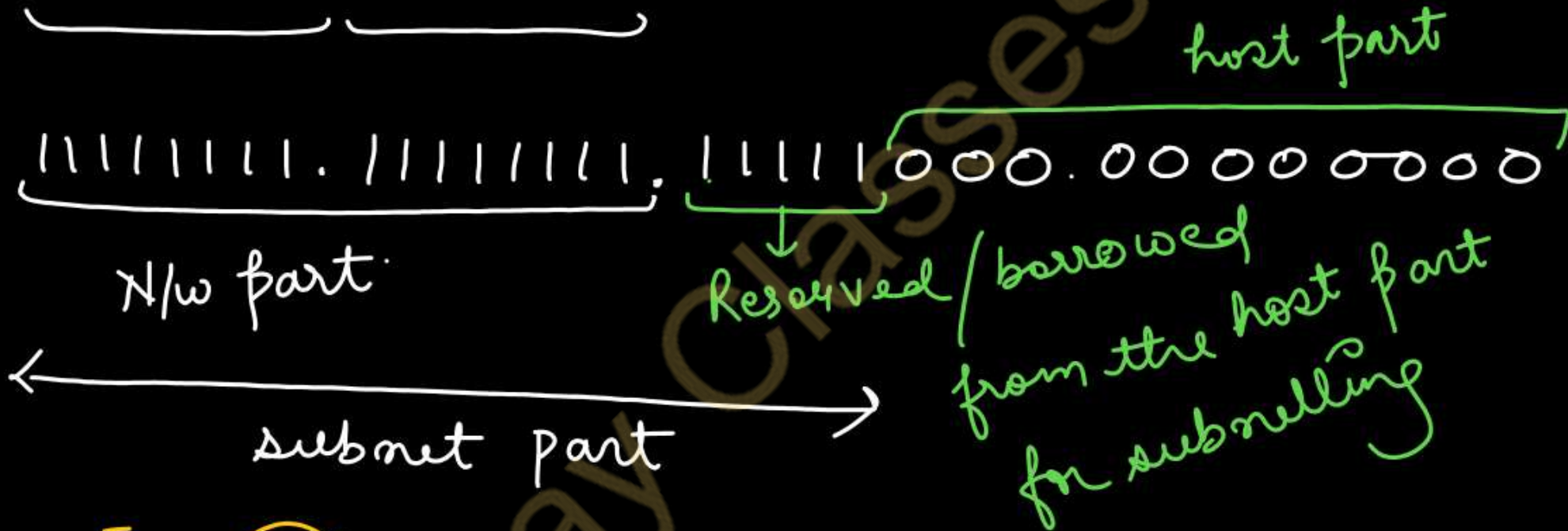
- The Limited Broadcast IP address is **255.255.255.255**.
- A host use this IP address to send a packet to every node on the Local Network.

Directed Broadcast Address

Hosts can use directed broadcast IP address of a foreign network to direct a broadcast to every node in that foreign network. Hence, the term directed broadcast (or sometimes targeted broadcast).

Q1: If a class B network on the internet has a subnet mask of 255.255.248.0. What is the maximum number of hosts per subnet? (AKTU 2018-19) 2 marks

Subnet Mask = $\underbrace{255.255.248.0}_{\text{N/w part}}$



$$\text{No. of subnets} = 2^5 = \underline{32}$$

$$\text{No. of hosts/subnet} = 2^{11} - 2 \Rightarrow 2048 - 2 = \underline{\underline{2046}}$$

Q-2: Given the network address 192.168.1.0 using subnet mask 255.255.255.192. How many subnets are there? How many valid hosts are there in each subnet?

Solⁿ

N/w address $\Rightarrow 192.168.1.0$

(AKTU 2022-23, 2018-19)

$\Downarrow$
class C IP address.

$\Downarrow$
n/w Mask $\Rightarrow 255.255.255.0$

Subnet Mask = $255.255.255.192$

$\Downarrow$
 $255.255.255.11000000$
 $\downarrow$ subnet bits

No. of subnets = $2^2 = 4$

Valid hosts / subnet = $2^{\text{host bits}} - 2 = 2^6 - 2 = 62 \checkmark$

Given the IP address 180.25.21.172 and the subnet mask 255.255.192.0, what is the subnet address?

AKTU 2017-18

IP address :- 180.25.21.172



class B IP address

Subnet Mask :- 255.255.192.0

AND

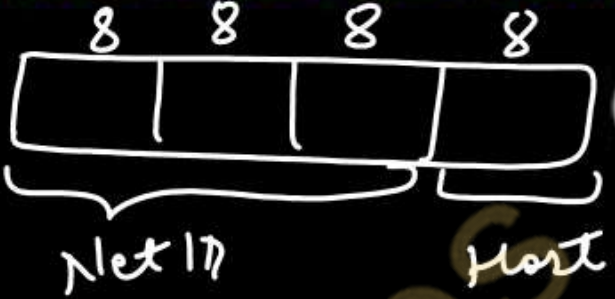
255.255.11000000.0

180.25.00010101.172

180.25.0.0 ⇒ subnet ID

Q4: The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Draw the subnets.

Solⁿ

$200.198.160.0 \Rightarrow$ class C N/w $\Rightarrow$  (AKTU 2022-23, 2018-19)

Subnet Mask = 255.255.255. 1 1 1 0 0 0 0 0

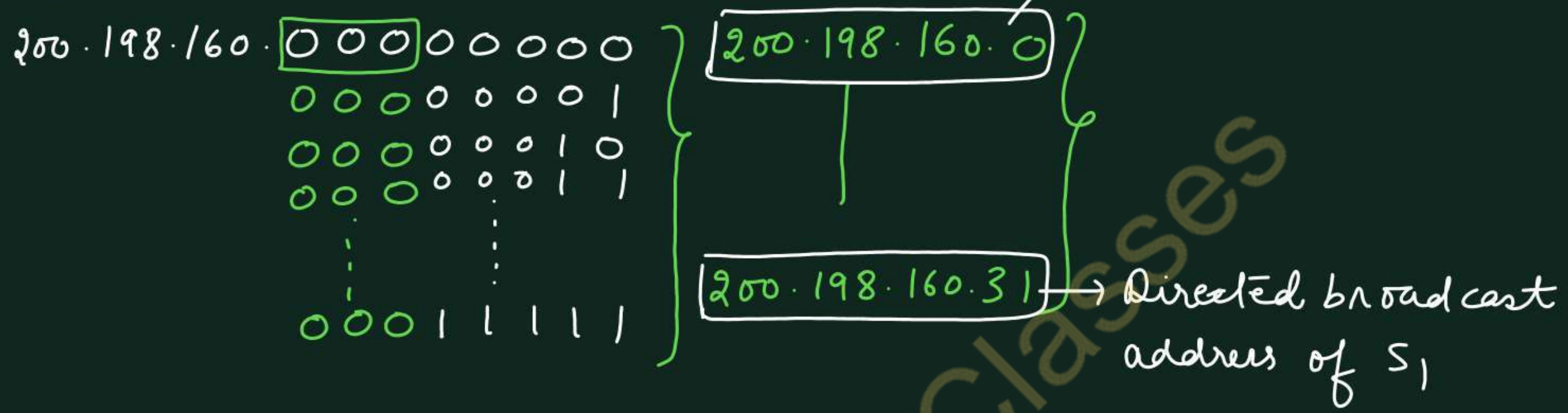
→ subnet bits borrowed from host bits

$$\text{No. of subnets} = 2^{\text{subnet bits}} = 2^3 = \textcircled{8}$$

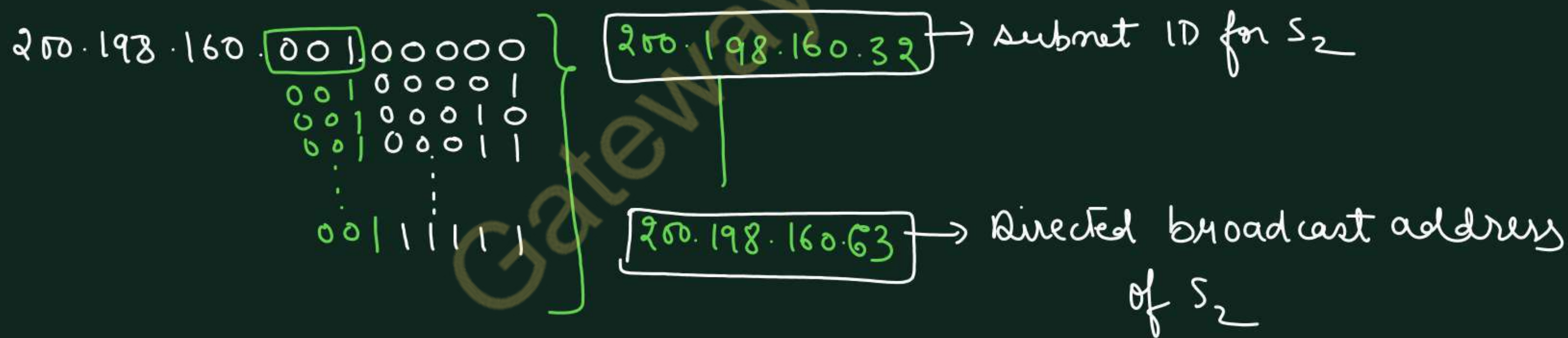
$$\text{No. of hosts/subnet} = 2^{\text{host bits}} = 2^5 = 32 - 2 = \textcircled{30}$$

000 → S₁
001 → S₂
010 → S₃
011 → S₄
100 → S₅
101 → S₆
110 → S₇
111 → S₈

For Subnet 1 (S₁)



For Subnet 2



Subnet $S_3 \rightarrow$ Range of IP addresses

200.198.160.64 ——— 200.198.160.95

subnet ID for S_3

directed broadcast address of S_3

Subnet $S_4 \rightarrow$ 200.198.160.96 ——— 200.198.160.127

Subnet $S_5 \rightarrow$ 200.198.160.128 ——— 200.198.160.159

Subnet $S_6 \rightarrow$ 200.198.160.160 ——— 200.198.160.191

Subnet $S_7 \rightarrow$ 200.198.160.192 ——— 200.198.160.223

Subnet $S_8 \rightarrow$ 200.198.160.224 ——— 200.198.160.255

Q5: Divide the network with IP address 200.1.2.0 into 5 subnets. (AKTU 2021-22) (AKTU 2018-19)

Solⁿ.

N/w IP address $\Rightarrow 200.1.2.0 \Rightarrow \text{class C n/w} \Rightarrow \underbrace{200.1.2}_{\text{net id}}; \underbrace{0}_{\text{host id}}$

class c n/w mark = 255.255.255.0

250.1.2. 0000 00000000

→ subnet bits

No. of subnets = 8

$$\text{valid hosts/subnet} = 2^5 - 2 = \underline{30}$$

Subnet $\rightarrow 200.1.2.0 - 200.1.2.31$

subset $\rightarrow 200.1.2.32 - 200.1.2.63$

$$\text{Subnet 3} \Rightarrow 250.1.2.64 - 250.1.2.95$$

Subnet 4 $\Rightarrow 200.1.2.96 - 200.1.2.127$

~~subnet 5~~ $\Rightarrow 200.1.2.128 - 200.1.2.159$

Reserve subnet 6 $\Rightarrow 200.1.2.160 - 200.1.2.191$

Subnet 7 $\Rightarrow 200.1.2.192 \rightarrow 200.1.2.223$

Subnet 8 $\Rightarrow 200.1.2.224 \rightarrow 200.1.2.255$

VLSM (Variable Length Subnet Masking)

IP address: 200.10.20.0

Aim → S_1, S_2, S_3
↓ ↓ ↓
 $128 + \underline{64} + 64 = \underline{256}$

for S_1

200.10.20. 0 0 0 0 0 0 0 0
↑ subnet bit
7 bits will provide 128 IP addresses
/0/1

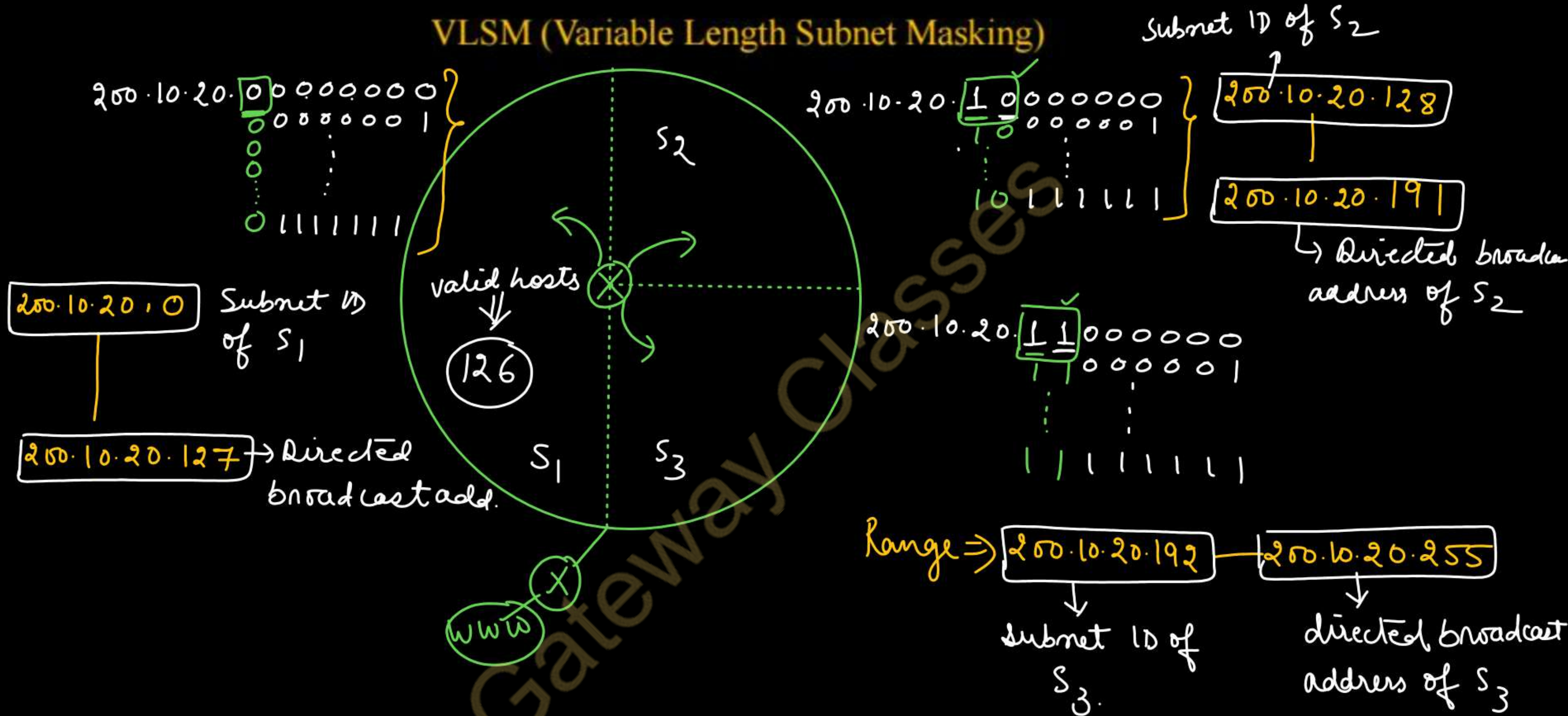
for S_2

200.10.20. 1 0 0 0 0 0 0 0
↑ subnet bits
6 bits will provide 64 IP addresses

for S_3

200.10.20. 1 1 0 0 0 0 0 0
↑ subnet bits
6 bits for 64 IP addresses

VLSM (Variable Length Subnet Masking)



Subnet Mask for $S_1 \Rightarrow$ 11111111 11111111 11111111 10000000
 $\Rightarrow$ 255.255.255.128

Subnet Mask for $S_2 \Rightarrow$ 11111111 11111111 11111111 11000000
255.255.255.192

subnet mask for $S_3 \Rightarrow$ 255.255.255.192

AKTU PYQs

Q: The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Draw the subnets.

(AKTU 2022-23, 2018-19)

Q: Divide the network with IP address 200.1.2.0 into 5 subnets. (AKTU 2021-22) (AKTU 2018-19)

Q: Given the IP address 180.25.21.172 and the subnet mask 255.255.192.0, what is the subnet address?

Q: What is IP Addressing? How it is classified? How is subnet addressing performed? (AKTU 2018-19)



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Computer Networks

Unit-3: Lecture 4

Today's Target

- Classless Addressing ✓
- CIDR Notation ✓
- AKTU PYQs

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Flaw in Classful Addressing

1. A block in class A address is too large for almost any organization. Most addresses were wasted.
2. A block in class B is too large for many of the organizations that received a class B block.
3. A block in class C is probably too small for many organizations.
4. Class D addresses were designed for multicasting. Each address in this class is used to define one group of hosts on the Internet. The Internet authorities wrongly predicted a need for 268,435,456 groups. This never happened and many addresses were wasted here too.
5. class E addresses were reserved for future use; another waste of addresses.

32
2

In classful addressing, a large part of the available addresses were wasted.

Classful addressing, which is almost obsolete, is replaced with classless addressing.

Classless Addressing

- Designed and implemented ^{to} overcome address depletion and give more organizations access to internet.

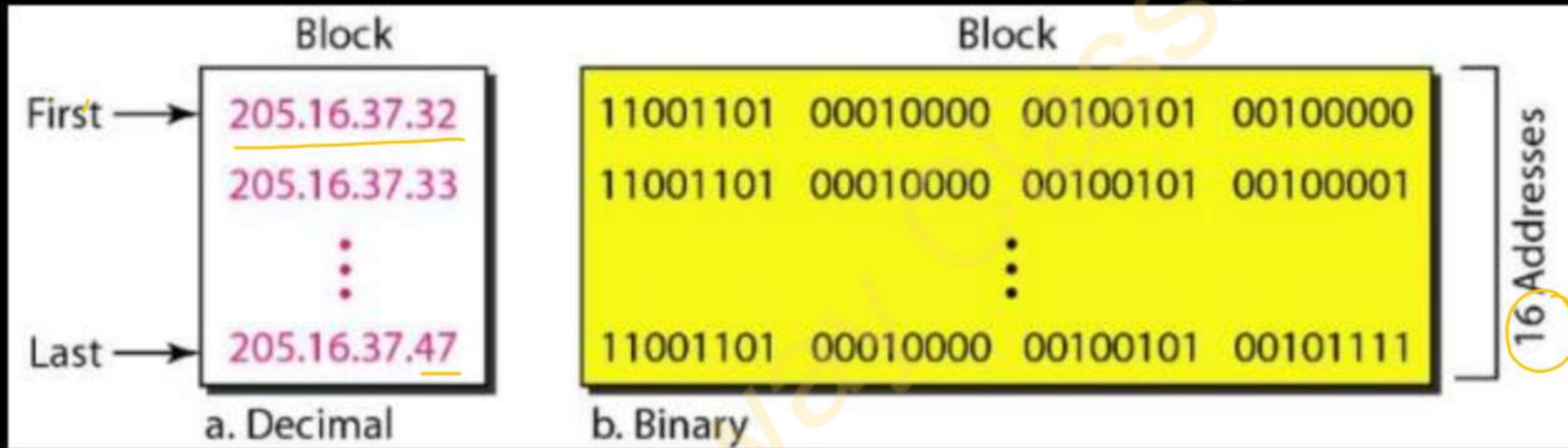
Address Blocks

- In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses.
- The size of the block (the number of addresses) varies based on the nature and size of the entity.
- For example, a household may be given only two addresses; a large organization may be given thousands of addresses. An ISP, as the Internet service provider, may be given thousands or hundreds of thousands based on the number of customers it may serve.

Restriction on classless address blocks:

1. The addresses in a block must be contiguous, one after another.
2. The number of addresses in a block must be a power of 2 (1, 2, 4, 8, ...).
3. The first address must be evenly divisible by the number of addresses.

Example:



205.16.37.32 ✓

205.16.37.00100000

4-bits are 0

↓
address is divisible by 2^4

✓
 $2^4 = 16$ ✓

classful Addressing

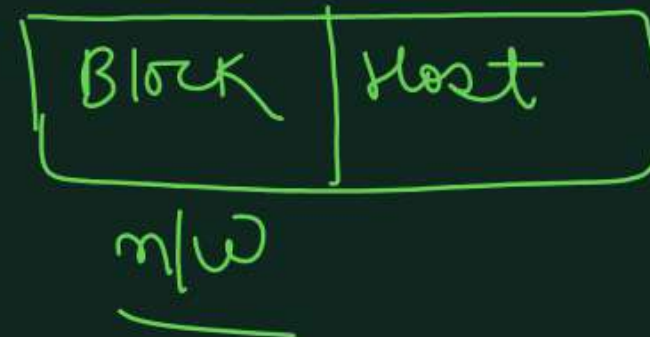
class A : $\overbrace{255}^{N/w} . \overbrace{0.0.0}^{host}$
class B : $\overbrace{255.255}^{N/w} . \overbrace{0.0}^{host}$
class C : $\overbrace{255.255.255}^{N/w} . \overbrace{0}^{host}$

classless Addressing

CIDR Notation

$x.y.z.w / n$ → Mask.
IP address

↓
It tells the number of n/w bits



Mask:

- Mask is a 32-bit number in which the n leftmost bits are 1s and the $32 - n$ rightmost bits are 0s.
- Mask represents how many bits are used for network part.
- In classless addressing, the mask for a block can take any value from 0 to 32.
- We give just the value of n preceded by a slash (**CIDR notation**).

In IPv4 addressing, a block of addresses can be defined as



- $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.
- The address and the $/n$ notation completely define the whole block (first address, last address, and the number of addresses).
- **First Address:** The first address in the block can be found by setting the $32 - n$ rightmost bits in the binary notation of the address to 0.

Last Address: The last address in the block can be found by setting the $32 - n$ rightmost bits in the binary notation of the address to 1s.

Number of Addresses: 2^{32-n}

Q: A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address, last address in the block and number of addresses?

Solⁿ.

205.16.37.39 / 28 $\rightarrow n$ (Mask)
IP address

To find first address

$32 - 28 = 4 \Rightarrow$ Set Rightmost 4-bits to 0.

1) First address:-

205.16.37.00100111

205.16.37.00100000

205.16.37.32 ✓✓

② Last Address:- $205.16.37.39/28$

$32-28=4 \Rightarrow$ To find last address set rightmost 4-bits to 1.

$205.16.37.00100111$
 $\Downarrow$
 $205.16.37.00101111$
 $\Downarrow$
 $205.16.37.47 \Rightarrow$ last address

③ No. of addresses:- $2^{32-n} = 2^{32-28} = 2^4 = 16$

Block:- $\sqrt{205.16.37.32/28} \text{ --- } 205.16.37.32/28$

Another way to find the first address, the last address, and the number of addresses:

1. The first address can be found by bit by bit ANDing the given addresses with the mask.

205.16.37.39/28 → mask

AND

11001101 00010000 00100101 00100111
11111111 11111111 11111111 11110000

11001101 00010000 00100101 00100000

205.16.37.32/28 ⇒ First address

2. The last address can be found by bit-by-bit ORing the given addresses with complement of mask.

OR

11001101 00010000 00100101 00100111
00000000 00000000 00000000 00001111

11001101 00010000 00100101 00101111

↓
205.16.37.47 ⇒ Last address

3. The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Address:- 205.10.37.39/ —

Mask = /28

Mask = 11111111 11111111 11111111 11110000

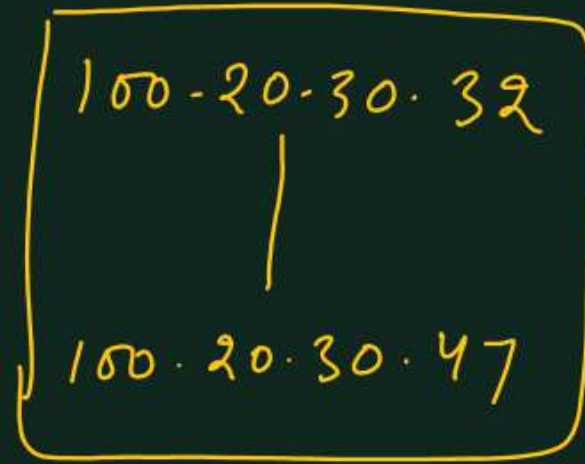
↓ complement of Mask

00000000 00000000 00000000 00001111

↓ convert into decimal

15 → 15 + 1 = 16 ✓

Example -



block is assigned
to a org.

1) First address -

Gateway Classes

Classless Inter Domain Routing (CIDR)

- CIDR is a method of IP address allocation for more efficient use of IP addresses.
- CIDR is based on the idea that IP addresses can be allocated based on their network prefix rather than their class.
- CIDR addresses are represented using a slash notation, which specifies the number of bits in the network prefix.
- For example, an IP address of 192.168.1.0 with a prefix length of 24 would be represented as 192.168.1.0/24. This notation indicates that the first 24 bits of the IP address are the network prefix and the remaining 8 bits are the host identifier.

Q: Suppose an organization is given block 17.12.40.0/26, which contains 64 addresses. The organization has three offices and needs to divide the addresses into three subblocks of 32, 16, and 16 addresses.

Solⁿ -

17.12.40.0 / 26 $\rightarrow n$

$$\text{no. of addresses} = 2^{32-n} \Rightarrow 2^{32-26} = 2^6 = 64$$

First Block :- No. of addresses = 32

✓ / 27

$$2^{32-n} = 32$$

$$\Rightarrow 2^{32-n} = 2^5 \Rightarrow 32-n=5$$

$$\Rightarrow n = 32-5$$

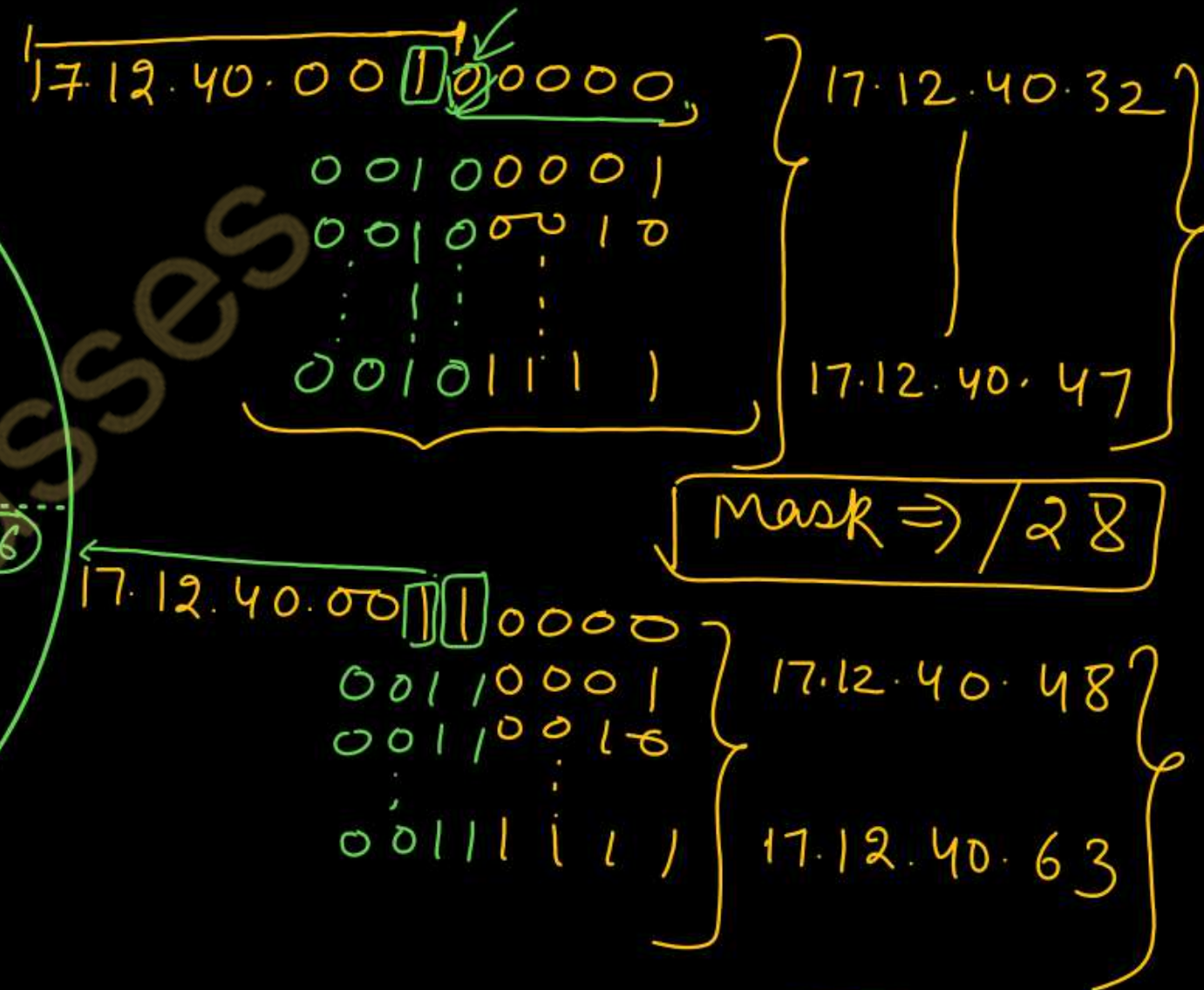
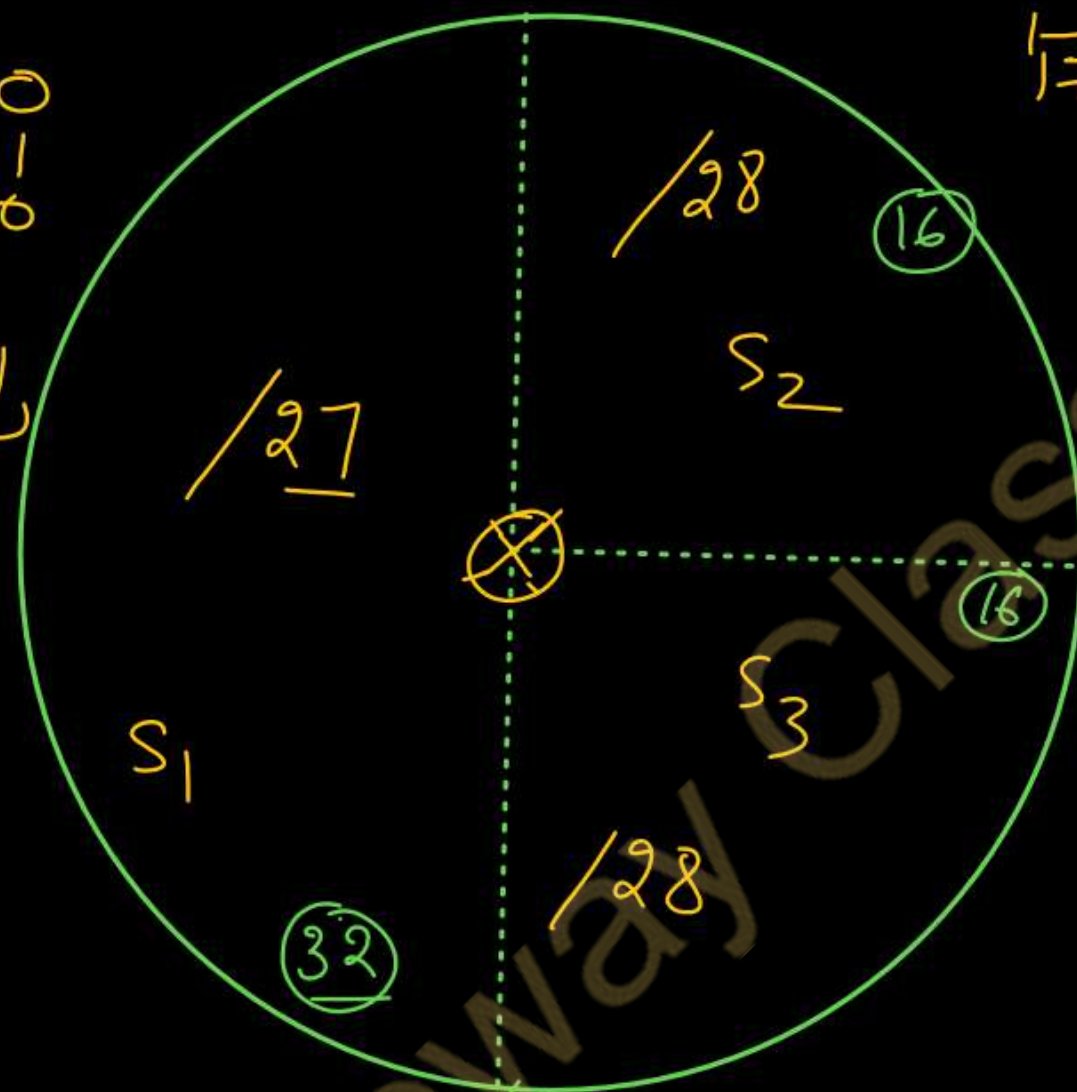
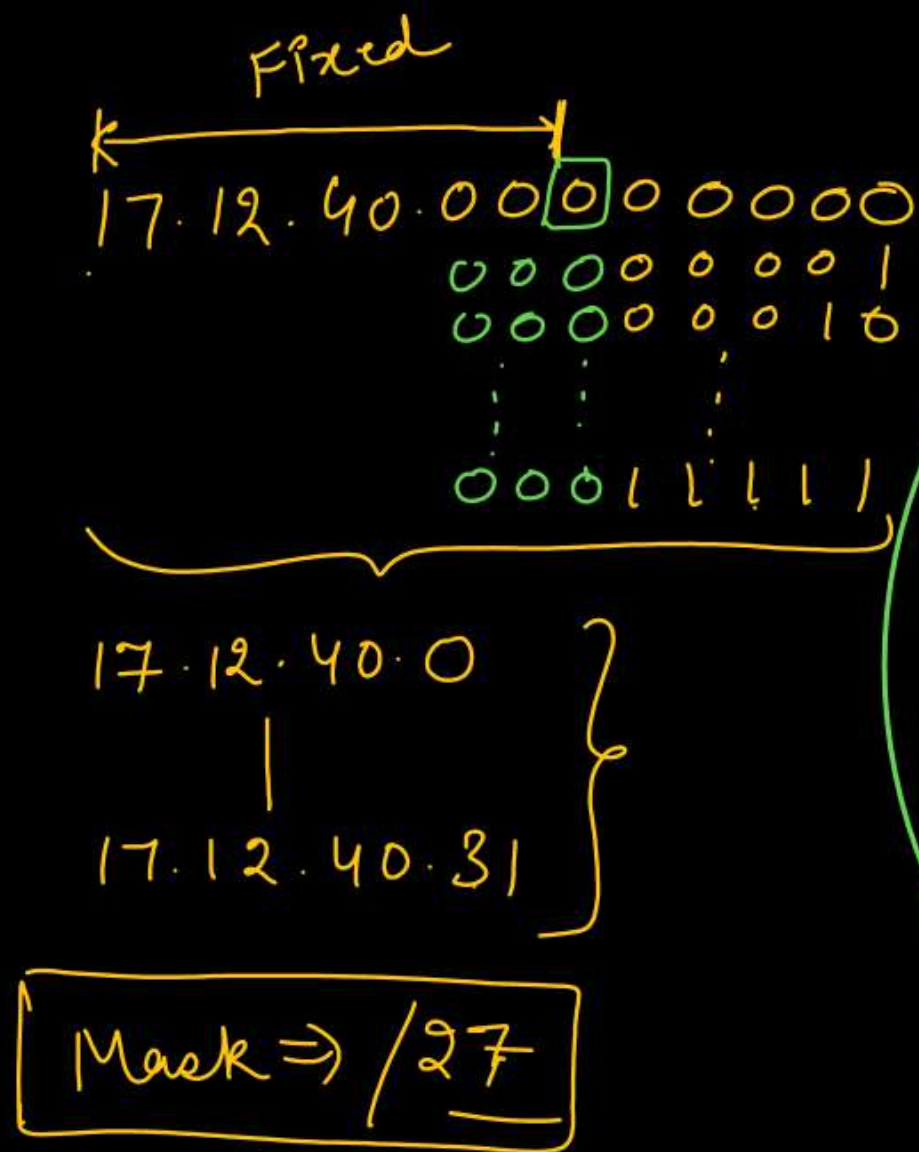
$$= 27 \Rightarrow \text{Mask for First block.}$$

Second Block - No. of addresses = 16

$$2^{32-n} = 2^4$$

$$\Rightarrow n = 28$$

Third Block $\Rightarrow$ / 28



X 17.12.40.0/26

www

32-26

= 6

Q- Given IP address :- 195.10.20.129/26 Divide into two equal size sub-blocks.

$$195.10.20.129 / \overset{n}{\textcircled{26}}$$

$$\text{No. of Addresses} = 2^{32-n} = 2^{32-26} = 2^6 = \underline{64}$$

$$\text{First address} \Rightarrow \underbrace{195.10.20.10}_{\text{n/w part}} \underbrace{000001}_{\downarrow}$$

$$195.10.20.10000000$$

$$\Rightarrow \boxed{195.10.20.128}$$

$195.10.20.10$ 000000
 10000001
 10000010
 $\vdots$
 10011111

$195.10.20.10$ 100000
 10100001
 $\vdots$
 10111111



$195.10.20.128$

$195.10.20.159$

Mask : $/27$

$195.10.20.160$

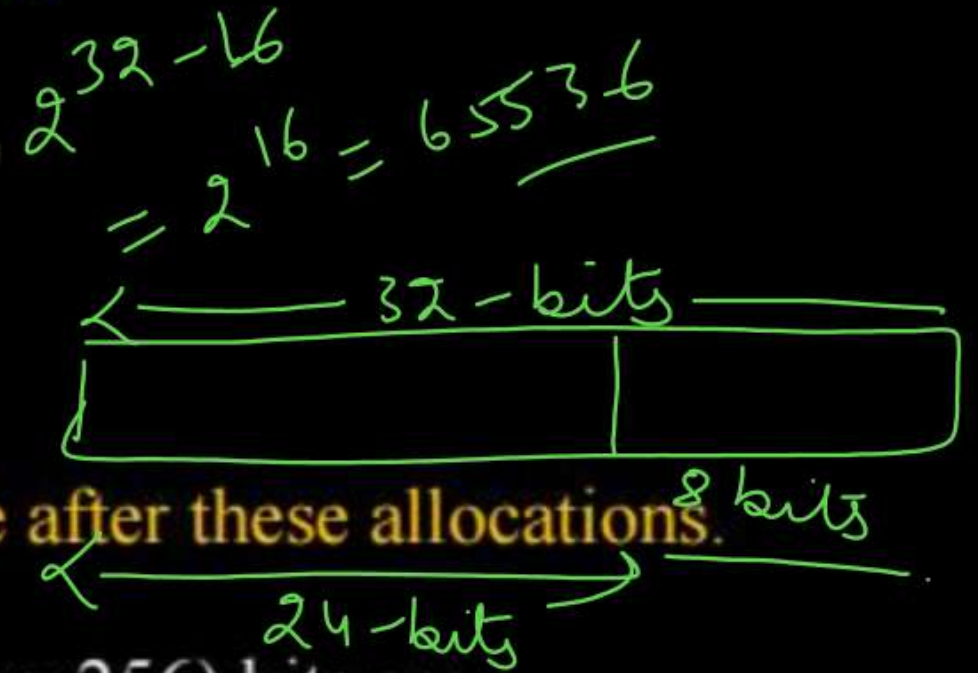
$195.10.20.191$

$/27$

X $195.10.20.128$ $/26$

AKTU
Q: An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.
- b. The second group has 128 customers; each needs 128 addresses.
- c. The third group has 128 customers; each needs 64 addresses.



Design the subblocks and find out how many addresses are still available after these allocations.

1. Group 1: Each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

1st Customer: 190.100.0.0/24 190.100.0.255/24

2nd Customer: 190.100.1.0/24 190.100.1.255/24

64th Customer: 190.100.63.0/24 190.100.63.255/24

$$\text{Total} = \underline{64} \times \underline{256} = \underline{16,384}$$

2. Group2: Each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$.

The addresses are

1st Customer:	190.100.64. <u>0</u> /25	190.100.64. <u>127</u> /25
2nd Customer:	190.100.64.128/25 <u>190.100.65.0/25</u>	190.100.64.255/25 <u>190.100.65.127/25</u>
128th Customer:	190.100.127.128/25	190.100.127.255/25

Total = 128 X 128 = 16,384

3. Group3: Each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host.

The prefix length is then $32 - 6 = 26$. The addresses are

1st Customer:	190.100.128. <u>0</u> /26	190.100.128.63/26
2nd Customer:	190.100.128.64/26	190.100.128.127/26
128th Customer:	190.100.159.192/26	190.100.159.255/26

Total = 128 X 64 = 8192

- Number of granted addresses to the ISP: 65,536
- Number of allocated addresses by the ISP: 40,960
- Number of available addresses: 24,576

Gateway Classes



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Computer Networks

Unit-3: Lecture 5

Today's Target

- Network Address Translation ✓
- IPv6 Header Format ✓
- Difference between IPv4 and IPv6 ✓
- Transition from IPv4 to IPv6 ✓
- AKTU PYQs

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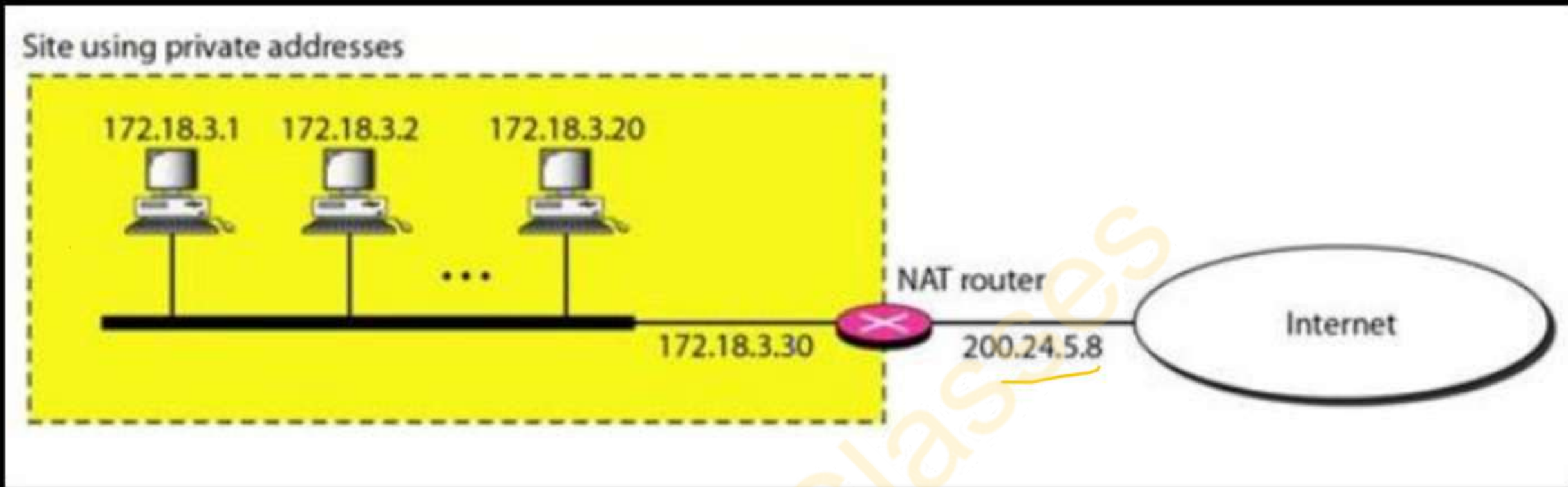
Network Address Translation (NAT)

- Today home users and small businesses have created small networks with several hosts and need an IP address for each host, so demand for IP addresses is ever increasing.
- With the shortage of addresses, quick solution to this problem is called network address translation (NAT).
- **NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally.**
- To separate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved three sets of addresses as private addresses.

Addresses for private networks

<i>Range</i>	<i>Total</i>
10.0.0.0 to 10.255.255.255	2^{24}
172.16.0.0 to 172.31.255.255	2^{20}
192.168.0.0 to 192.168.255.255	2^{16}

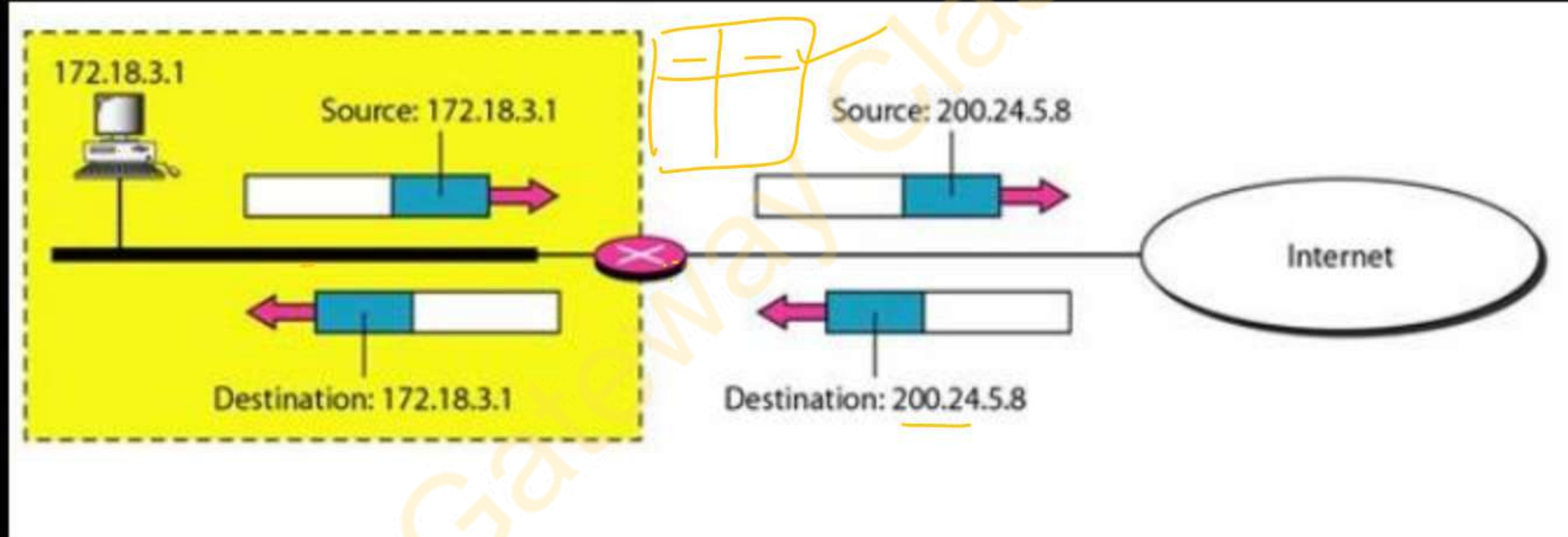
- Any organization can use an address out of this set without permission from Internet authorities.
- Everyone knows that these reserved addresses are for private networks.
- They are unique inside the organization, but they are not unique globally.
- No router will forward a packet that has one of these addresses as the destination address.



- The site must have only one single connection to the global Internet through a router that runs the NAT software.
- That router uses one private address and one global address.
- The private network is transparent to the rest of Internet; the rest of the Internet sees only the NAT router with the address 200.24.5.8.

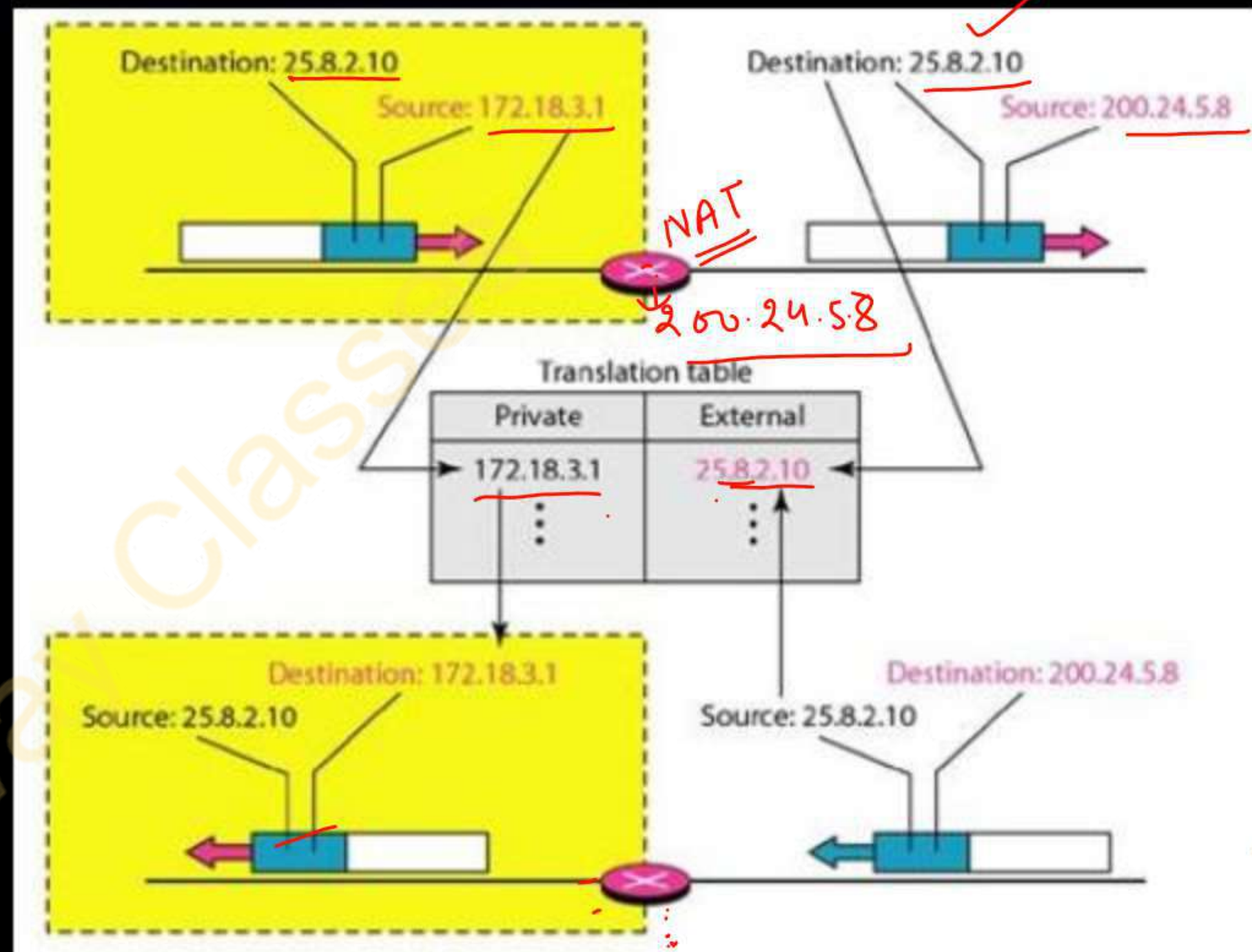
Address Translation

- NAT router replaces the source address in all outgoing packets with the global NAT address.
- NAT router also replaces the destination address in all incoming packets (the NAT router global address) with the appropriate private address.



Translation Table

- NAT router has a translation table with private' address and external address (DA of the packet).
- When the router translates the SA of outgoing packet, it also makes note of the DA the packet is going.
- When response comes back from destination, the router uses the SA of the packet (as external address) to find the private address of the packet.



IPv6 ADDRESSES

- Despite all short-term solutions, such as classless addressing, DHCP and NAT, **address depletion** is still a long-term problem for the Internet.
- Another problem is the **lack of accommodation** for real-time audio and video transmission, and encryption and authentication of data for some applications.

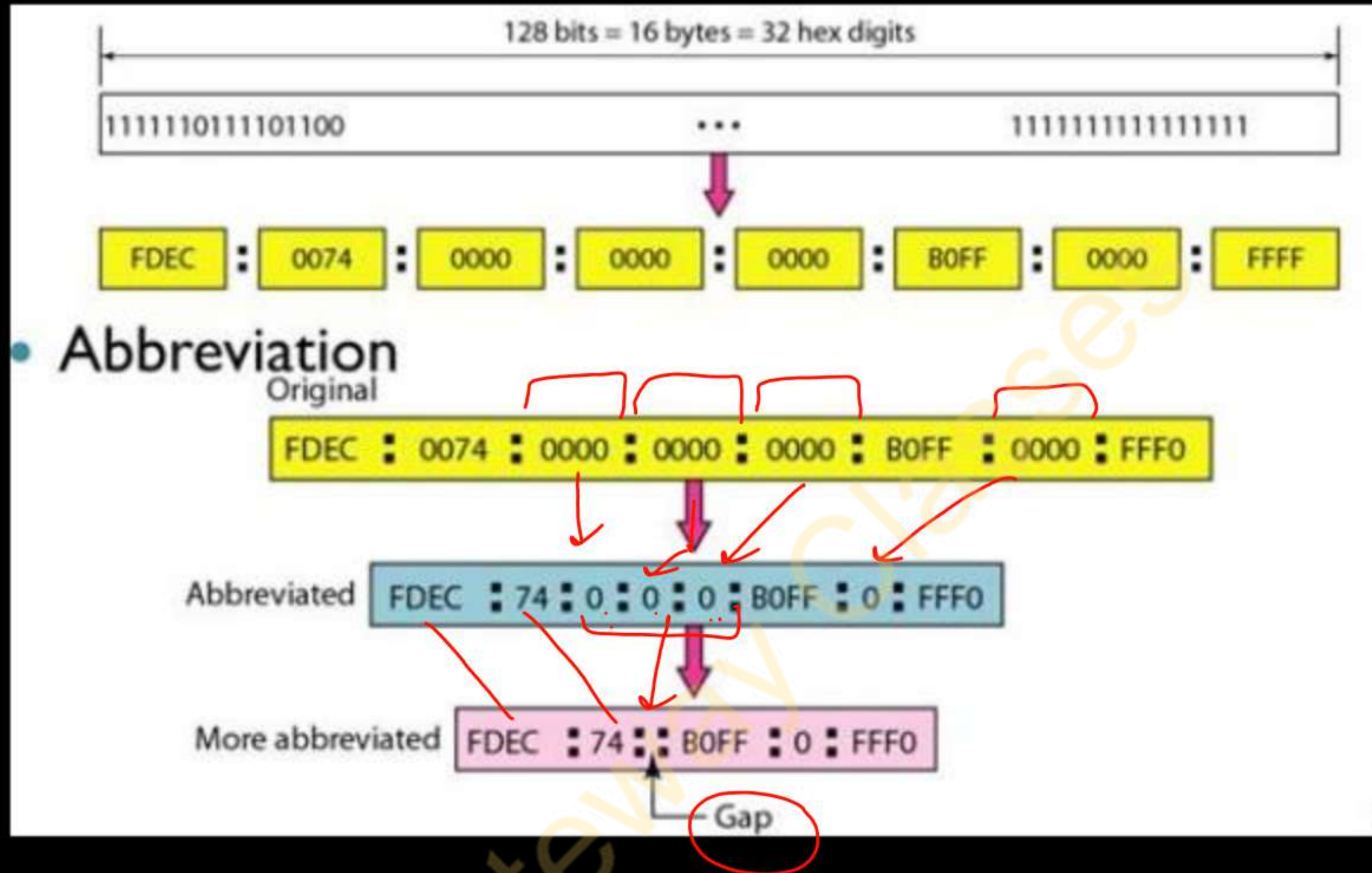
switch to
IPv6

An IPv6 address consists of 16 bytes (octets); it is 128 bits long.

Hexadecimal Colon Notation:

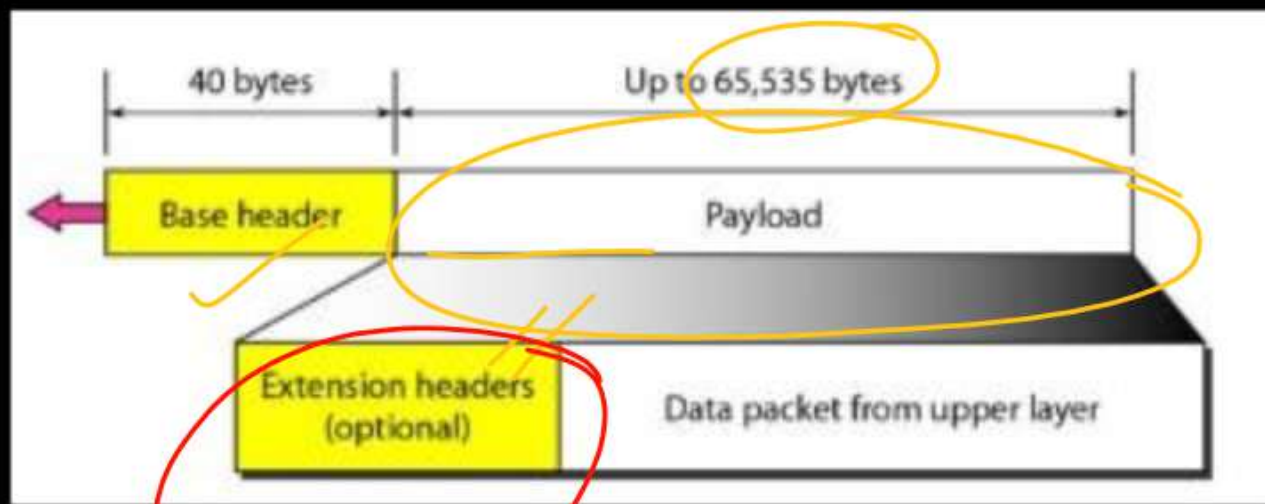
The address consists of 32 hexadecimal digits, with every four digits separated by a colon.

2^{128}

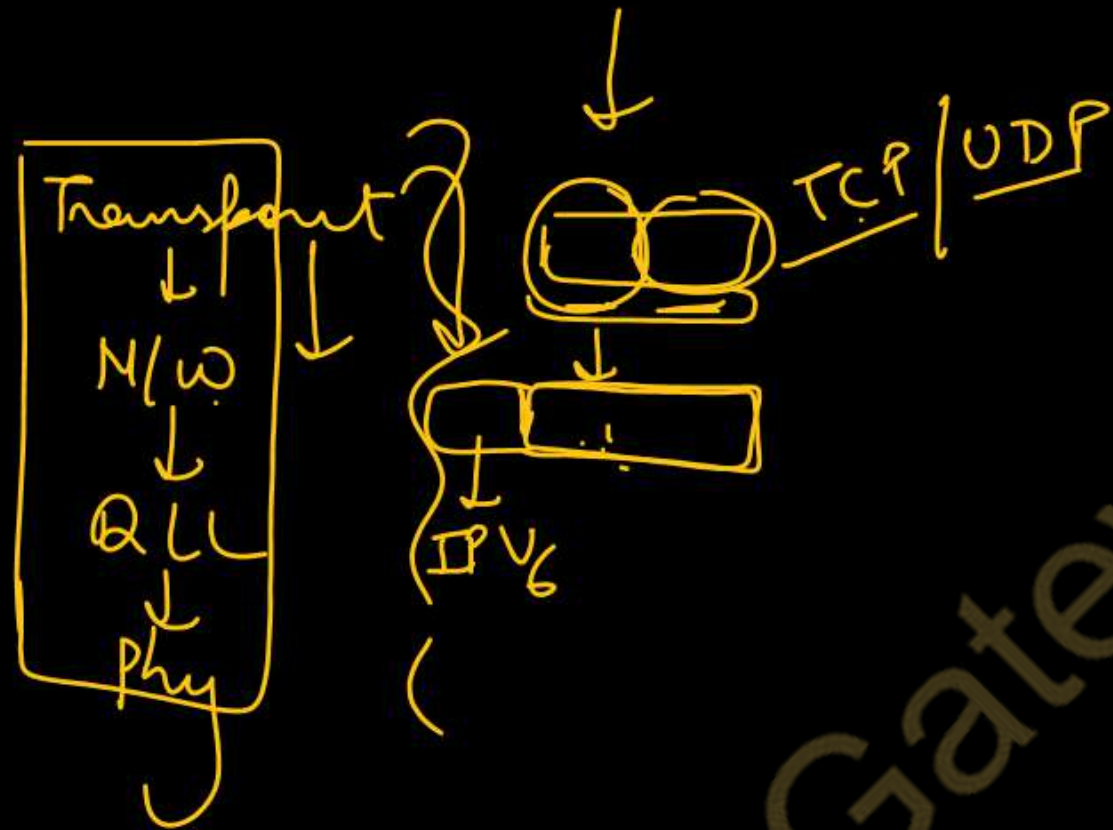
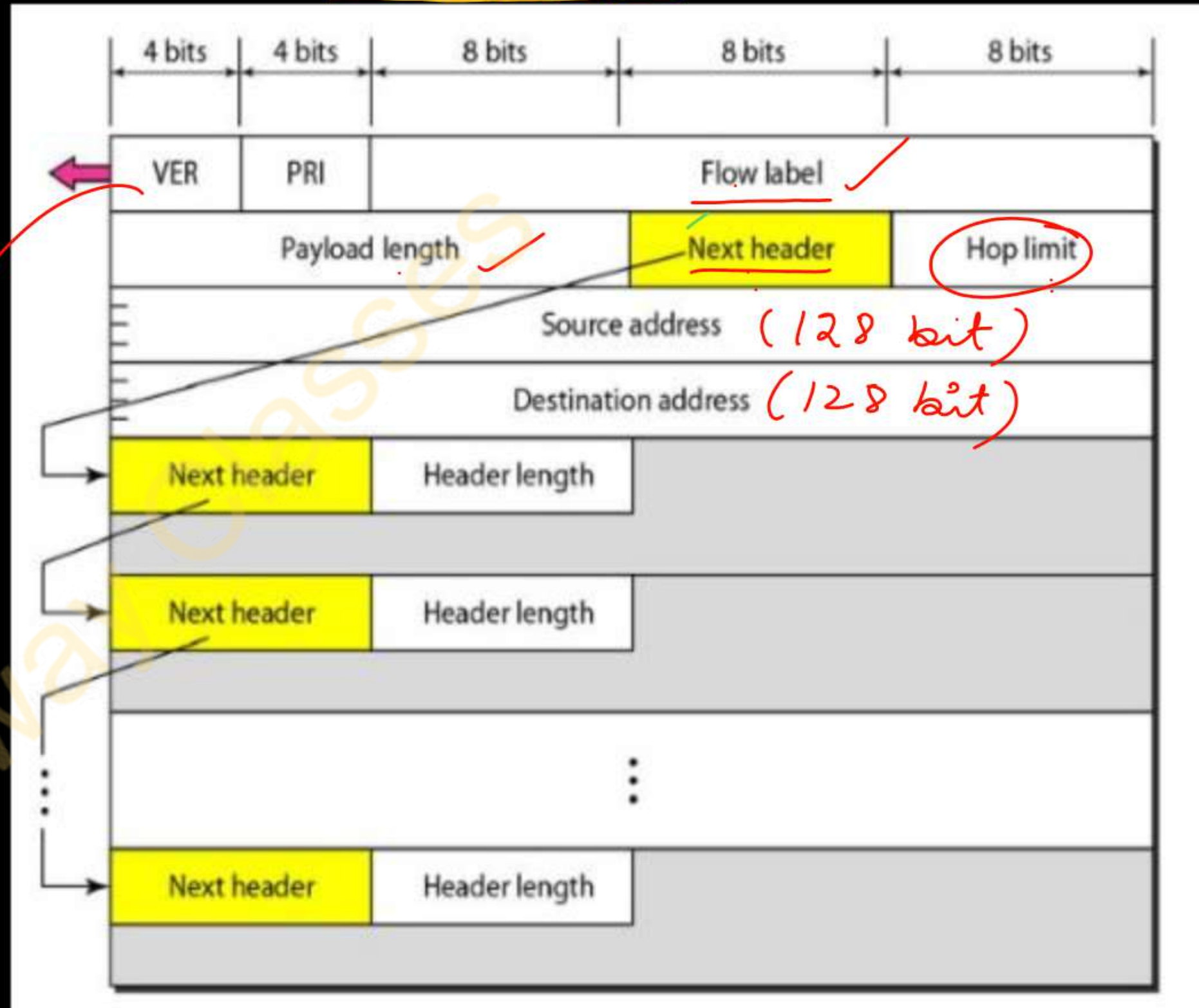


Address Space:

- IPv6 has a much larger address space; 2^{128} addresses are available.



IPv6 Header



Version: For IPv6, the value is 6 (0110)

Priority: 4-bit field defines the priority of packet with respect to traffic congestion.

Flow label: 3-byte field provide special handling for a particular flow of data.

Payload length: 2-byte field defines length of the IP datagram excluding the base header.

Next header:

- 8-bit field defining the header that follows the base header in the datagram. The next header is either one of the optional extension headers used by IP or the header of an encapsulated packet such as UDP or TCP. Each extension header also contains this field. Note that this field in version 4 is called the *protocol*.

Hop limit: This 8-bit hop limit field serves the same purpose as the TTL field in IPv4.

Source address and Destination address : 16-byte (128-bit) IP Address of source and destination.

Flow label: Provide special handling for a particular flow of data.

➤ A sequence of packets, sent from a particular source to a particular destination, that needs special handling by routers is called a flow of packets.

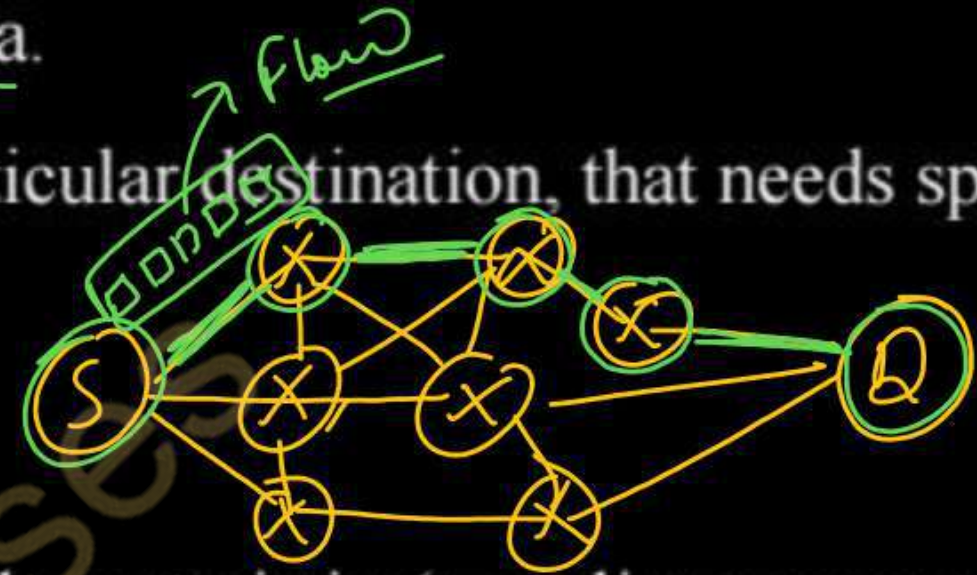
➤ **SA + flow label** uniquely defines a flow of packets.

➤ To a router, a flow is a sequence of packets that share same characteristics (traveling same path, same resources, same kind of security).

➤ A router has a flow label table with an entry for each active flow label.

➤ When router receives a packet, it consults its flow label table to find entry for flow label value defined in packet and provides mentioned services to the packet.

➤ flow label speed up processing of a packet by a router and support transmission of real-time audio and video as they require high bandwidth, large buffers, long processing time, and a process can reserve these resources to guarantee efficient real-time data transmission.



Next header codes for IPv6:

Code	Next Header
------	-------------

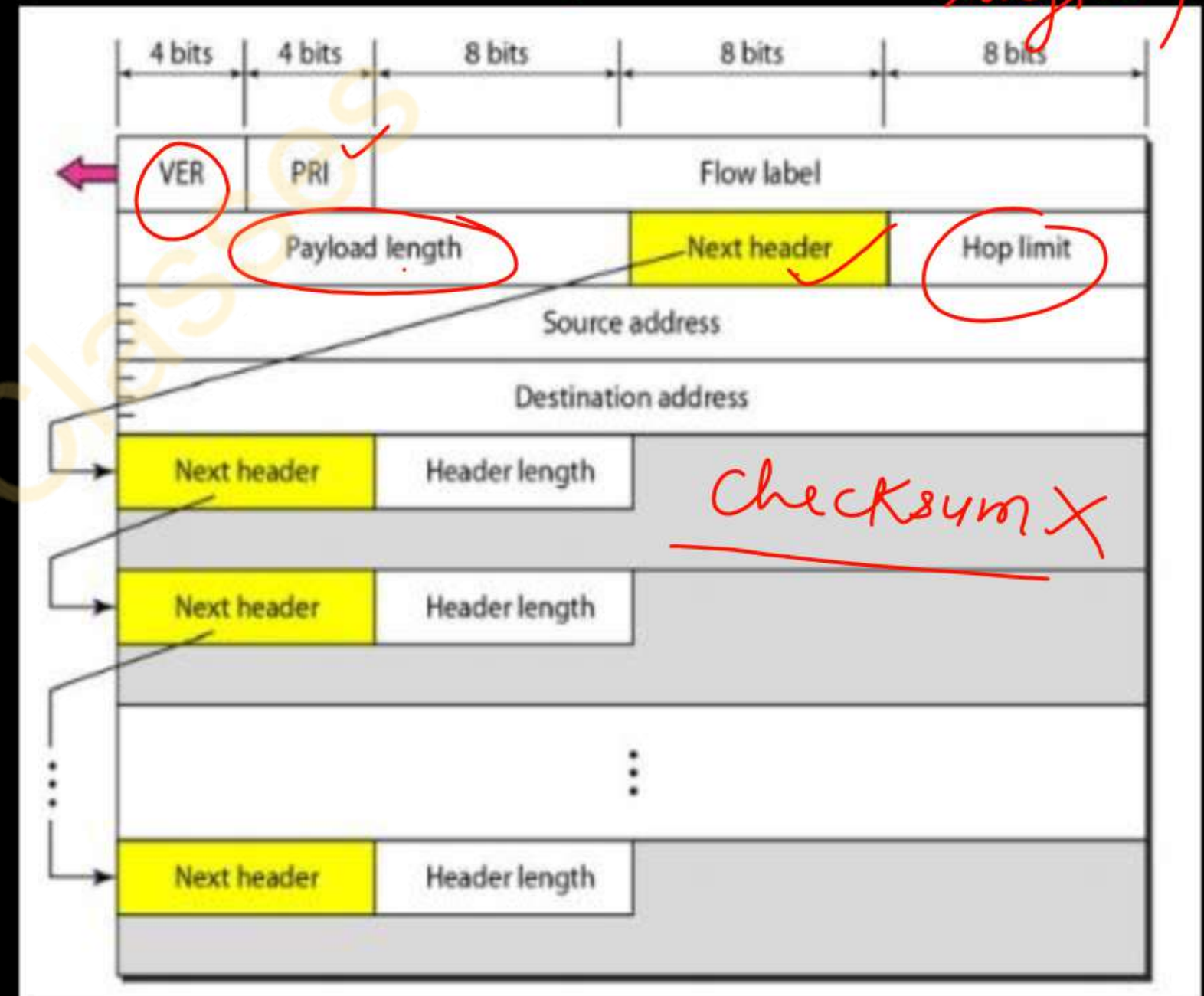
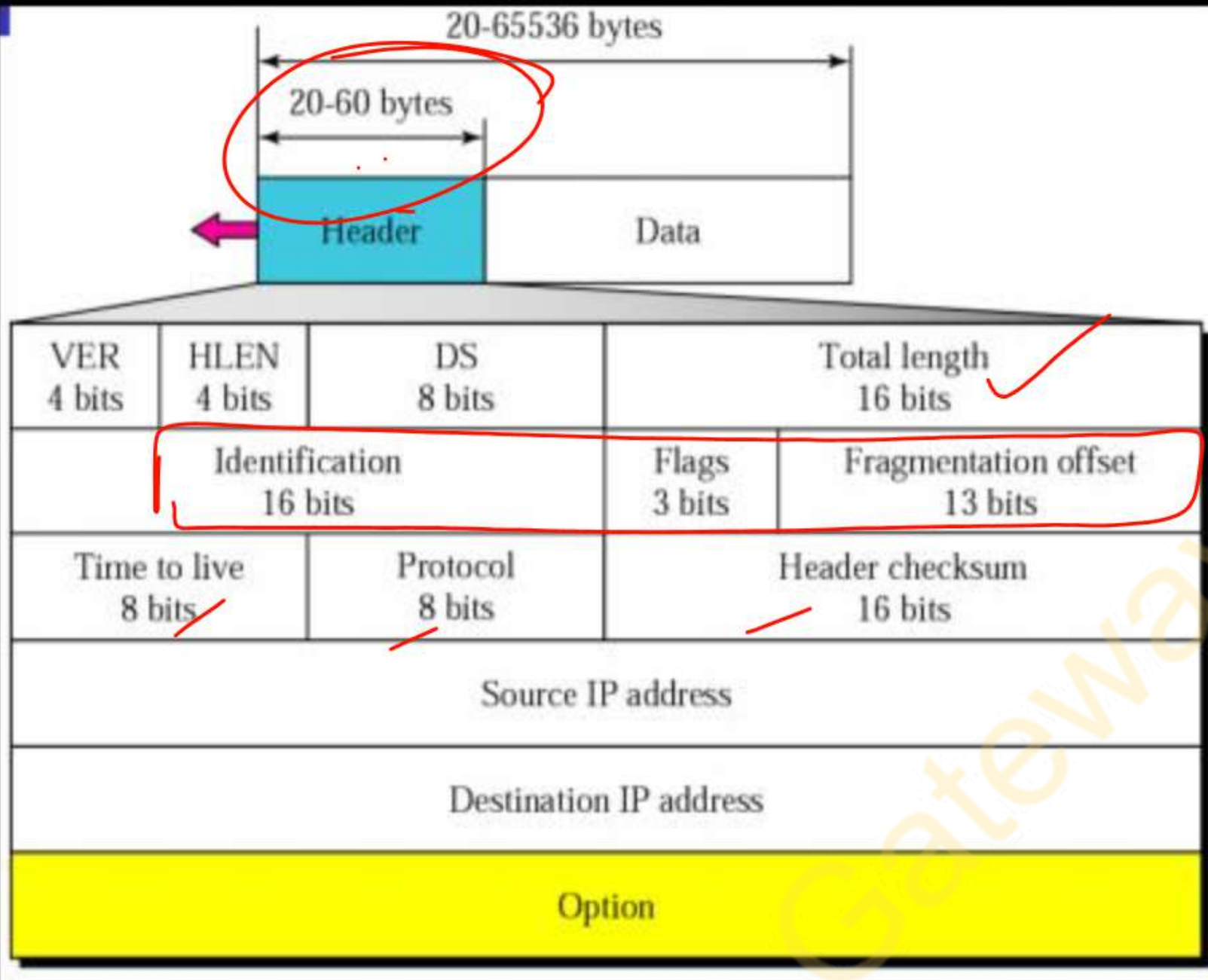
0	Hop-by-hop option: used when source needs to pass info to all routers visited by datagram.
2	ICMP
6	TCP
17	UDP
43	<u>Source routing</u> : combines concepts of strict source route and loose source route options of IPv4.
44	<u>Fragmentation</u> : used if datagram is fragmented. Only original source can fragment.
50	Encrypted security payload: Provides confidentiality and guards against eavesdropping.
51	Authentication: Validates the message sender and ensures the integrity of data.
<u>59</u>	<u>Null (no next header)</u>
60	Destination option: used when the source needs to pass information to the destination only.

Intermediate routers are not permitted access to this information.

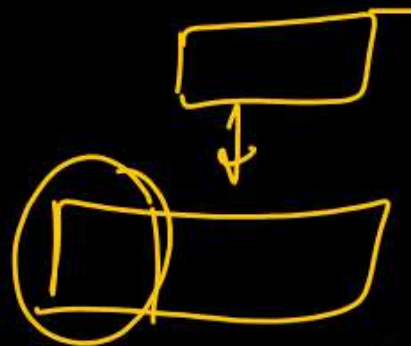
Imp

Comparison Between IPv4 and IPv6 Headers

40 byte Header (fixed length)



Imp Comparison Between IPv4 and IPv6 Headers

1. The header length field is eliminated in IPv6 because the length of header is fixed in this version.
2. Service type (Differentiated services) in IPv4 is replaced with priority + flow label.
3. The total length field is eliminated in IPv6 and replaced by the payload length field.
4. The identification, flag, and offset fields are eliminated from base header in IPv6 and included in fragmentation extension header. Option fields are implemented as extension headers in IPv6.
5. The TTL field is called hop limit in IPv6.
6. The protocol field is replaced by the next header field.
7. The header checksum is eliminated as checksum is provided by upper-layer protocols. That saves processing time and speeds up the packet forwarding. The logic is that the layer-2 and layer-4 protocols each already have a checksum. The layer-2 checksum covers the entire IPv6 packet, and the layer-4 checksum covers the transport datagram.

Advantages of IPv6 over IPv4

1. Larger address space: An IPv6 address is 128 bits long, resulting in huge increase in address space.
2. Better header format:
 - Options are separated from the base header and inserted, when needed, between the base header and the upper-layer data. This speeds up the routing process as most of the options do not need to be checked by routers.
3. New options: to allow for additional functionalities.
4. Allowance for extension:
 - Allow extension of protocol if required by new technologies or applications.
5. Support for resource allocation:
 - In IPv6, the type-of-service field has been removed, but flow *label* enable source to request special handling of packet to support real-time audio and video.
6. Support for more security of packet: Through encryption and authentication options.

Difference between IPv4 and IPv6

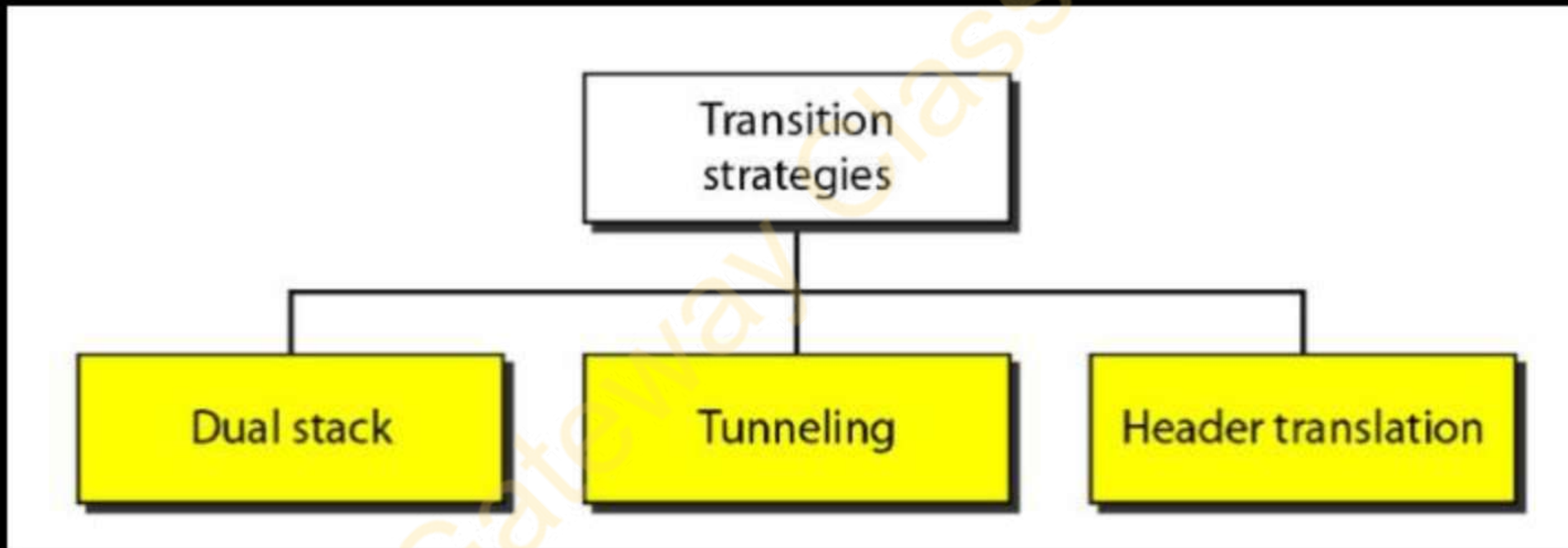
IPv4	IPv6
IPv4 has a 32-bit address length	IPv6 has a 128-bit address length
Supports Manual and <u>DHCP</u> address configuration	It supports Auto and renumbering address configuration
End to end connection integrity is Unachievable	End-to-end, connection integrity is Achievable
It can generate 4.29×10^9 address space	It can produce 3.4×10^{38} address space
Security feature is dependent on the application	IPSEC is an inbuilt security feature
Address representation is in decimal	Address representation is in hexadecimal

Difference between IPv4 and IPv6

IPv4	IPv6
Fragmentation performed by Sender and routers	Fragmentation is performed only by the sender
Packet flow identification is not available	Packet flow identification uses the flow label field
Checksum field is available	Checksum is not available
broadcast Message Transmission Scheme	multicast and anycast message transmission scheme
Encryption and Authentication not provided	Encryption and Authentication are provided
IPv4 has a header of 20-60 bytes.	IPv6 has a header of 40 bytes fixed
IPv4's IP addresses are divided into Classes.	Does not have any classes of the IP address.

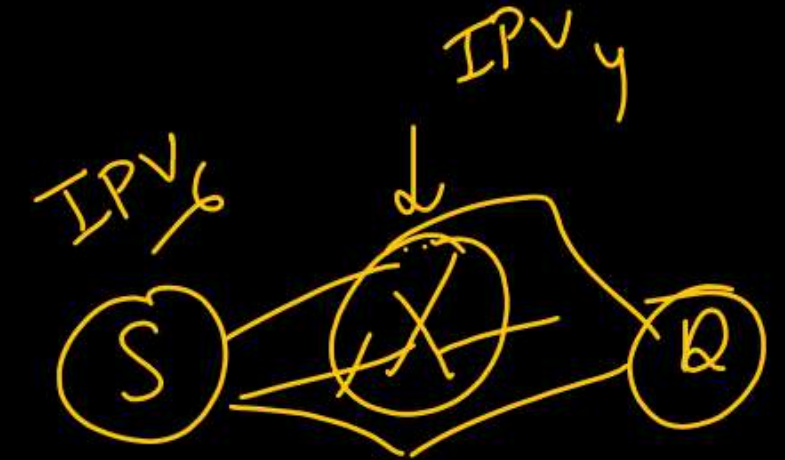
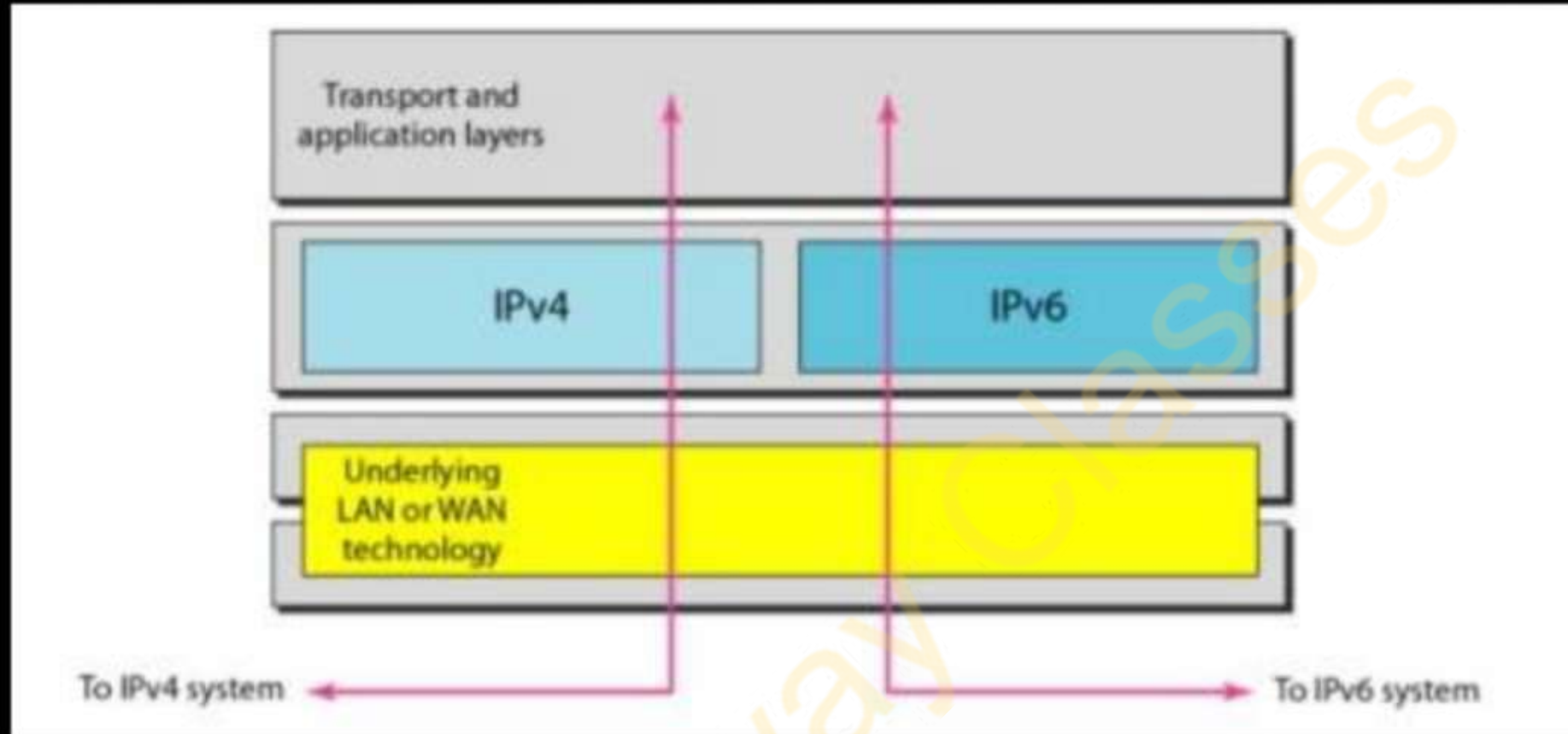
Transition from IPv4 TO IPv6

- Due to huge number of systems on Internet, the transition from IPv4 to IPv6 cannot happen suddenly.
- It takes a considerable amount of time before every system in the Internet can move from IPv4 to IPv6.
- The transition must be smooth to prevent any problems between IPv4 and IPv6 systems.



Dual Stack:

A station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.

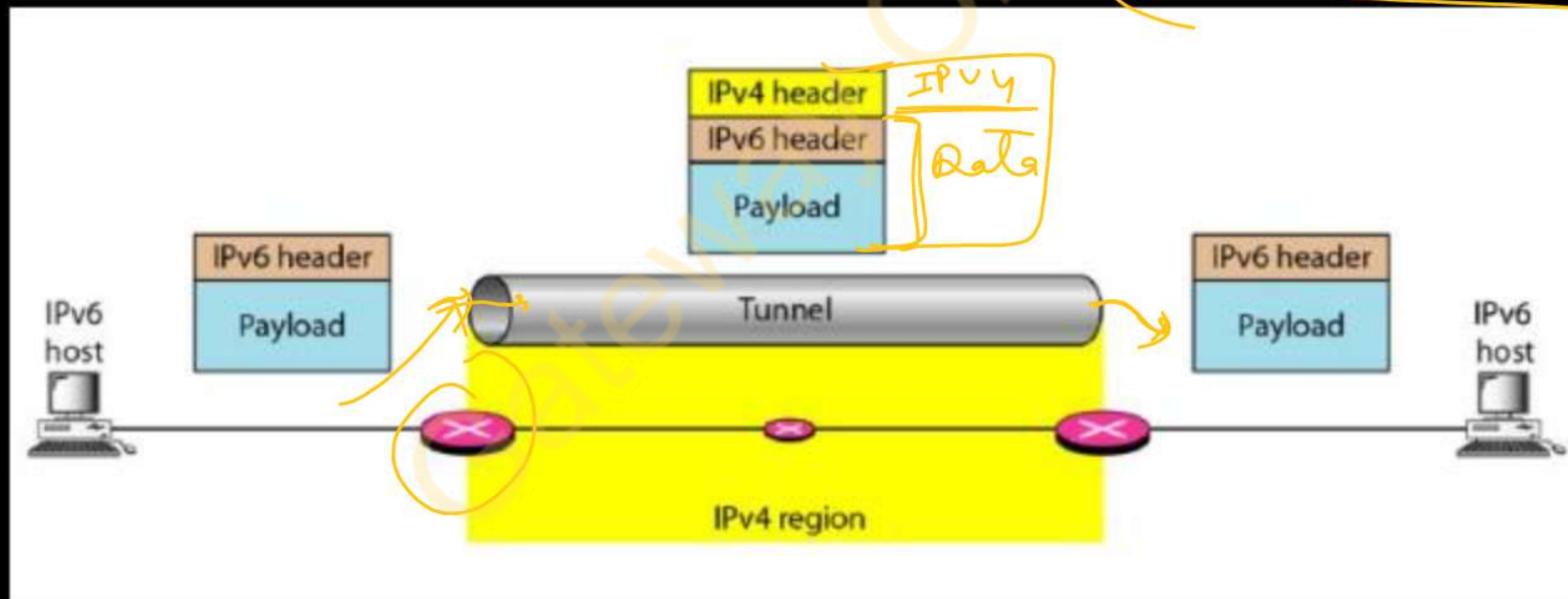


To determine which version to use when sending a packet to a destination, source host queries DNS.

- If the DNS returns an IPv4 address, the source host sends an IPv4 packet.
- If the DNS returns an IPv6 address, the source host sends an IPv6 packet.

Tunneling: (Imp)

- When two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4, the IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region.
- It seems as if the IPv6 packet goes through a tunnel at one end and emerges at the other end. To make it clear that the IPv4 packet is carrying an IPv6 packet as data, the protocol value is set to 41.



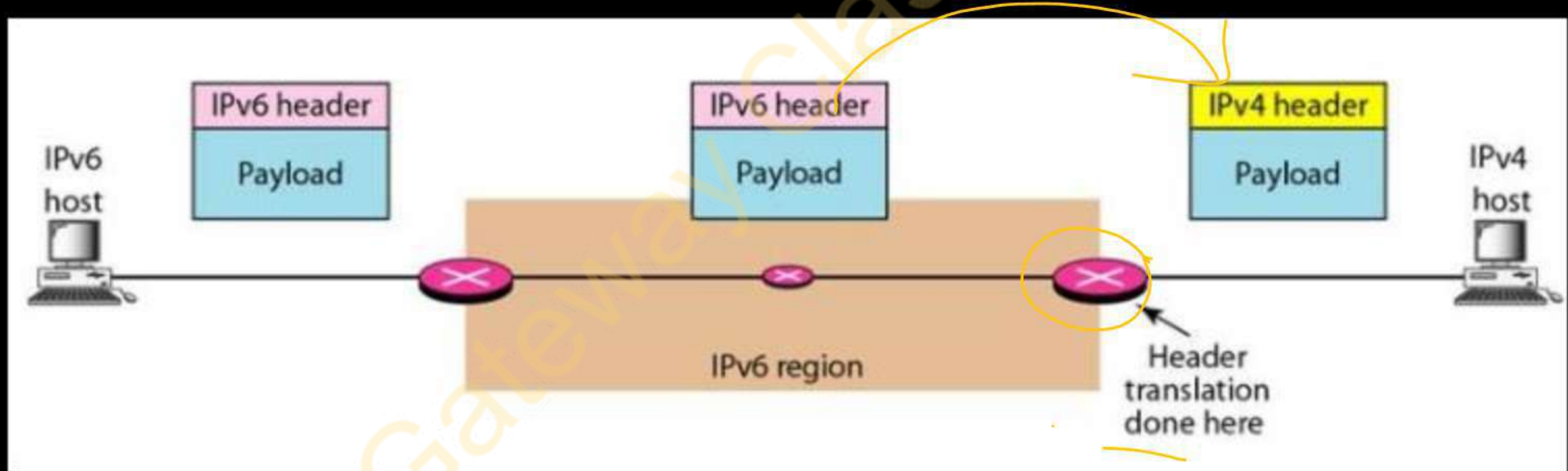
Header Translation:

The sender wants to use IPv6, but the receiver does not understand IPv6.

Tunneling doesn't work here because packet must be in the IPv4 format to be understood by the receiver.

So, header format must be totally changed through header translation.

The header of the IPv6 packet is converted to an IPv4 header



AKTU PYQs

Q: Illustrate the difference between IPv4 and IPv6.(AKTU 2022-23)

Q: Discuss advantages of IPv6 over IPv4.(AKTU 2021-22)

Gateway Classes



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Computer Networks

Unit-3: Lecture 6

Today's Target

- Address Mapping Protocols
 - ARP ✓
 - RARP ✓
- BOOTP ✓
- DHCP ✓
- ICMP ✓
- AKTU PYQs

By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



AKTU PYQs

Q: Explain ICMP protocol and its application in real-world scenarios. (AKTU 2022-23)

Q: Explain ARP. (AKTU 2022-23)

Q: Discuss in detail about ICMP role in network layer. (AKTU 2021-22)

Q: How is BOOTP different from DHCP? (AKTU 2018-19)

Gateway Classes

Address Resolution Protocol (ARP)

- Delivery of a packet to a host or a router requires two levels of addressing: logical and physical.
- We need to map a logical address to its corresponding physical address and vice versa using either static or dynamic mapping.

Static mapping:

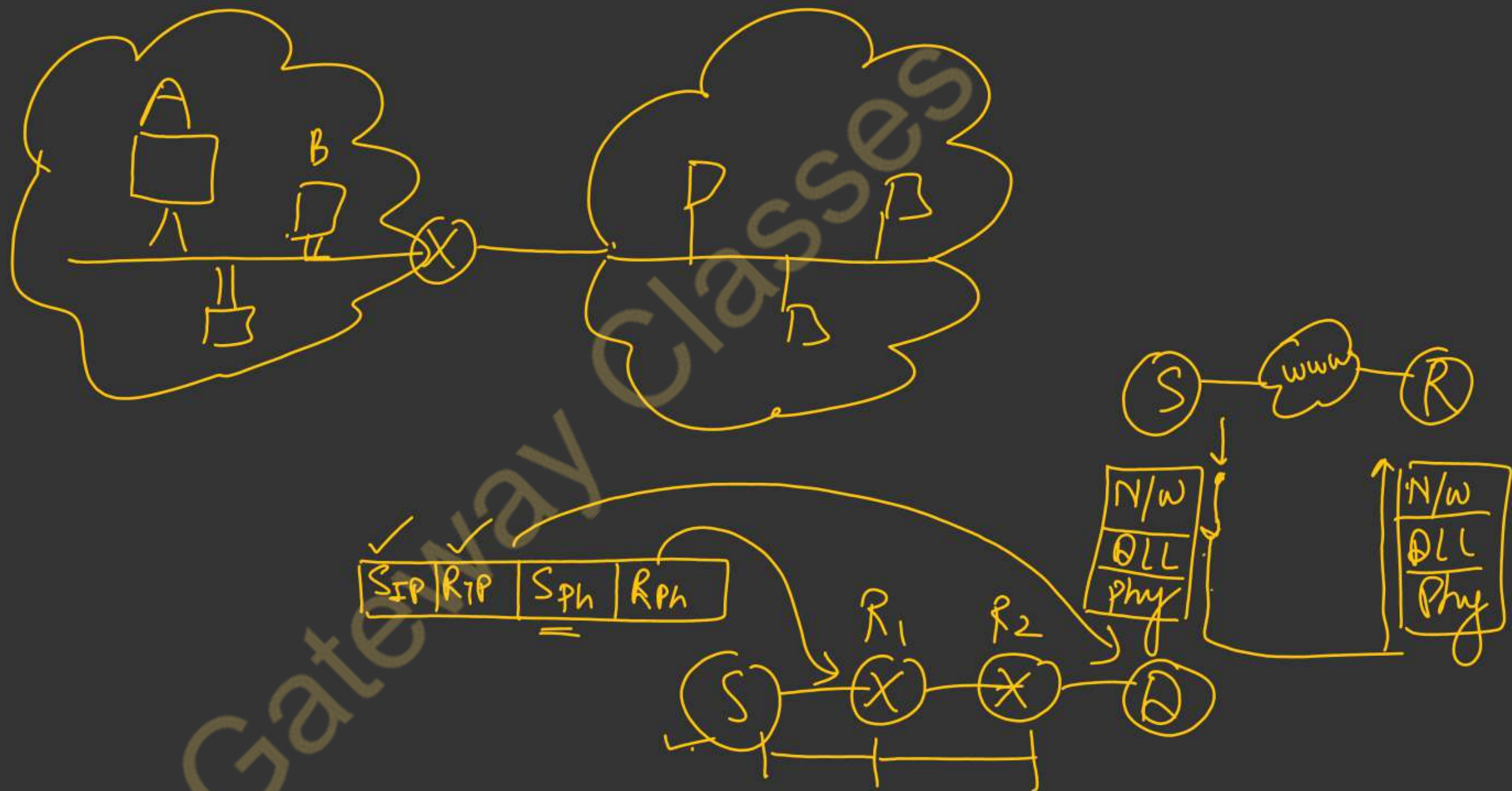
IP	Phy
—	—
—	—
—	—

Each machine carry a table that associates a logical address with a physical address. But physical addresses may change if a machine change its NIC, or move from one physical network to another.

- So, a static mapping table must be updated periodically(Overhead).

Dynamic mapping:

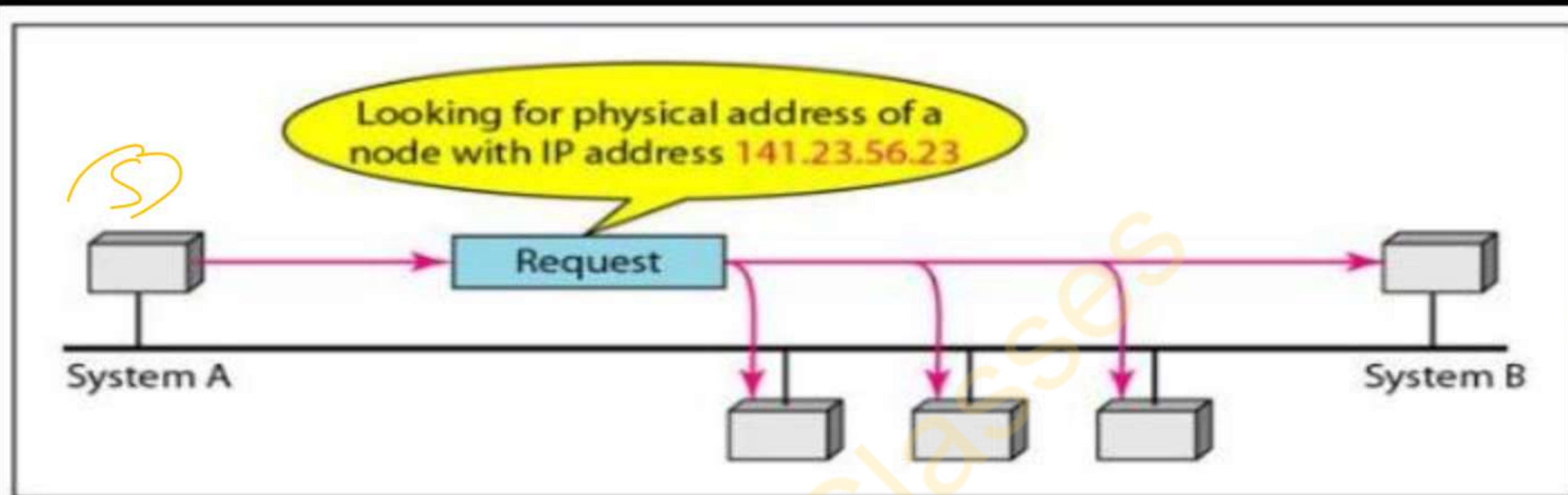
- Each time a machine knows one of the two addresses (logical or physical), it use a protocol to find the other one.



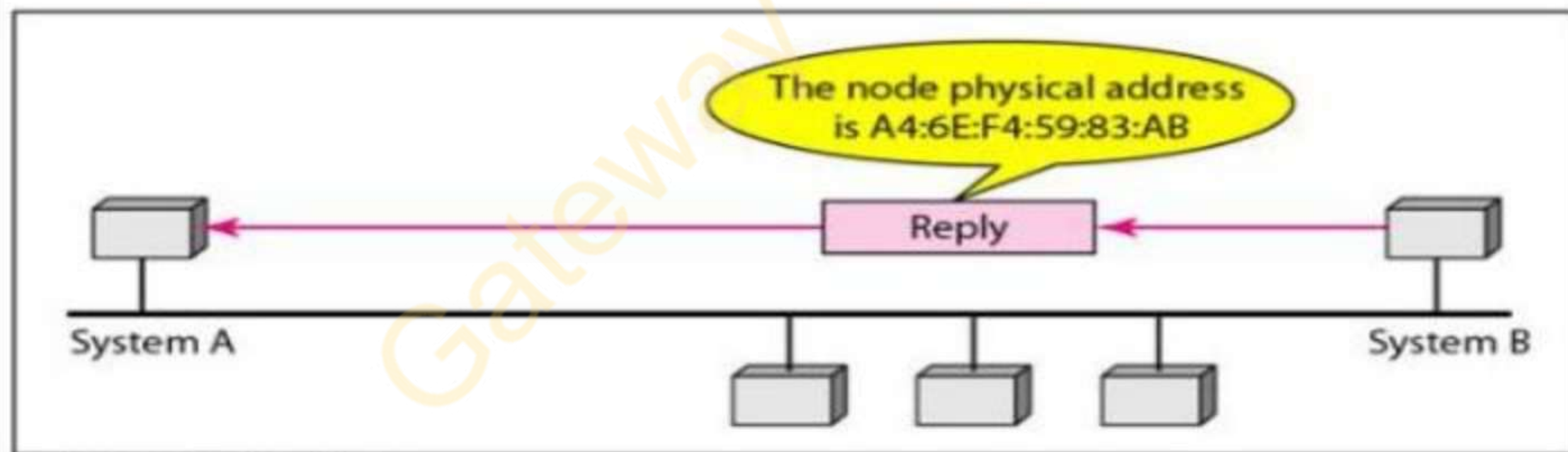
Mapping Logical to Physical Address: ARP

Anytime a host or router has an IP datagram to send to another host or router, it has IP address of receiver but IP datagram must be encapsulated in a frame to pass through physical network, So sender needs the physical address of the receiver also.

- The host or the router broadcasts an ARP query packet (includes physical and IP addresses of sender and the IP address of the receiver).
- The intended recipient recognizes its IP address and sends back an ARP response packet containing its IP and physical addresses.
- The packet is unicast directly to the inquirer using physical address received in the query packet.



a. ARP request is broadcast



b. ARP reply is unicast

ARP Packet Format

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Hardware type(16-bit):

- Type of network on which ARP is running. Each LAN has been assigned an integer based on its type (e.g, Ethernet is given type 1).

Protocol type(16-bit): ARP can be used with any higher-level protocol. For IPv4, it is 080016.

Hardware length(8-bit): Define length of the physical address in bytes (For Ethernet the value is 6).

Protocol length(8-bit): Define length of the logical address in bytes. For IPv4, it is 4.

Operation(16-bit): Type of packet(ARP request (1) or ARP reply (2)).

Sender hardware address: physical address of the sender. For Ethernet, it is 6 bytes long.

Sender protocol address: logical address of the sender. For IP protocol, it is 4 bytes long.

Target hardware address: physical address of the target. For an ARP request, this field is all 0s.

Target protocol address: logical address of the target.

Mapping Physical to Logical Address:

➤ RARP ✓

➤ BOOTP

➤ DHCP ↓

There are occasions in which a host knows its physical address, but needs to know its logical address.

This may happen in two cases:

- A diskless station is just booted. It can find its physical address by checking its interface, but it does not know its IP address. (IP configuration file is generally stored on disk)
- An organization does not have enough IP addresses to assign to each station; it needs to assign IP addresses on demand. The station can send its physical address and ask for a short time lease.

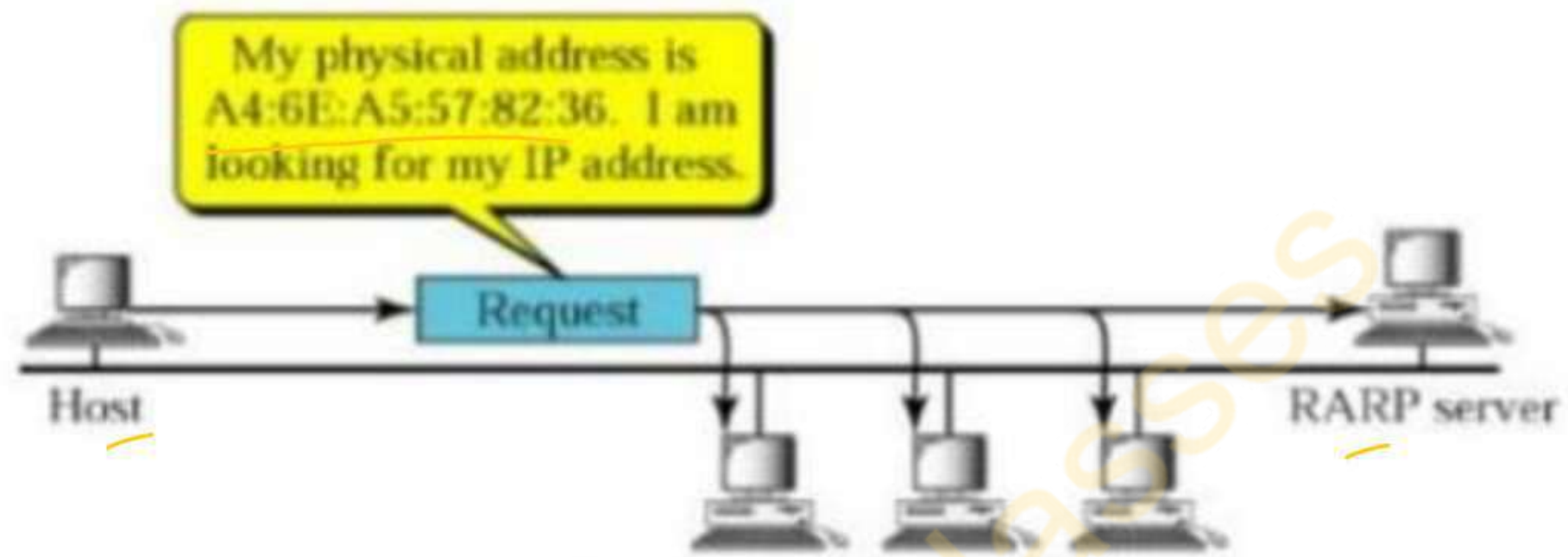
① Reverse Address Resolution Protocol (RARP)

- Finds the logical address for a machine that knows only its physical address.
- It can then use the physical address to get the logical address by using the RARP protocol.
- A RARP request is created and broadcast on the local network.
- Another machine on the local network that knows all the IP addresses will respond with a RARP reply. Requesting machine runs a RARP client program; the responding machine runs a RARP server.

FF:FF:FF:FF:FF:FF

Problem with RARP:

- The physical broadcast address (in the case of Ethernet), does not pass the boundaries of a network.
- This means that if an administrator has several networks or subnets, it needs to assign a RARP server for each network or subnet.



a. RARP request is broadcast



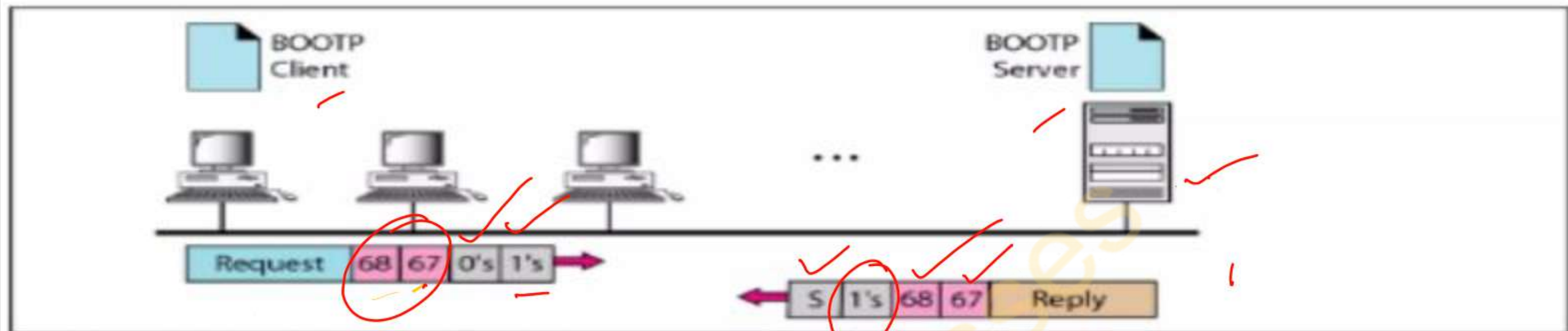
b. RARP reply is unicast

Bootstrap Protocol (BOOTP) (Application layer protocol)

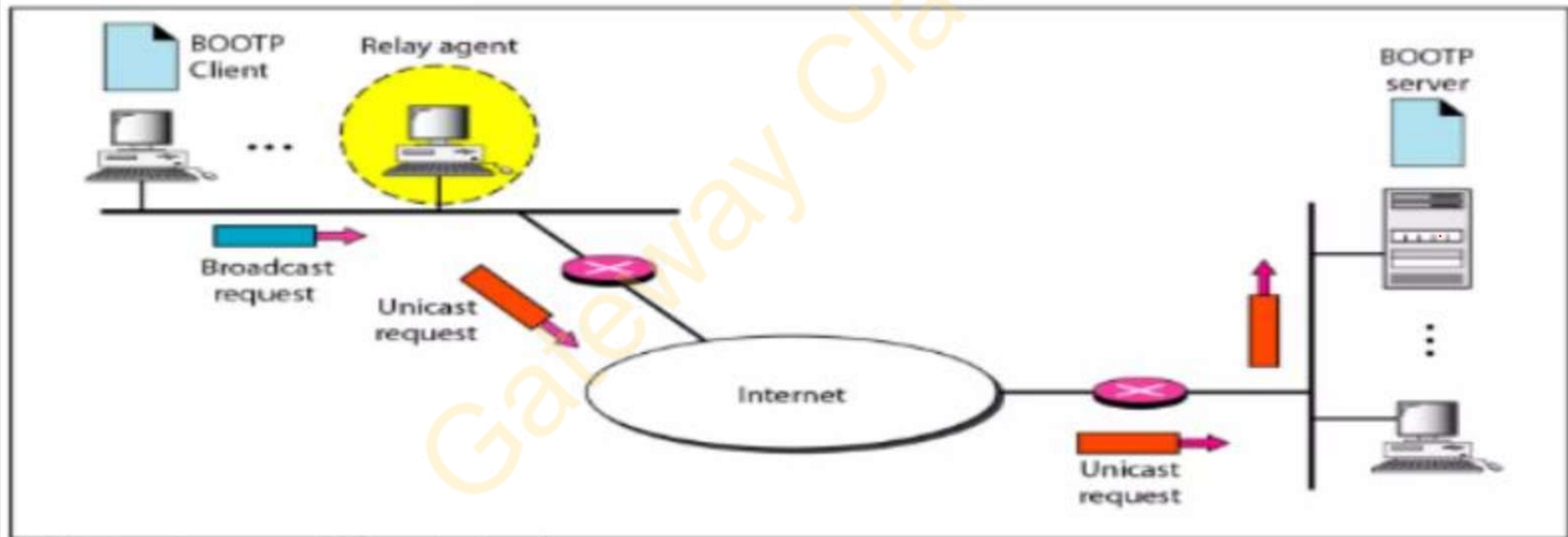
- A client/server protocol designed to provide physical address to logical address mapping.
- The administrator may put the client and the server on the same network or on different networks.

How client can send IP datagram without knowing its own IP address and server's IP address:

- The client uses all 0's as source address and all 1's as the destination address.
- The BOOTP request is broadcast as client does not know the IP address of the server.
- A broadcast IP datagram cannot pass through any router so an intermediary is required.
- One of the hosts or router is configured to operate at application layer and used as a relay agent.
- The relay agent knows the unicast address of a BOOTP server. When it receives this type of packet, it encapsulates the message in a unicast datagram and sends the request to the BOOTP server.
- BOOTP server knows the message comes from a relay agent because request message defines the IP address of relay agent. The relay agent, after receiving the reply, sends it to the BOOTP client.



a. Client and server on the same network



b. Client and server on different networks

Dynamic Host Configuration Protocol (DHCP)

Why DHCP?

- BOOTP is a static configuration protocol.
- (Server has a static table filled by administrator to match physical address of client with its IP address).

What if a host moves from one physical network to another?

What if a host wants a temporary IP address?

- BOOTP cannot handle these situations

DHCP provide static and dynamic address allocation that can be manual or automatic.

Static Address Allocation:

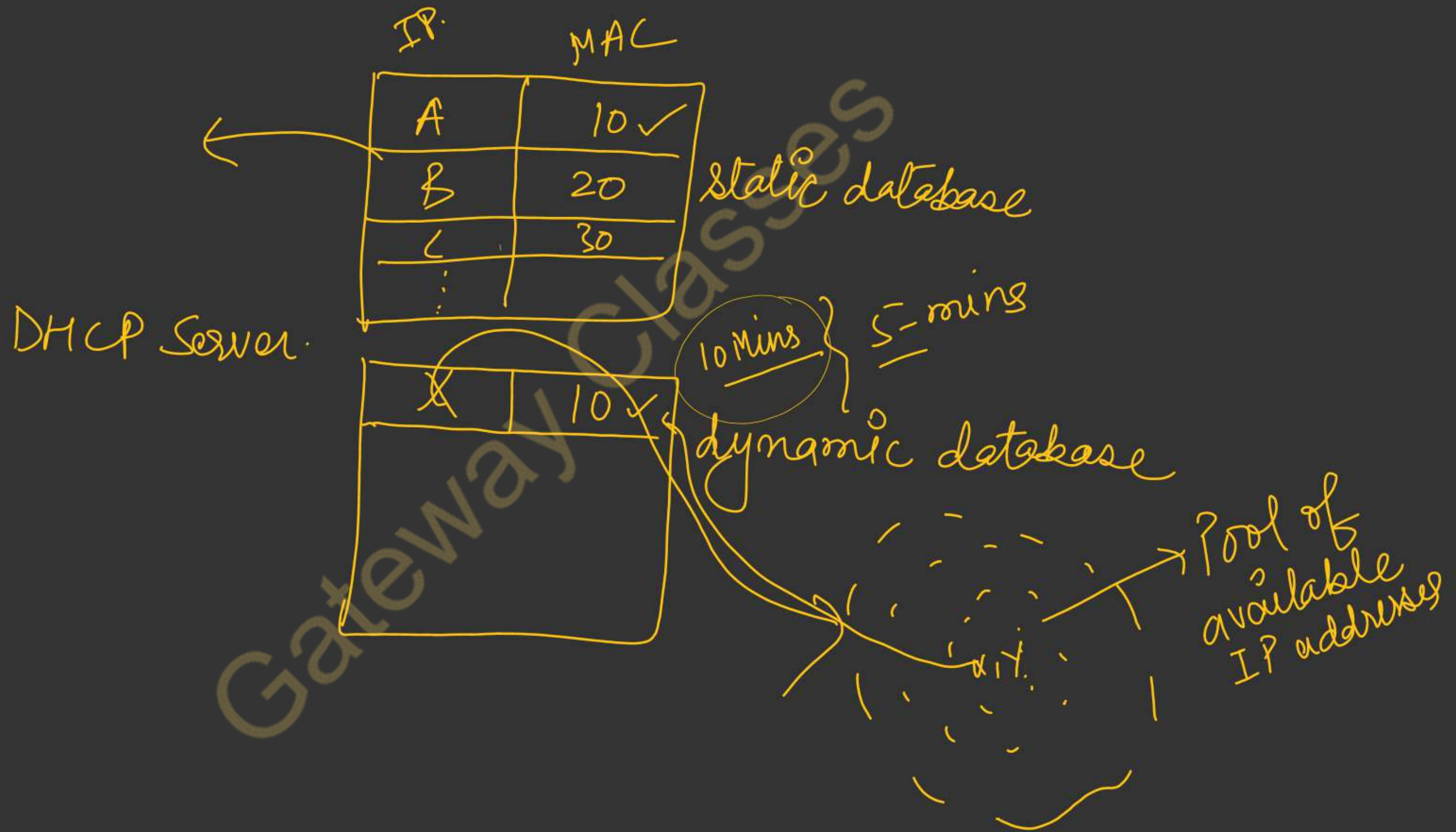
- Host running the BOOTP client can request a static address from a DHCP server.
- A DHCP server has a database that statically binds physical addresses to IP addresses.

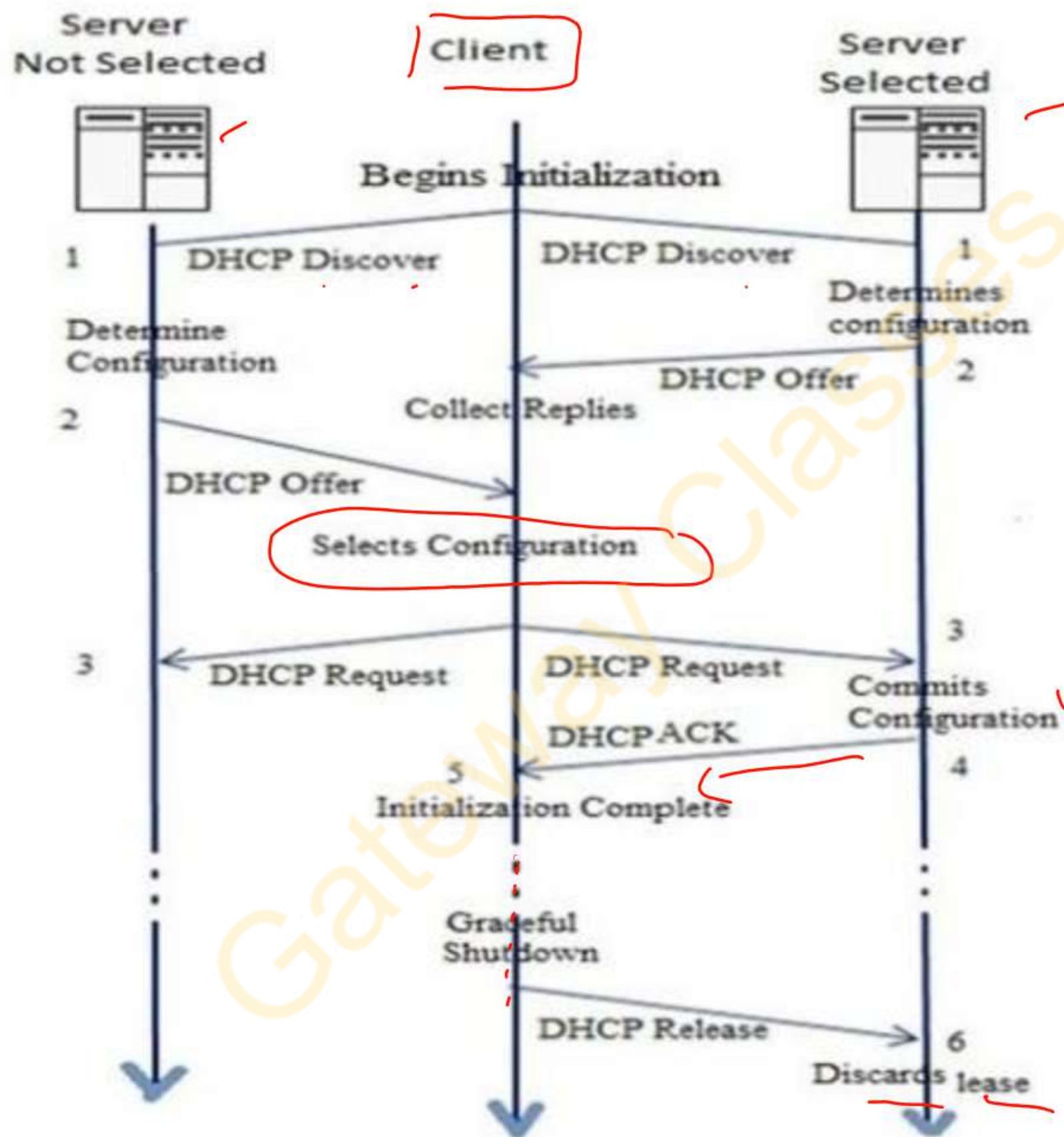
Dynamic Address Allocation:

- DHCP has a second database with a pool of available IP addresses.
- When a DHCP client sends a request to a DHCP server, the server first checks its static database.
- If an entry with the requested physical address exists in the static database, the permanent IP of the client is returned otherwise DHCP server assigns an IP address from pool of available IP addresses for a negotiable period of time and adds the entry to the dynamic database.

DHCP provides temporary IP addresses for a limited time

- When the lease time expires, the client must either stop using the IP address or renew the lease.
- The server has the option to agree or disagree with the renewal.
- If the server disagrees, the client stops using the address.





DHCP Discover:

Client broadcasts DHCP discover to see if there are any DHCP servers present in a network or not.

The message might request subnet mask, domain name server, and domain name, etc.

DHCP Offer:

DHCP Server offers unleashed IP address and other TCP configuration information to Client

The DHCP server/s respond to the host specifying the server ID.

DHCP Request: Client Accepts one of the DHCP Server Offer and sends DHCP request.

DHCP Ack:

Selected DHCP server acknowledges client and leases the IP address and additional configuration info.

Client may use this IP address and configuration till its lease expires or client sends a DHCP Release message to server to end the lease. Client requests renewal of lease when half lease time has passed by sending a DHCP Request to server. After lease expiry, client must start DHCP Discover process again.

- IP provides unreliable and connectionless datagram delivery.
 - The IP protocol has no error-reporting or error-correcting mechanism.
 - The IP protocol also lacks a mechanism for host and management queries.

Internet Control Message Protocol (ICMP)

It is a companion to the IP protocol.

Types of Messages

1. Error-reporting messages
2. query messages ✓

(ICMP does not correct errors)

Why ICMP?

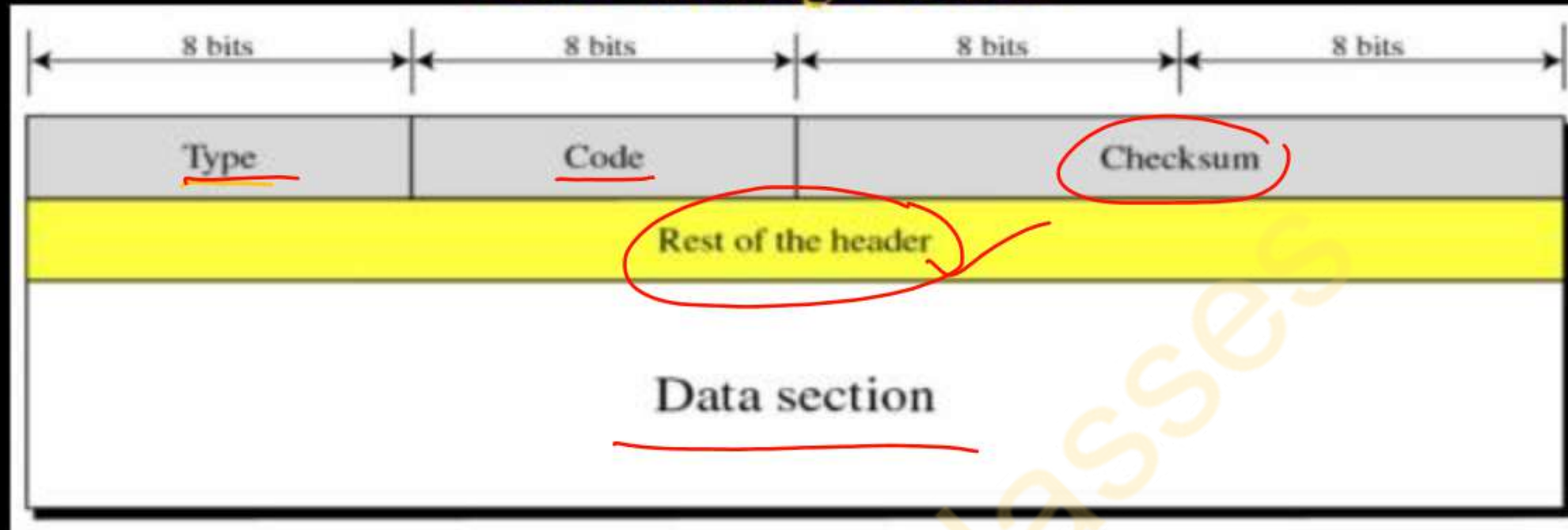
Error Reporting
↓

- 1) what happens if anything goes wrong?
- 2) what if a router has to discard a packet as it can't help to enroute the packet to its' destination?
- 3) what happens if destination discards the whole datagram if any of its' fragment is arrived after the time limit.

host / Management query
↓

- To check if a particular host is alive or not.

ICMP Message Format



An ICMP message has 8-byte header (first 4 bytes are common to all) and a variable-size data section.

Type: defines the type of the message.

Code: reason for the particular message type.

Checksum: calculated over the entire message (header and data).

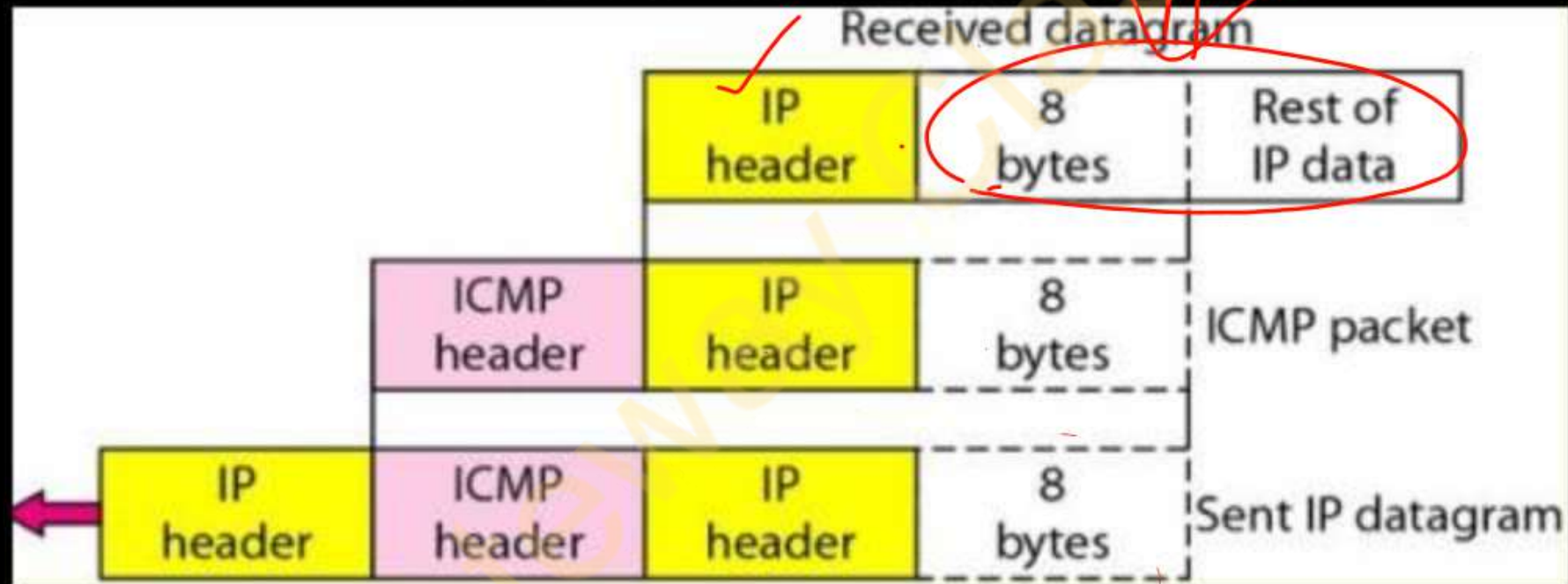
The rest of the header is specific for each message type.

Data section: In error messages carries info for finding the original packet that had the error.

In query messages, the data section carries extra information based on the type of the query.

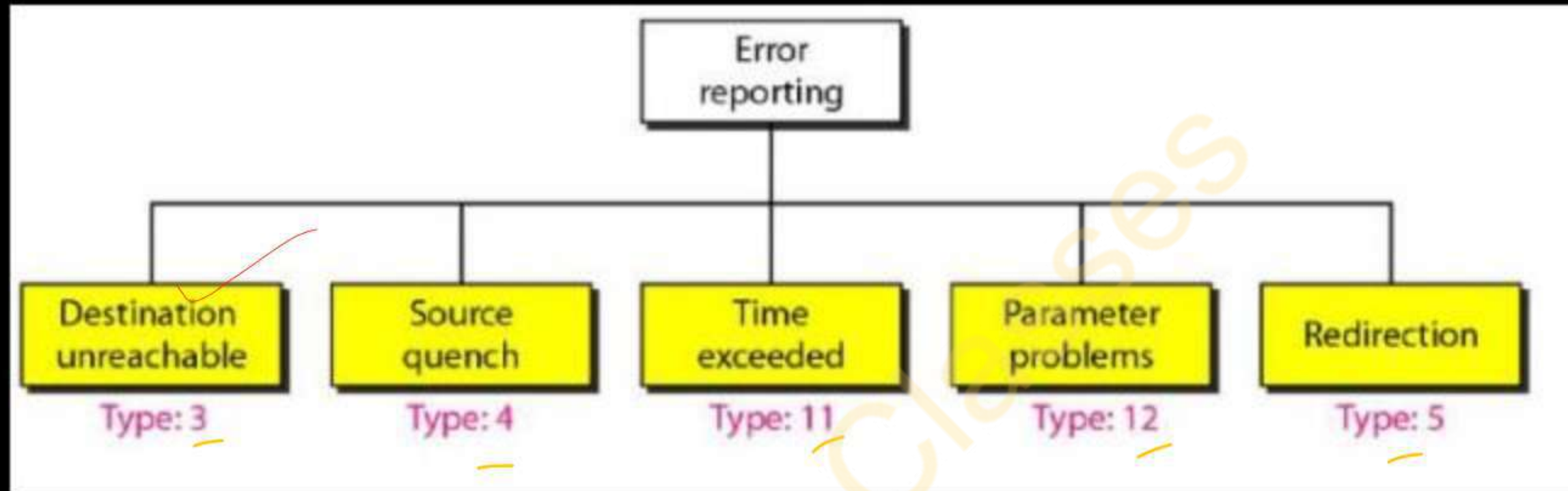
Error Reporting:

- ICMP does not correct errors-it simply reports them.
- Error correction is left to the higher-level protocols.
- Error messages are always sent to the original source as the only information available in the datagram about the route is source and destination IP addresses.



- The 8 bytes of data provide information about the port numbers (UDP and TCP) and sequence number (TCP) to inform the protocols (TCP or UDP) about the error.

Five types of errors are handled:



1. Destination Unreachable:

- When a router cannot route a datagram or a host cannot deliver it, datagram is discarded.
- Router or the host sends a destination-unreachable message back to the source host that initiated the datagram.

2. Source Quench:

- IP does not have a flow control mechanism.
- If source produce datagrams faster than can be forwarded by routers or processed by the destination, it create congestion in routers or destination so it discard some of the datagrams and sends a **source-quench** message to the sender of datagram to inform that datagram has been discarded and warns the source that there is congestion somewhere in the path and that the source should slow down the sending process.

3. Time Exceeded:

- If there are errors in one or more routing tables, a packet can travel in a loop endlessly.
- When the TTL value reaches 0, router discards the datagram and sends a time-exceeded message to the original source.
- This time-exceeded message is also generated when not all fragments that make up a message arrive at the destination host within a certain time limit.

3. Parameter Problem:

If a router or the destination host discovers an ambiguous/missing value in any field of datagram, it discards the datagram and sends a parameter-problem message back to the source.

4. Redirection

Routers are supposed to update their routing tables dynamically.

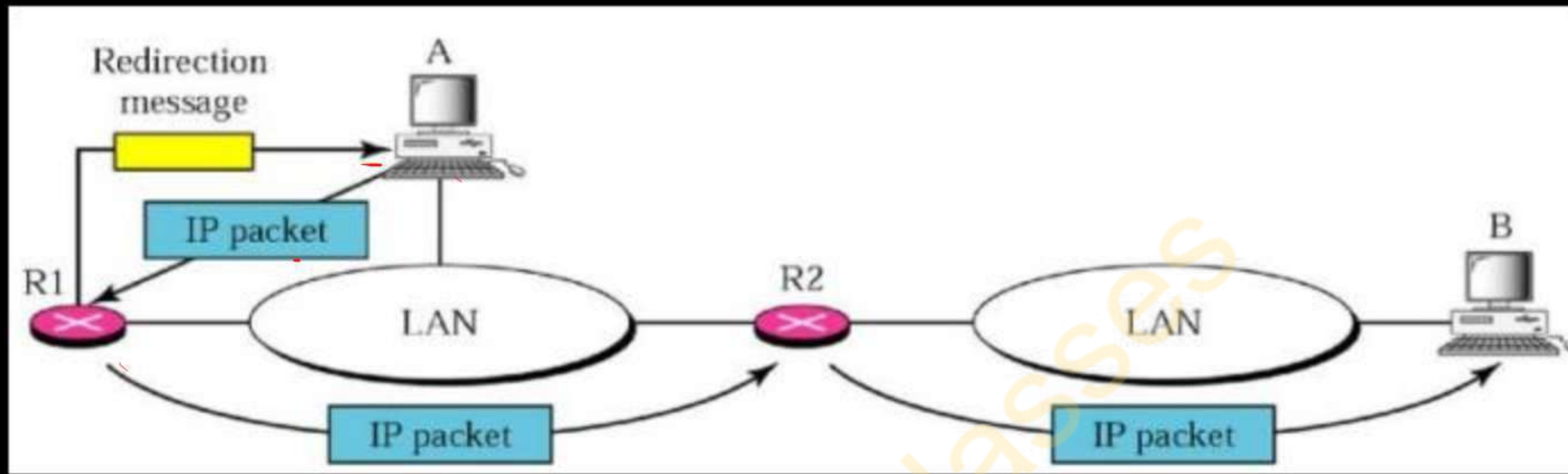
When a host comes up, its routing table has a limited number of entries.

It usually knows the IP address of only the default router.

So, host may send a datagram, destined for another network, to the wrong router.

Now, the router that receives the datagram will forward the datagram to the correct router.

To update the routing table of the host, it sends a redirection message to the host.

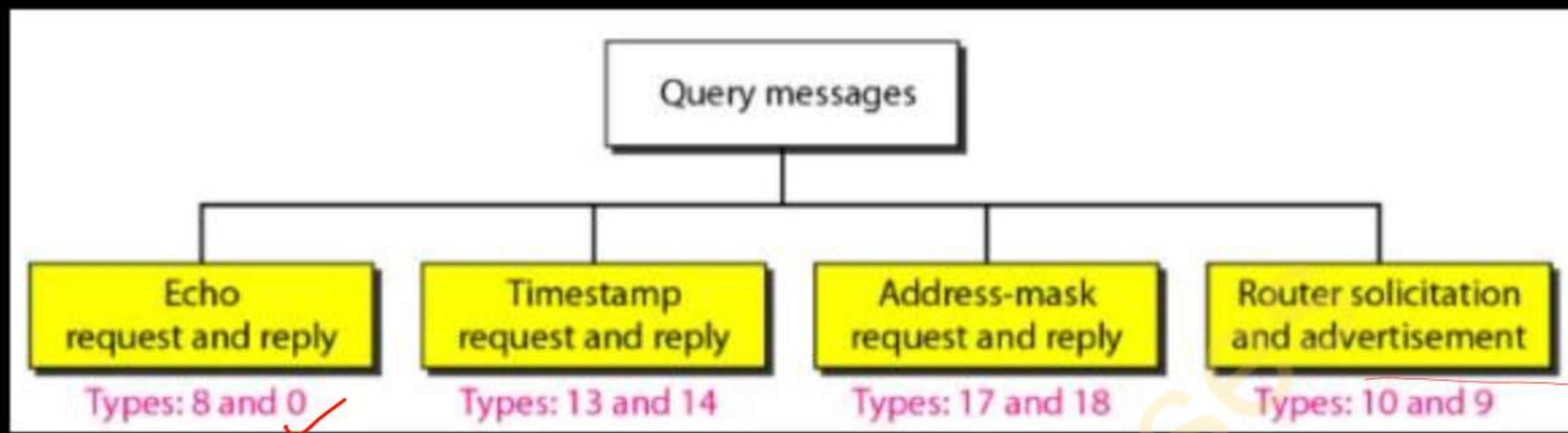


Query:

ICMP can diagnose some network problems through the query messages.

In this ICMP message, a node sends a message that is answered in a specific format by destination node.

A query message is encapsulated in an IP packet, then encapsulated in a data link layer frame.

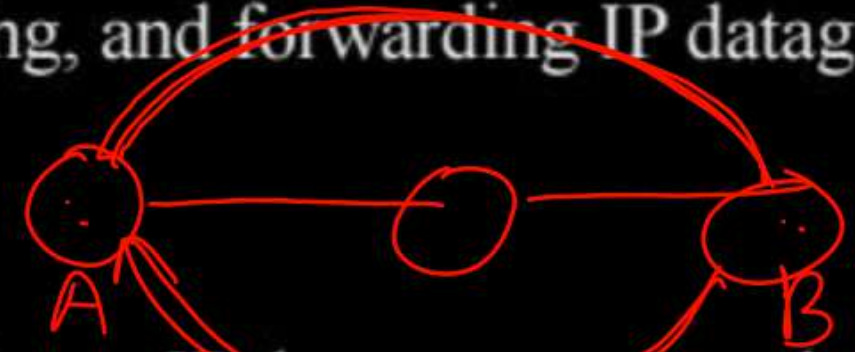


Echo Request and Reply:

- Determines whether two systems (hosts or routers) can communicate with each other.
- Also, it is proof that the intermediate routers are receiving, processing, and forwarding IP datagrams.

Timestamp Request and Reply:

- Hosts or routers use these to determine the round-trip time needed for an IP datagram to travel between them.
- It can also be used to synchronize the clocks in two machines.



Address-Mask Request and Reply:

A host may know its IP address, but not the corresponding mask.

To obtain its mask, it sends an address-mask-request message to a router on the LAN.

The router receiving the address-mask-request message responds with an address-mask-reply message.

Router Solicitation and Advertisement:

A host needs to know the address of routers connected to its own network; it can broadcast a router-solicitation message.

In response, router/s broadcast their routing information using the router advertisement message.

A router can also periodically send router-advertisement messages even if no host has solicited.

Checksum :- Calculated over (header + data)

AKTU PYQs

Q: Explain ICMP protocol and its application in real-world scenarios. (AKTU 2022-23)

Q: Explain ARP. (AKTU 2022-23)

Q: Discuss in detail about ICMP role in network layer. (AKTU 2021-22)

Q: How is BOOTP different from DHCP? (AKTU 2018-19)

Gateway Classes



AKTU

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Computer Networks

Unit-3: Lecture 7

Today's Target

- Forwarding and Delivery ✓
- Unicast Routing Protocols ✓
 - Distance Vector Routing ✓
- AKTU PYQs

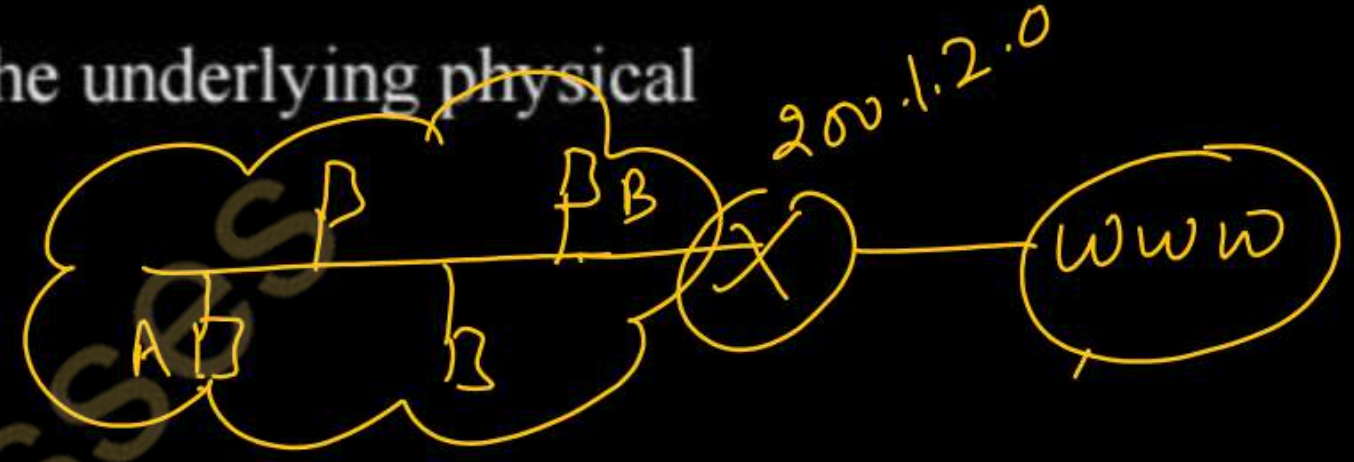
By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Delivery: [Explain delivery in n/w layer]

The network layer supervises the handling of the packets by the underlying physical networks. (host to host delivery)

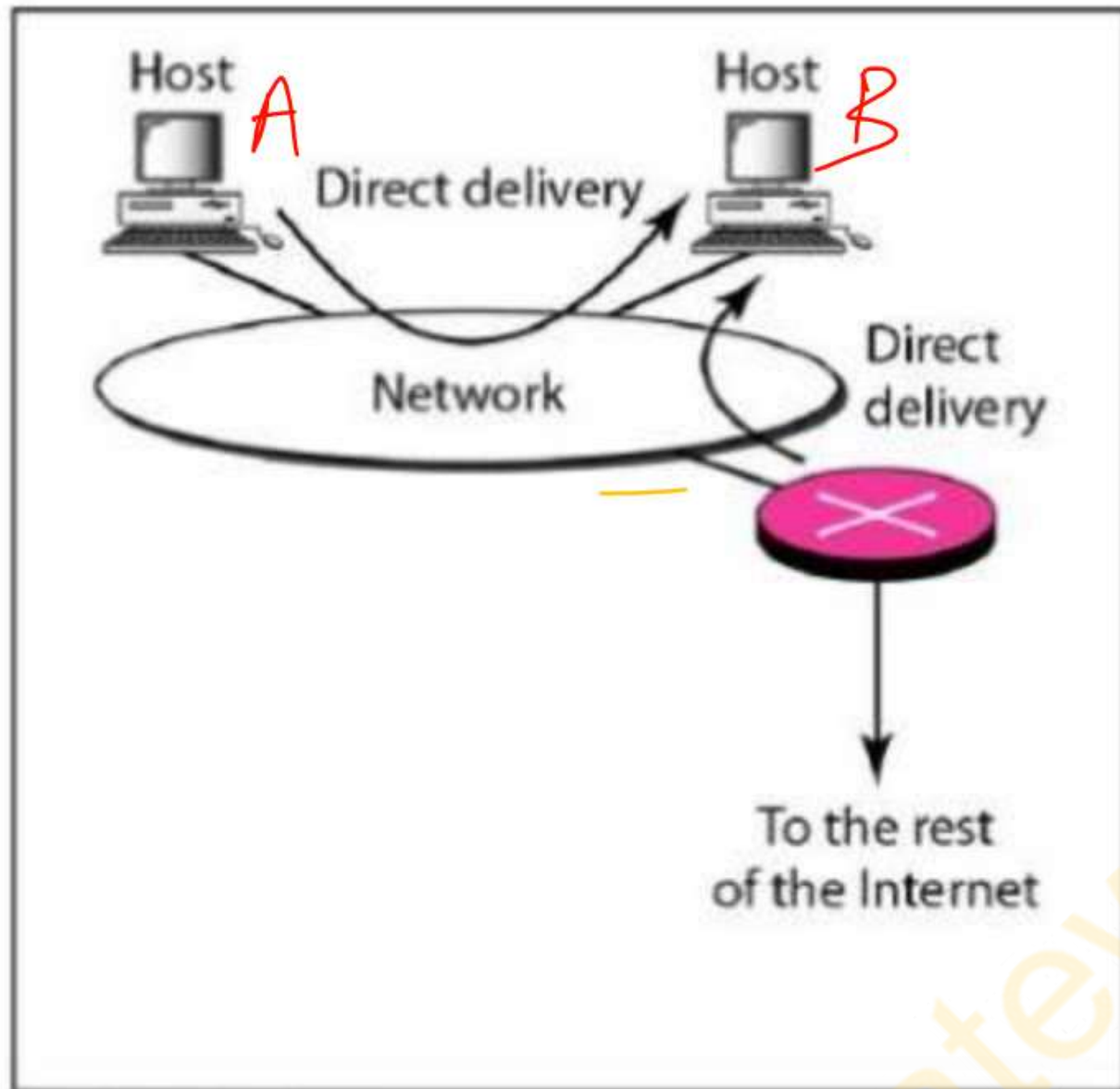


Direct Delivery:

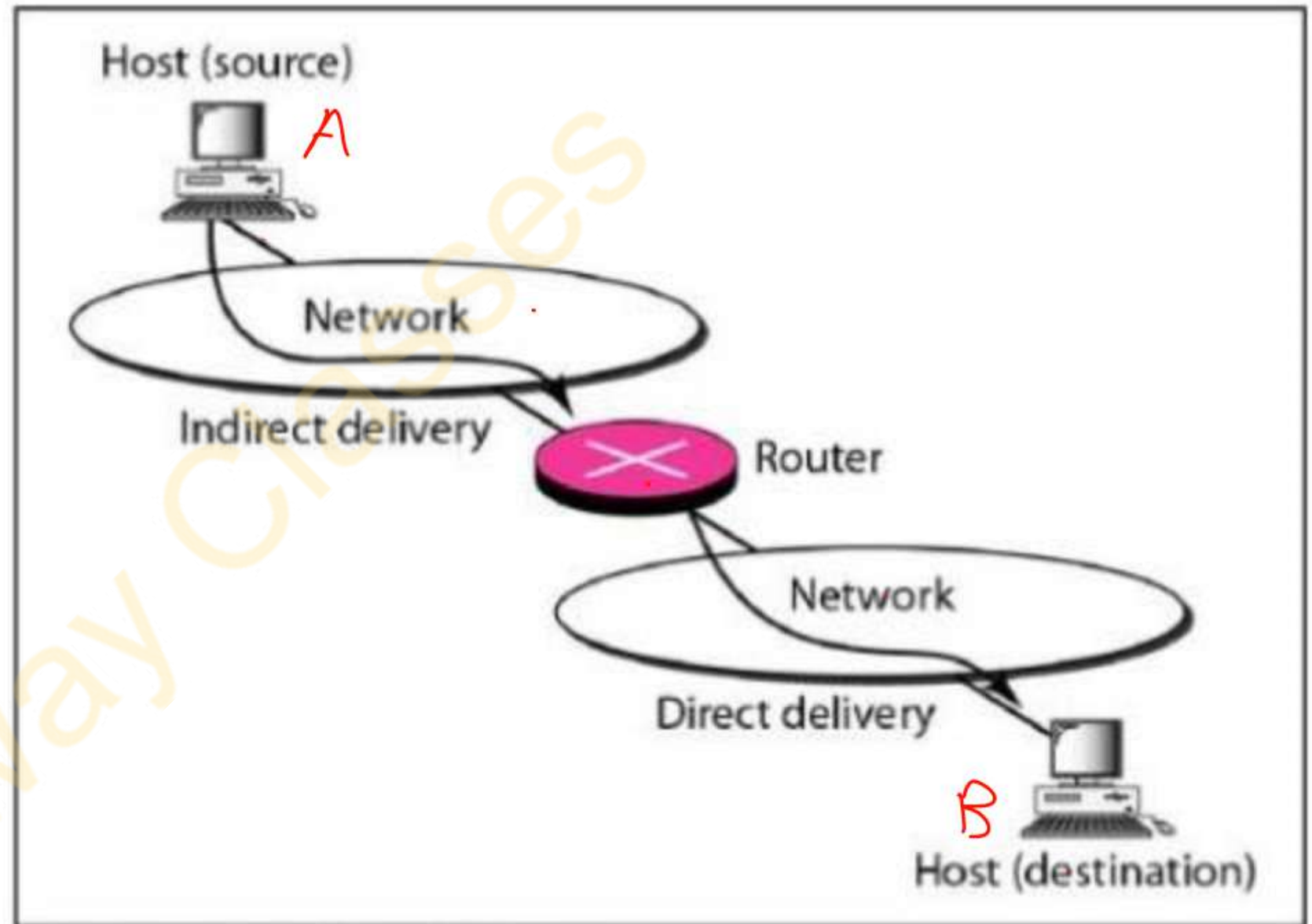
- Source and destination of the packet are located on the same physical network.
- To determine if the delivery is direct, sender can extract network address of the destination (using the mask) and compare this address with its network address. If a match is found, the delivery is direct.

Indirect Delivery:

- If destination host is not on the same network, the packet is delivered indirectly.
- The packet goes from router to router until it reaches its final destination.



a. Direct delivery



b. Indirect and direct delivery

Forwarding:

- Forwarding means to place the packet in its route to its destination.
- When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this routing table to find the route to the final destination.

Forwarding Techniques:

- To make size of routing table manageable and also handle issues such as security.

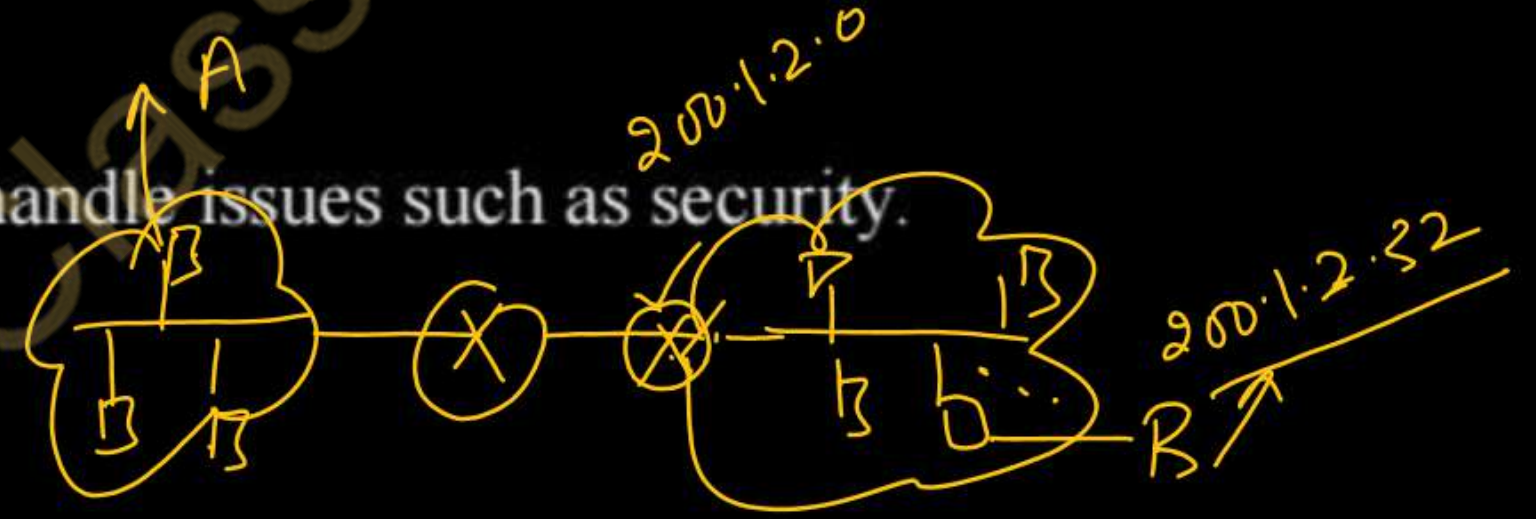
①

Next-Hop Method Versus Route Method:

- Routing table holds only the address of the next hop instead of complete route (Route method).

Network-Specific Method Versus Host-Specific Method:

- Instead of having an entry for every destination host connected to the same physical network (host-specific method), we have only one entry that defines the address of the destination network itself.



a. Routing tables based on route

Destination	Route
Host B	R1, R2, host B

Routing table
for host A

Destination	Route
Host B	R2, host B

Routing table
for R1

Destination	Route
Host B	Host B

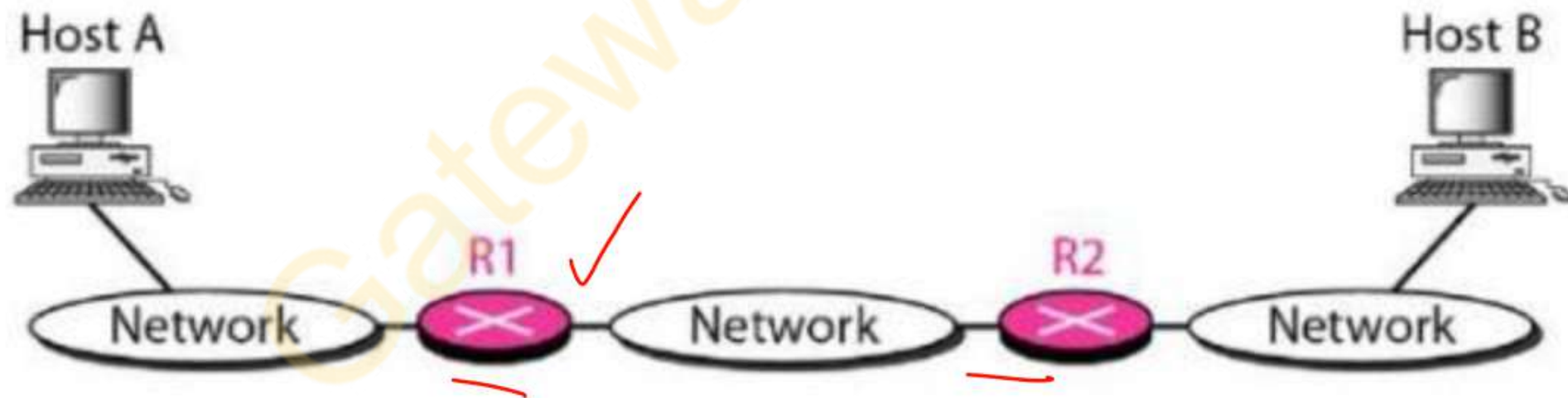
Routing table
for R2

b. Routing tables based on next hop

Destination	Next hop
Host B	R1 ✓

Destination	Next hop
Host B	R2 ✓

Destination	Next hop
Host B	---

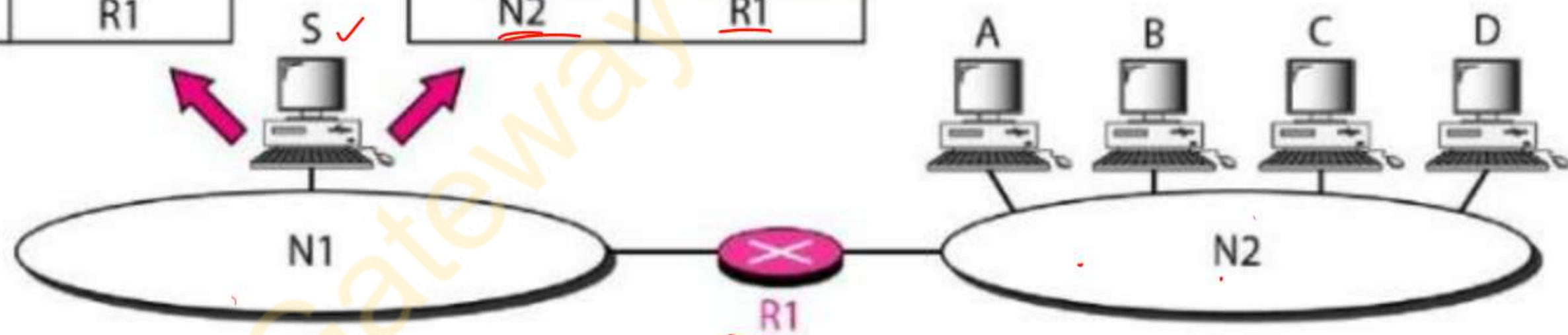


Routing table for host S based on host-specific method

Destination	Next hop
<u>A</u>	R1
<u>B</u>	R1
<u>C</u>	R1
<u>D</u>	R1

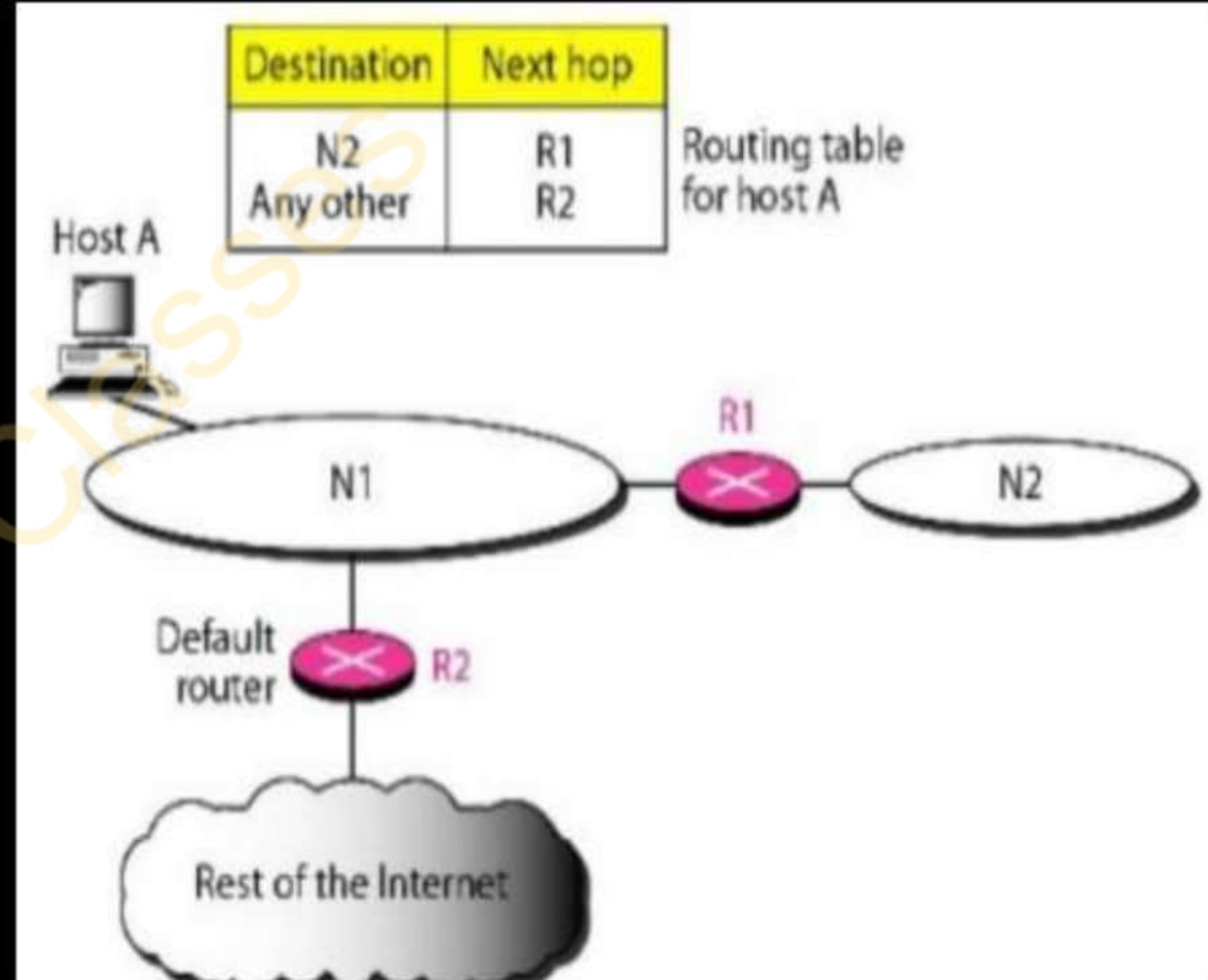
Routing table for host S based on network-specific method

Destination	Next hop
<u>N2</u>	<u>R1</u>



Default Method:

- Host A is connected to a network with two routers.
- Router R1 routes the packets to hosts connected to network N2. However, for the rest of the Internet, router R2 is used.
- So, instead of listing all networks in the entire Internet, host A can just have one entry called the default (normally defined as network address



Routing Table:

The routing table can be either static or dynamic.

Static Routing Table:

- Route for each destination entered manually by the administrator.
- Doesn't update automatically when there is a change in the Internet. The table must be manually altered by the administrator.
- A static routing table can be used in a small internet that does not change very often.

Dynamic Routing Table:

- Updated periodically using any dynamic routing protocols such as RIP, OSPF, or BGP.
- Whenever there is a shutdown of router or breaking of a link, the dynamic routing protocols update all the routing tables automatically.
- The routers in Internet need to be updated dynamically for efficient delivery of IP packets.

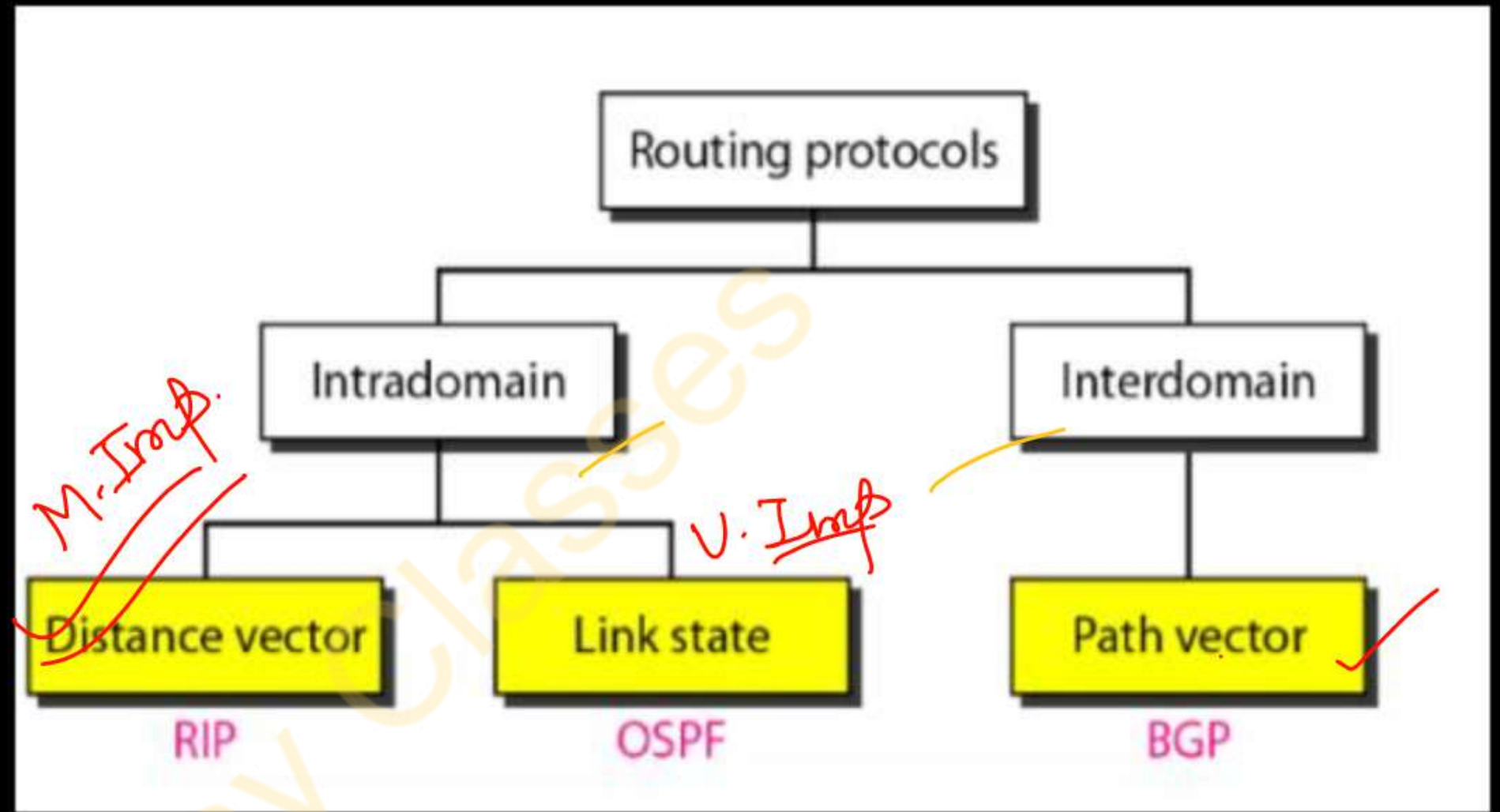
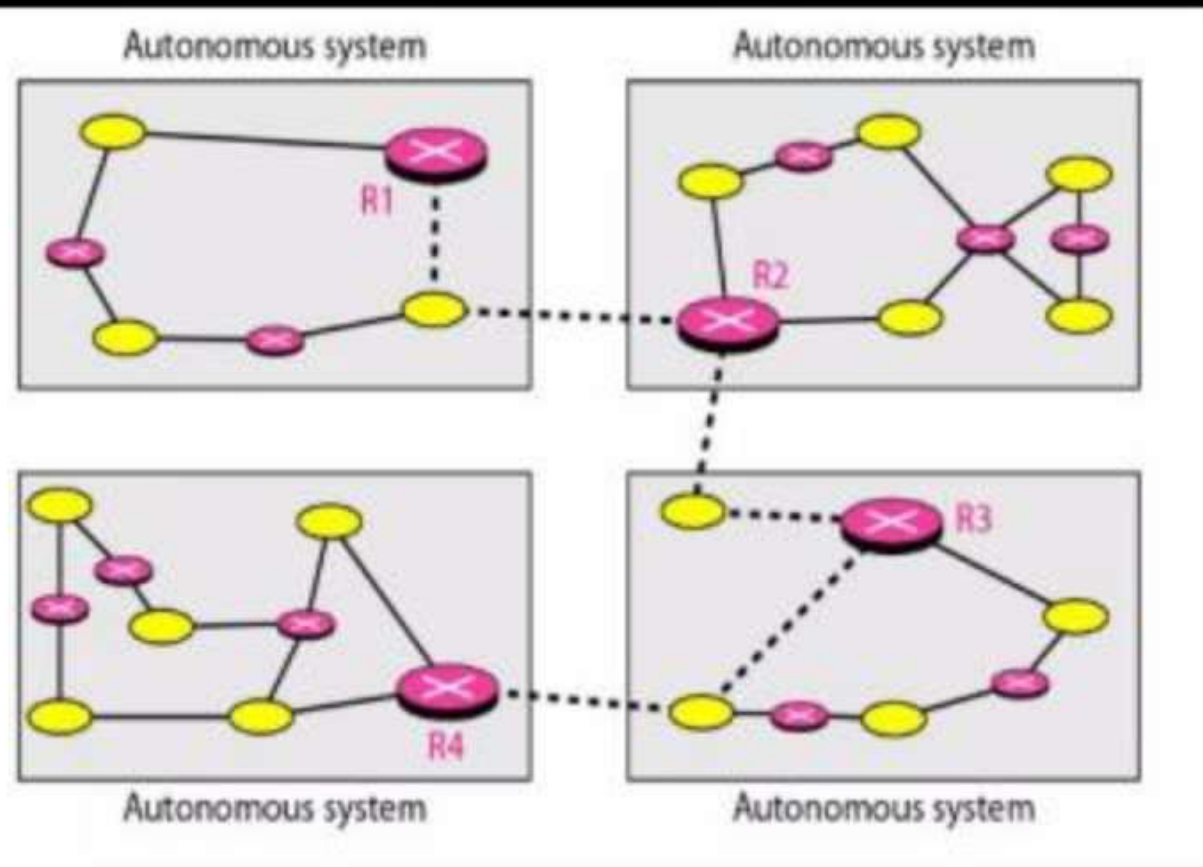
- A router receives a packet from a network and passes it to another network.
 - A router is usually attached to several networks.
- ① When it receives a packet, to which network should it pass the packet?
 - ② Which of the available pathways is the optimum pathway?

Intra- and Interdomain Routing

Internet is divided into autonomous systems (group of networks and routers under the authority of a single administration).

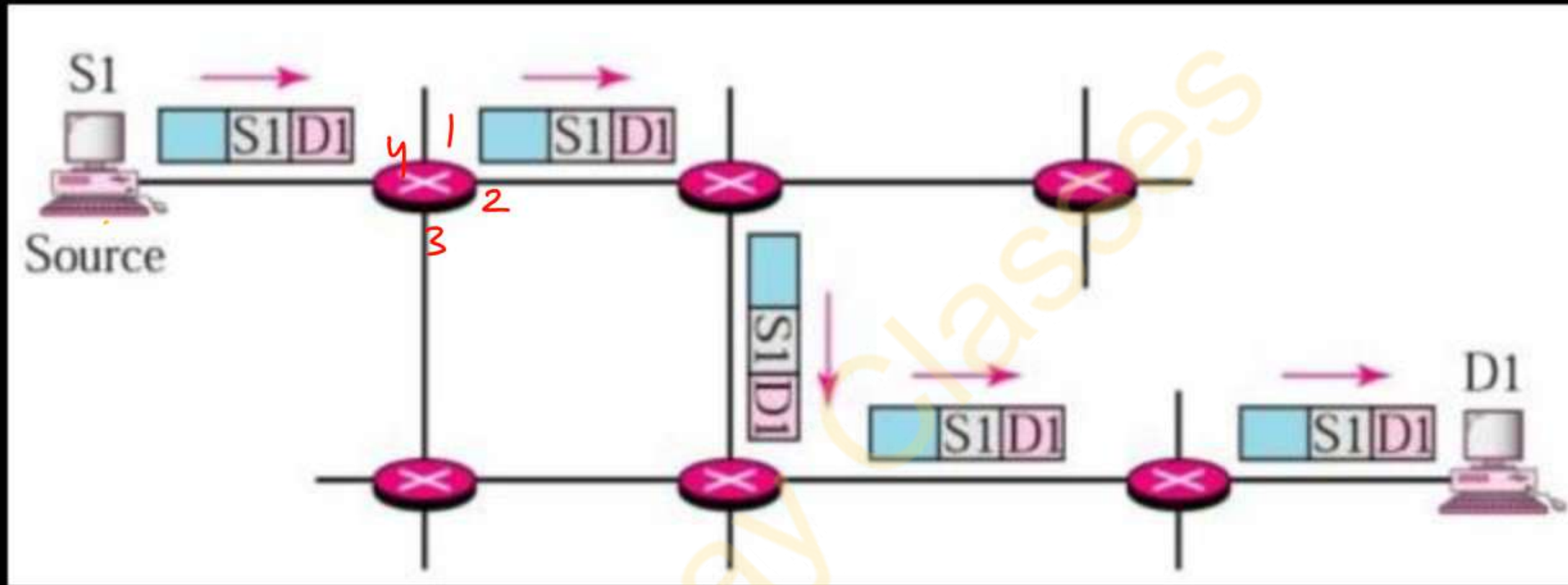
Two types of Routing - 1) Intra-domain ✓
2) Inter-domain ✓

- Routing inside an autonomous system is referred to as **intradomain routing**.
- Routing between autonomous systems is referred to as **interdomain routing**.
- Each autonomous system can choose one or more intradomain routing protocols.
- Only one interdomain routing protocol handles routing between autonomous systems.



- Routing Information Protocol (RIP) is an implementation of the distance vector protocol.
- Open Shortest Path First (OSPF) is an implementation of the link state protocol.
- Border Gateway Protocol (BGP) is an implementation of the path vector protocol.

Unicast Routing: Transmission from a single sender to a single receiver and router forwards the received packet through one of its ports.



Major Protocols of Unicast Routing:

1. **Distance Vector Routing:** uses a distributed algorithm to compute their routing tables.
2. **Link-State Routing:** uses link-state routers to exchange messages to learn the entire network topology.
3. **Path-Vector Routing:** It is a routing protocol that maintains the path that is updated dynamically.

Distance Vector Routing

- Least-cost route between any two nodes is the route with minimum distance.
- Each node maintains a vector (table) of minimum distances to every node.
- The table guides packets to the desired node by showing the next stop in the route (next-hop routing).

Initialization:

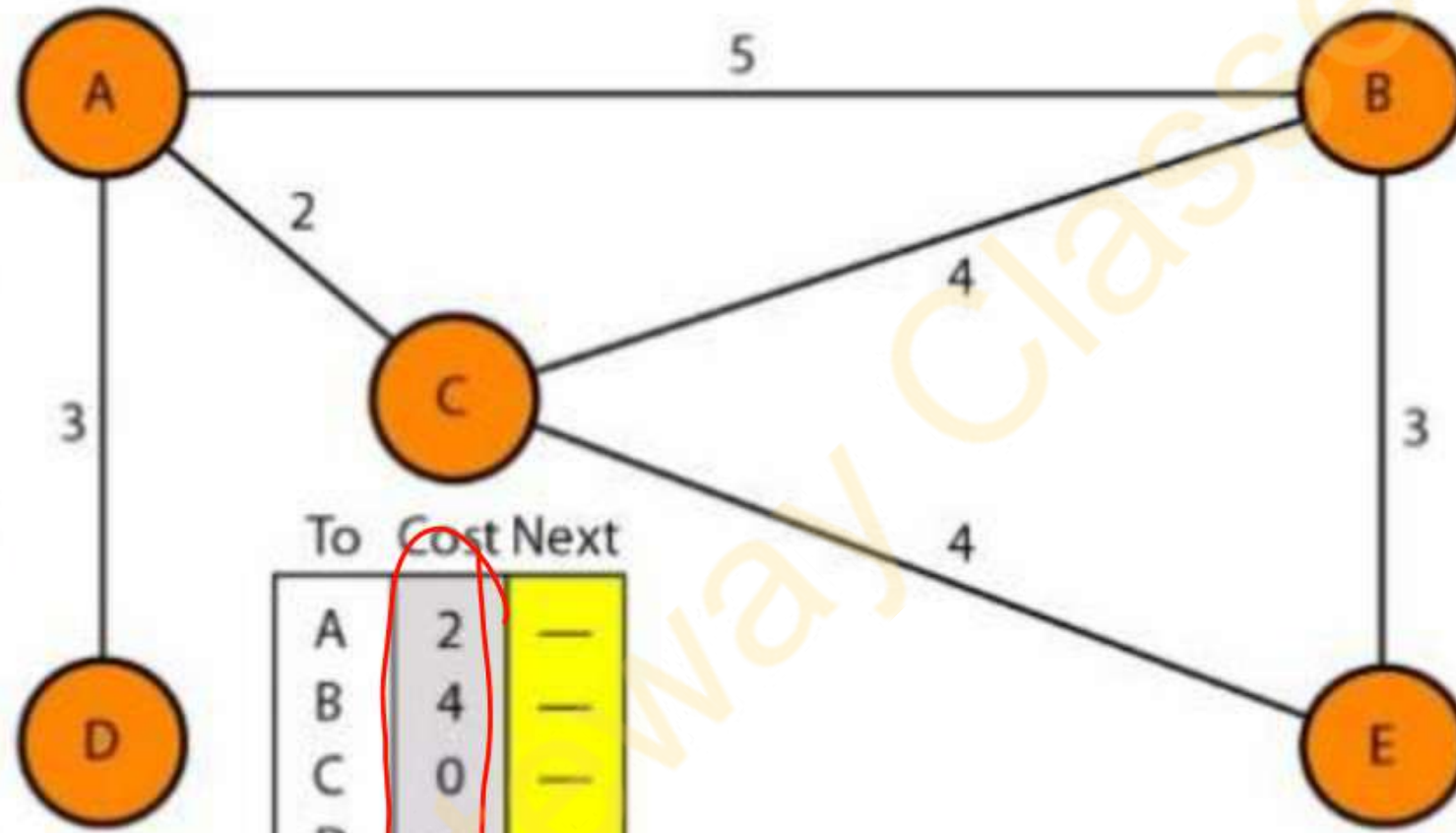
- At the beginning, each node can know ~~only know~~ the distance between itself and its immediate neighbors, those directly connected to it.
- We assume that each node can send a message to the immediate neighbors and find the distance between itself and these neighbors.
- The distance for any entry that is not a neighbor is marked as infinite (unreachable).

To	Cost	Next
A	0	—
B	5	—
C	2	—
D	3	—
E	∞	—

A's table

To	Cost	Next
A	3	—
B	∞	—
C	∞	—
D	0	—
E	∞	—

D's table



To	Cost	Next
A	2	—
B	4	—
C	0	—
D	∞	—
E	4	—

C's table

To	Cost	Next
A	5	—
B	0	—
C	4	—
D	∞	—
E	3	—

B's table

To	Cost	Next
A	∞	—
B	3	—
C	4	—
D	∞	—
E	0	—

E's table

Dest^m Distance vector

Sharing:

- Idea of distance vector routing is the sharing of information between neighbors.
- Although node A does not know about node E, node C does. So, if node C shares its routing table with A, node A can also know how to reach node E.
- Nodes can improve their routing tables if they help each other.
- A node send only the first two columns of its table to its immediate neighbors periodically and when there is a change.

Updating:

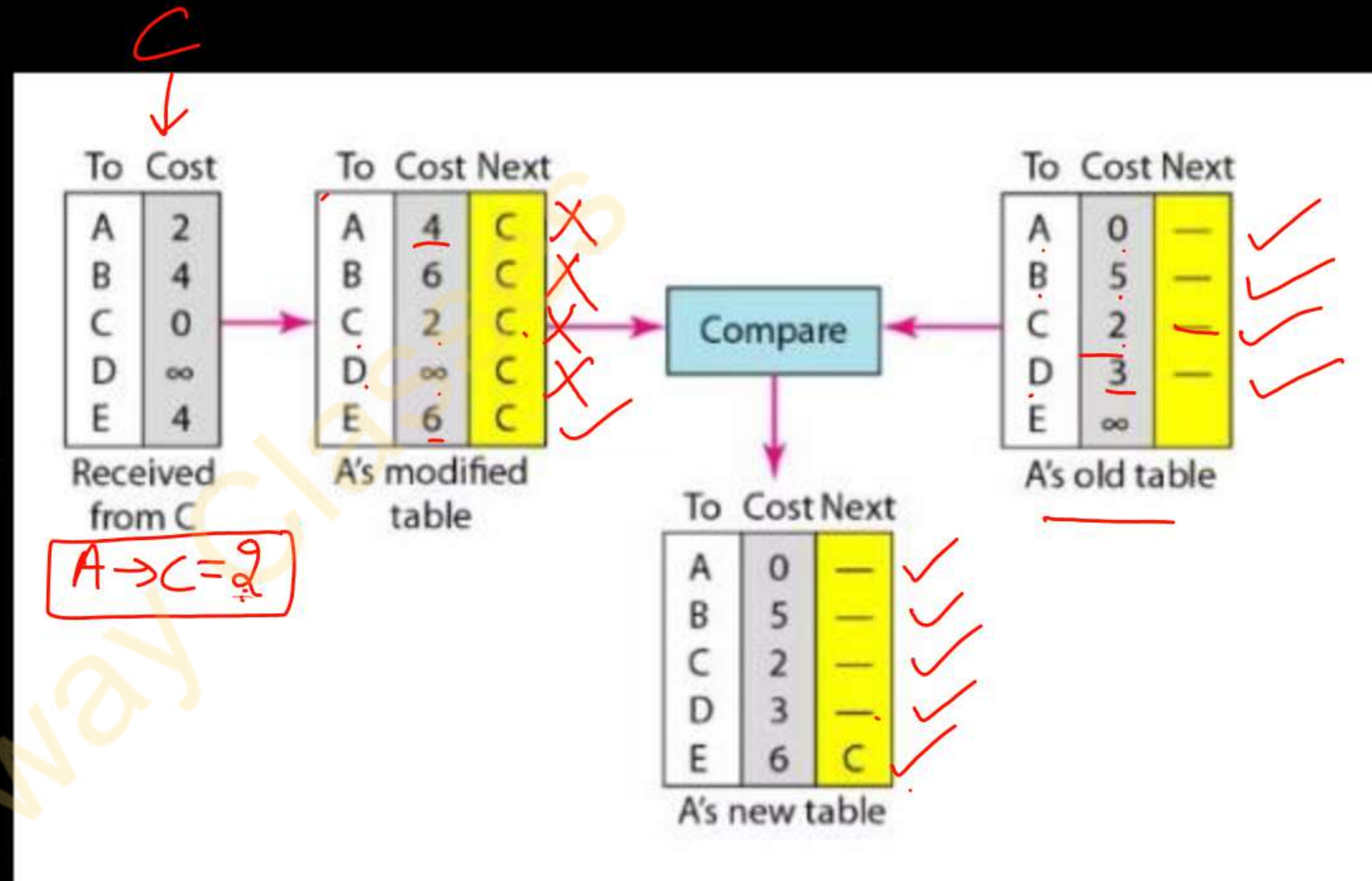
When a node receives a two-column table from a neighbor, it needs to update its routing table.

Updating takes three steps:

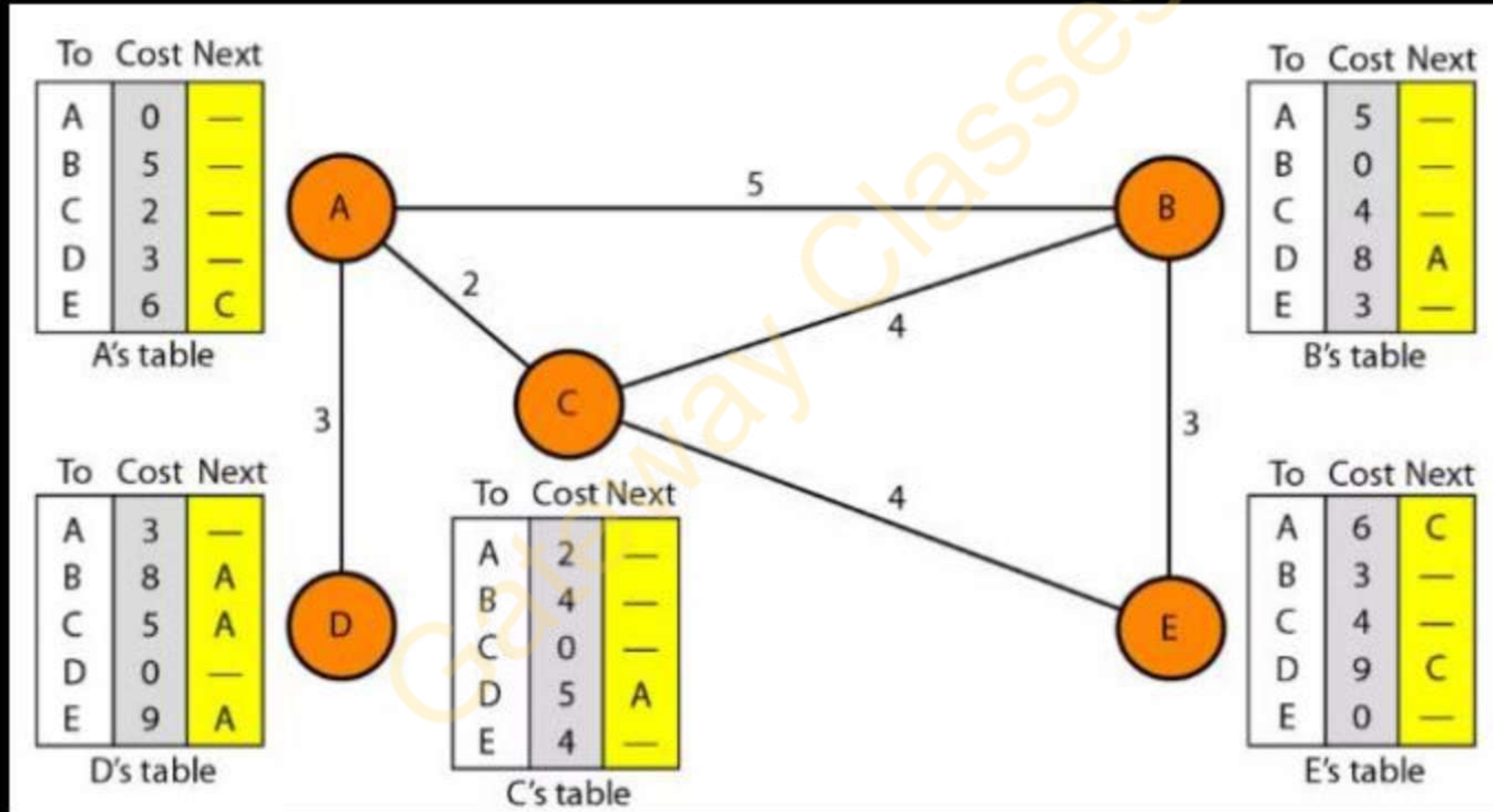
1. The receiving node needs to add the cost between itself and the sending node to each value in the second column.
2. The receiving node needs to add the name of the sending node to each row as the third column if the receiving node uses information from any row. The sending node is the next node in the route.
3. The receiving node needs to compare each row of its old table with the corresponding row of the modified version of the received table.
 - a. If the next-node entry is different, the receiving node chooses the row with the smaller cost. If there is a tie, the old one is kept.
 - b. If the next-node entry is the same, the receiving node chooses the new row.

Example:

- Suppose node C has previously advertised a route to node X with distance 3.
- Suppose now there is no path between C and X; node C now advertises this route with a distance of infinity.
- Node A must not ignore this value even though its old entry is smaller. The old route does not exist any more. The new route has a distance of infinity.



- Each node can update its table by using the tables received from other nodes.
- In a short time, if there is no change in the network itself, such as a failure in a link, each node reaches a stable condition in which the contents of its table remains the same.

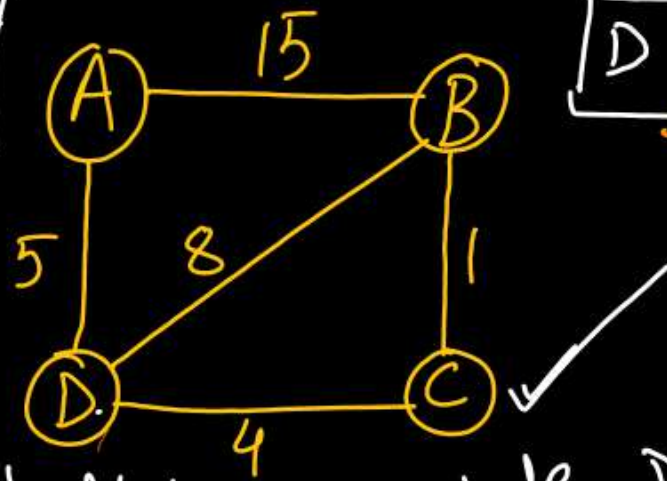


node Dist. Next hop

Example

A	0	—
B	15	—
C	∞	—
D	5	—

A	15	—
B	0	—
C	1	—
D	8	—



Node Dist. Next hop

A	5	—
B	8	—
C	4	—
D	0	—

Node Dist. Next hop

A	∞	—
B	1	—
C	0	—
D	4	—

C's Routing table updation

→ C will get distance vectors from B and D

	(B)	(D)
A	15	5
B	0	8
C	1	4
D	8	0

CB=1

CD=4



node Dist. next hop

A	?	?
B	?	?
C	?	?
D	?	?

C's updated table

for C → A Entry

$$C \rightarrow B \rightarrow A \Rightarrow 1 + 15 = \textcircled{16}$$

$$C \rightarrow D \rightarrow A \Rightarrow 4 + 5 = \textcircled{9} \checkmark$$

for C → B Entry

$$C \rightarrow B \rightarrow B \Rightarrow 1 + 0 = \textcircled{1} \checkmark$$

$$C \rightarrow D \rightarrow B \Rightarrow 4 + 8 = \textcircled{12}$$

C → C Entry

C → D Entry

$$C \rightarrow B \rightarrow D \Rightarrow 1 + 8 = \textcircled{9}$$

$$C \rightarrow D \rightarrow D \Rightarrow 4 + 0 = \textcircled{4} \checkmark$$

C's Routing table updation

→ C will get distance vectors from (B) and (D)

	(B)	(D)
A	15	5
B	0	8
C	1	4
D	8	0

CB = 1

CD = 4



node	Dist.	next hop
A	9	D
B	1	—
C	0	—
D	4	—

C's updated table

A → B Entry

$$A \rightarrow B \rightarrow B \Rightarrow 15 + 0 \Rightarrow \textcircled{15}$$

$$A \rightarrow D \rightarrow B \Rightarrow 5 + 8 \Rightarrow \textcircled{13} \checkmark$$

A → C Entry

$$A \rightarrow B \rightarrow C \Rightarrow 15 + 1 = \textcircled{16}$$

$$A \rightarrow D \rightarrow C \Rightarrow 5 + 4 = \textcircled{9} \checkmark$$

A → D Entry

$$A \rightarrow B \rightarrow D \Rightarrow 15 + 8 = \textcircled{23}$$

$$A \rightarrow D \rightarrow D \Rightarrow 5 + 0 = \textcircled{5} \checkmark$$

A's Routing table updation

→ A will get distance vectors from $\textcircled{B}$ and $\textcircled{D}$

	$\textcircled{B}$	$\textcircled{D}$
A	15	5
B	0	8
C	1	4
D	8	0

AB = 15

AD = 5



	node	dist.	next hop
A	0		—
B	13		D
C	9		D
D	5		—

A's updated table

Similarly (B) and (C) will also update their routing tables.

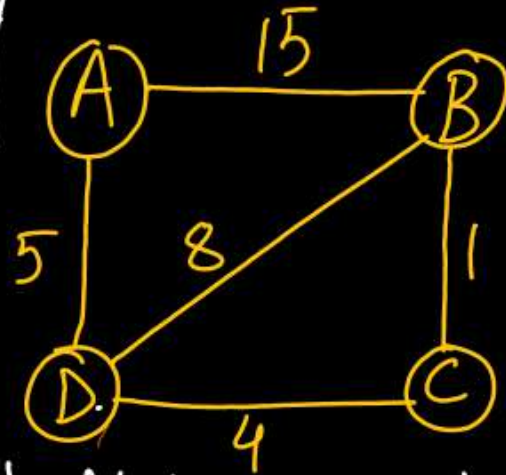
⇒ In Max $(n-1)$ exchanges, the all routers will reach to optimal routing table info.

↓
No. of nodes.

node Dist. Next hop

A	0	—
B	15	—
C	∞	—
D	5	—

A	15	—
B	0	—
C	1	—
D	8	—



Node Dist. Next hop

A	5	—
B	8	—
C	4	—
D	0	—

Node Dist. Next hop

A	∞	—
B	1	—
C	0	—
D	4	—

Final Routing tables

A's table

node Dist. Next hop

A	0	—
B	10	D
C	9	D
D	5	—

B's table

node Dist. Next hop

A	9	C
B	0	—
C	1	—
D	5	C

C's table

node Dist. Next hop

A	9	D
B	1	—
C	0	—
D	4	—

D's table

node Dist. Next hop

A	5	—
B	5	C
C	4	—
D	0	—

Imp

Count to Infinity Problem in DVR

Distance vector routing reacts rapidly to good news, but leisurely to bad news.

Case-I : Suppose A is down initially, all other routers know this and set delay to A as infinity.

- When A comes up, in the first exchange, B learns that its left-hand neighbor has zero delay to A . B now makes an entry in its routing table indicating that A is one hop away to the left.
- All the other routers still think that A is down.
- On next exchange, C learns that B has a path of length 1 to A , so it updates its routing table and so on...



Case-II

A	B	C	D	E	
●	●	●	●	●	Initially
	1	<u>2</u>	3	4	
	3	<u>2</u>	3	4	After 1 exchange
	3	4	3	4	After 2 exchanges
	5	4	5	4	After 3 exchanges
	5	6	5	6	After 4 exchanges
	7	6	7	6	After 5 exchanges
	7	8	7	8	After 6 exchanges
	⋮				
●	●	●	●		

Case-II All the links and routers are initially up. Routers B , C , D , and E have distances to A of 1, 2, 3, and 4 hops, respectively. Suddenly, either A goes down or the link between A and B is cut.

- **At the first packet exchange**, B does not hear anything from A . Fortunately, C says “Do not worry; I have a path to A of length 2.”
- Little does B suspect that C 's path runs through B itself. For all B knows, C might have ten links all with separate paths to A of length 2. As a result, B thinks it can reach A via C , with a path length of 3. D and E do not update their entries for A on the first exchange.
- **On the second exchange**, C notices that each of its neighbors claims to have a path to A of length 3. It picks one of them at random and makes its new distance to A 4 and so on....
- Gradually, all routers work their way up to infinity, this problem is known as the **count-to-infinity problem**.

AKTU PYQs

Q: Explain all Interdomain and Intradomain routings algorithms. (AKTU 2023-24)

Q: Describe “count to infinity problem” with an example (AKTU 2022-23, 2018-19, 2017-18, 2016-17)

Q: Explain distance vector routing (DVR) with working example in detail. (AKTU 2022-23)

Q: What is unicast routing? Explain all unicast routing protocols in detail. (AKTU 2018-19)

Gateway Classes



AKTU

B.Tech 6th Semester
CS IT & Allied



Computer Networks

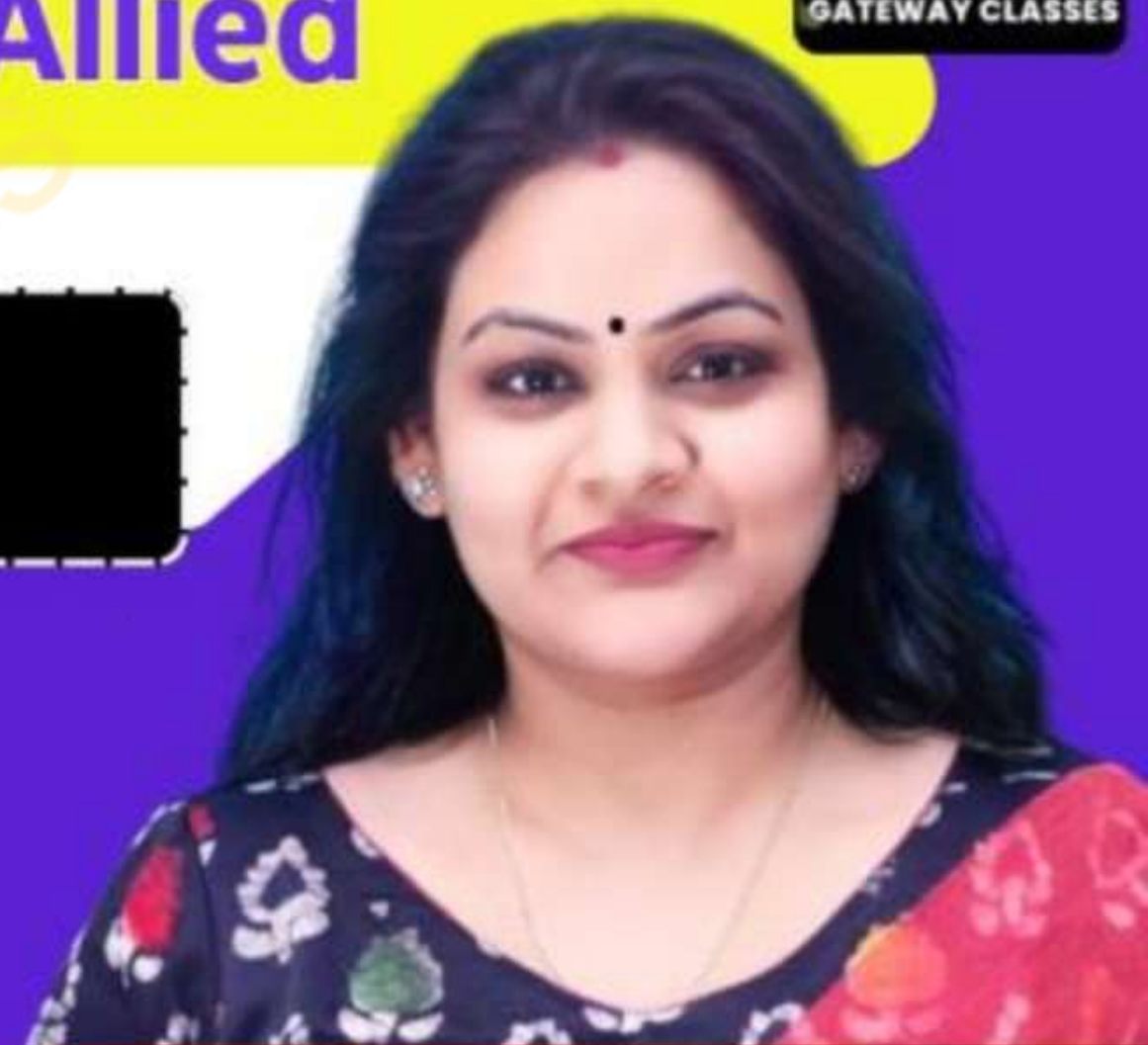
Unit-3: Lecture 8

Today's Target

- Link state Routing ✓
- Path Vector Routing ✓
- Hierarchical Routing ✓
- AKTU PYQs ✓

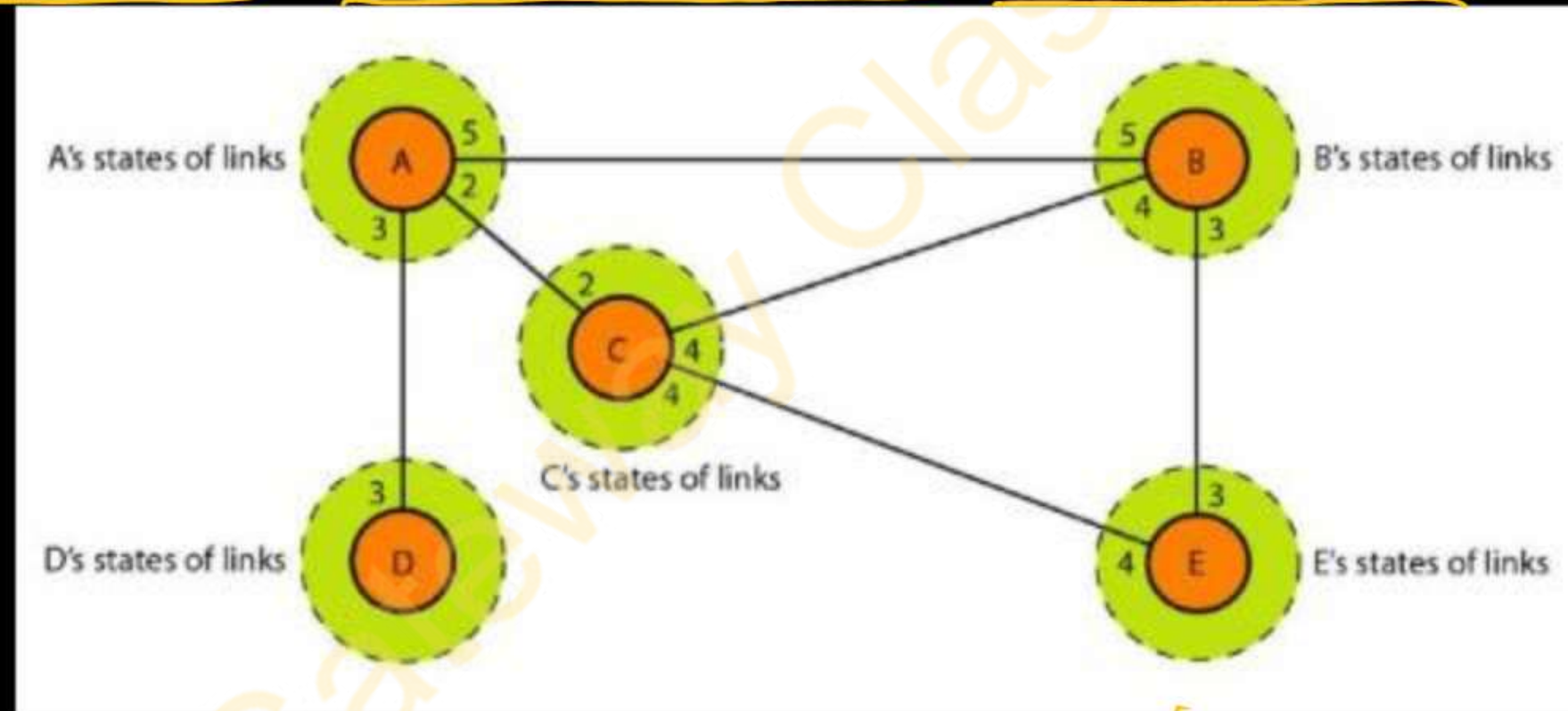
By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Link State Routing

- Based on the assumption that, although global knowledge about the topology is not clear, each node knows the state (type, condition, and cost) of its links. Each router constructs its own internal map of network topology. It provides information about whether the link to reach the router is active or not.
- Whole topology can be compiled from the partial knowledge of each node.



- Node A knows that it is connected to B with metric 5, to C with metric 2, and to D with metric 3. Node C knows that it is connected to node A with metric 2, to node B with metric 4, and to node E with metric 4. And so on.

Building Routing Tables

Four actions are required to ensure that each node has routing table with least-cost node to every other node.

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called **flooding**, in an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree.

1. Creation of Link State Packet (LSP):

Every router will create Link state packets. In the first round every node creates link state packets with the help of Hello packets. Link state packet carries **node identity, list of links, sequence number, and age**.

- **node identity and the list of links**, are needed to make the topology.
- **Sequence number** facilitates flooding and distinguishes new LSPs from old ones.
- **Age** prevents old LSPs from remaining in the domain for a long time.

LSPs are generated on two occasions:

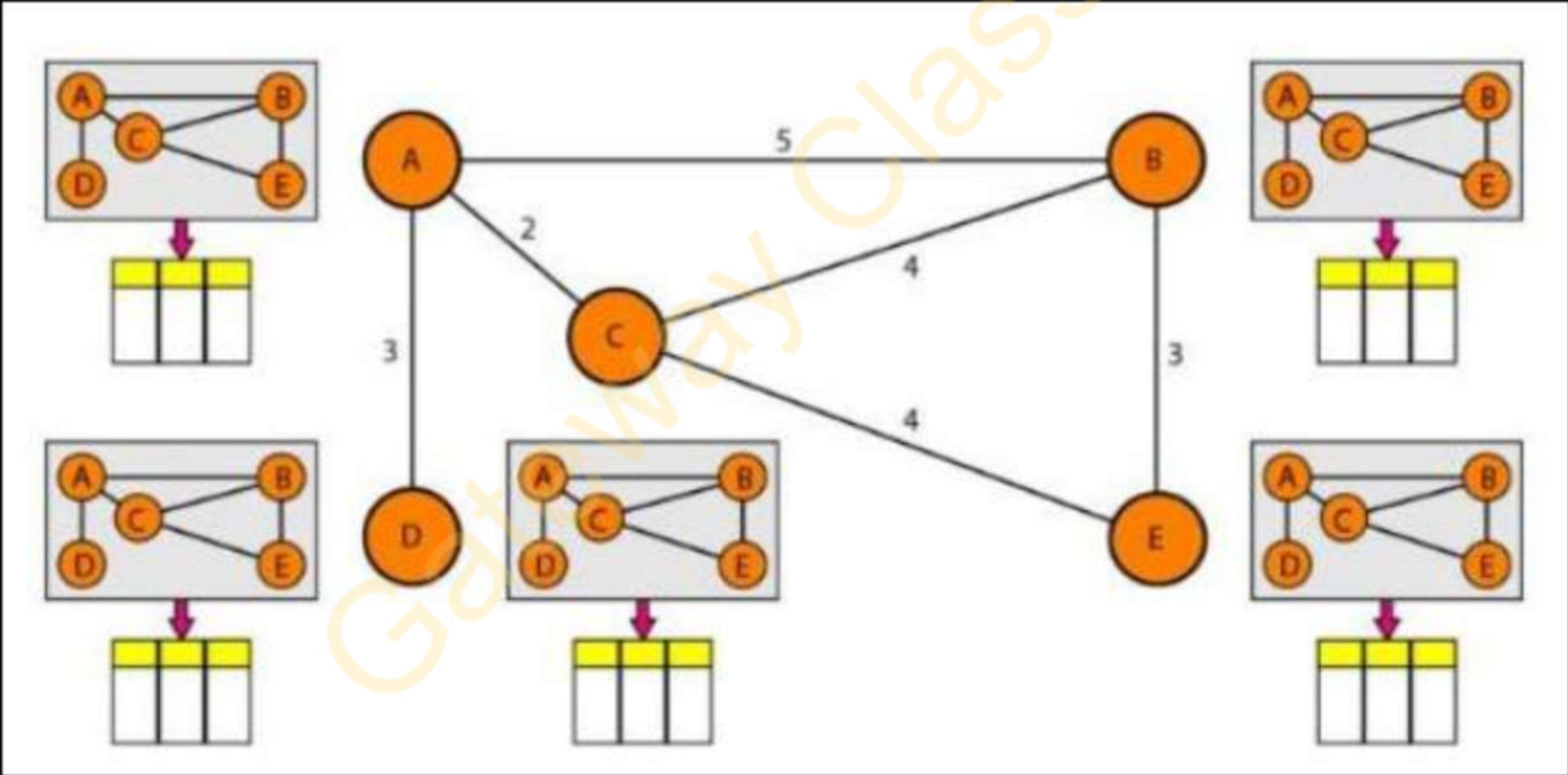
- a) When there is a change in the topology of the domain.
- b) On a periodic basis. A longer period ensures, flooding does not create too much traffic on network.

2. Flooding of LSPs :

After a node has prepared an LSP, it must be flooded to all other nodes, not only to its neighbors.

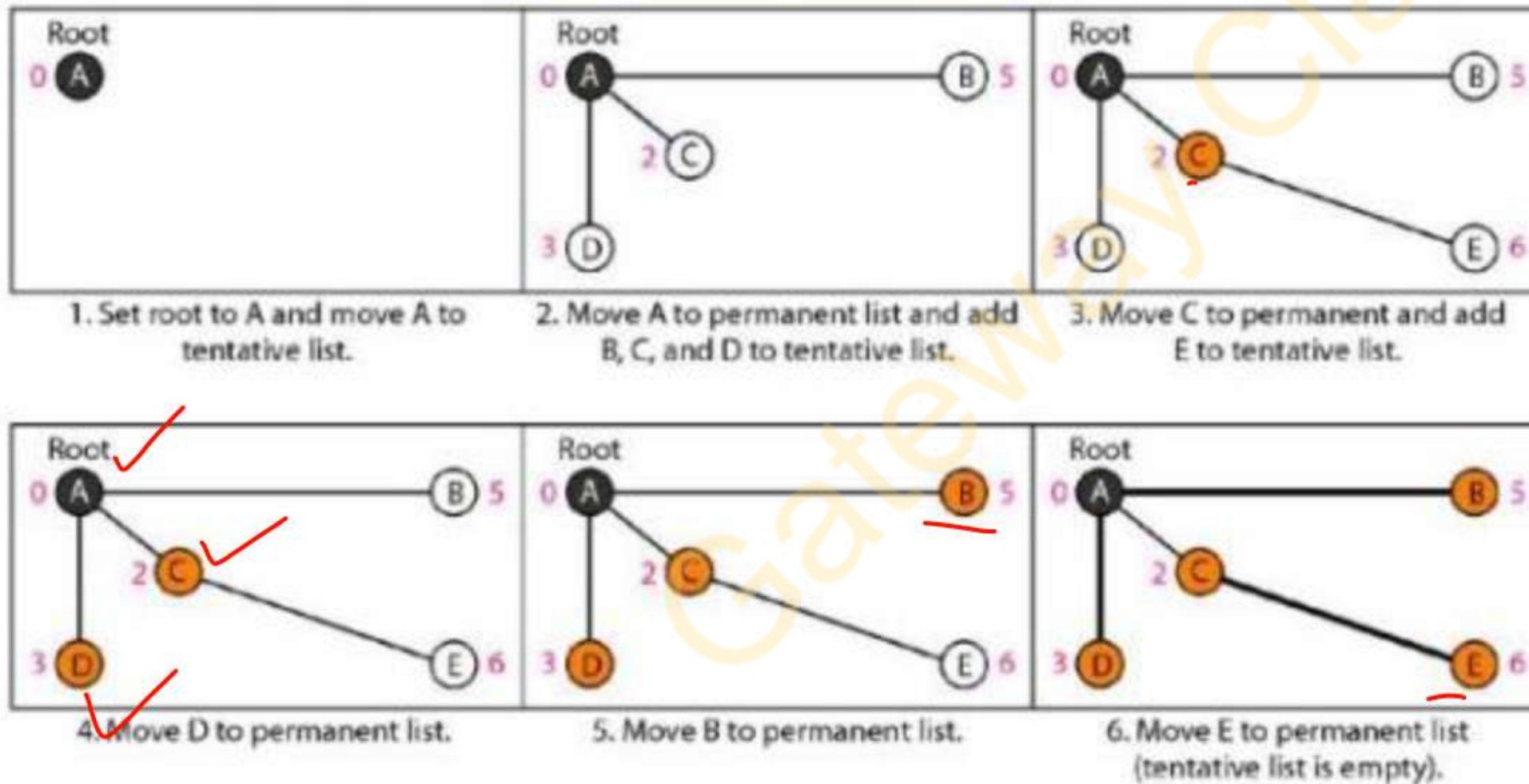
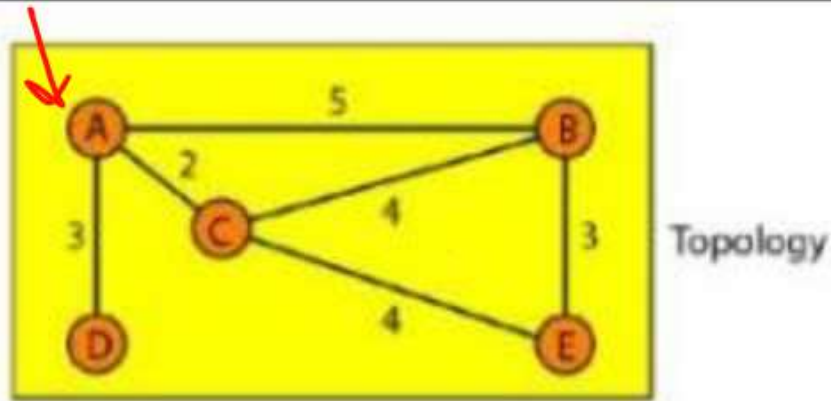
1. Every node sends a copy of LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have. If newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
 - a. It discards old LSP and keeps the new one.
 - b. It sends a copy of it out of each interface except the one from which the packet arrived.

Now, each node in the domain has the entire topology of the domain---the list of nodes and links, how they are connected including the type, cost (metric), and condition of the links (up or down)-the node can use Dijkstra's algorithm to build a routing table.



3. Formation of Shortest Path Tree:

The Dijkstra algorithm at every node finds the least cost path to every other node.



Example

✓ Step-1. Each Router will create a LSP.

LSP of A

A	
Seq. No	
Age (TTL)	
B	15
D	5

LSP of B

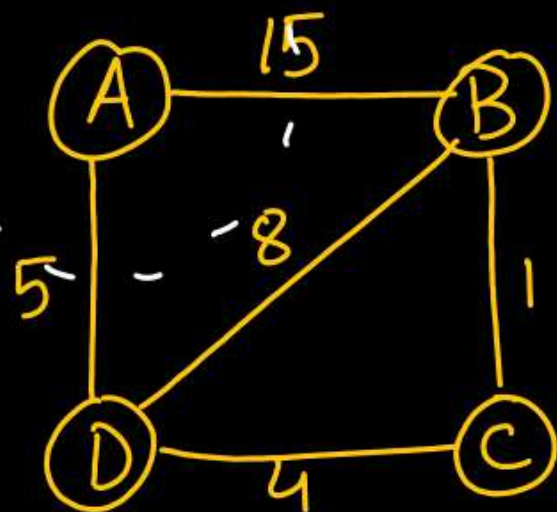
B	
Seq. no	
Age	
A	15
C	1
D	8

LSP of D

D	
Seq. no.	
Age	
A	5
B	8
C	4

LSP of C

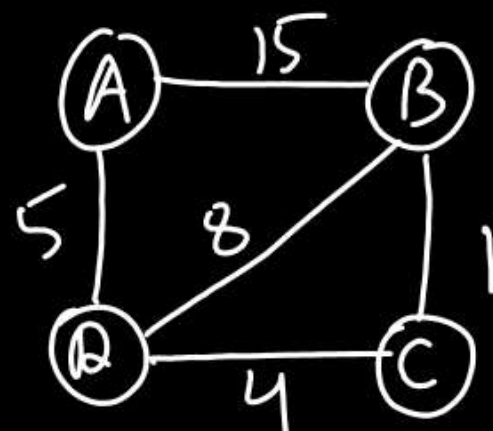
C	
Seq. no.	
Age	
B	1
D	4



Step-2. All routers will flood their LSPs.

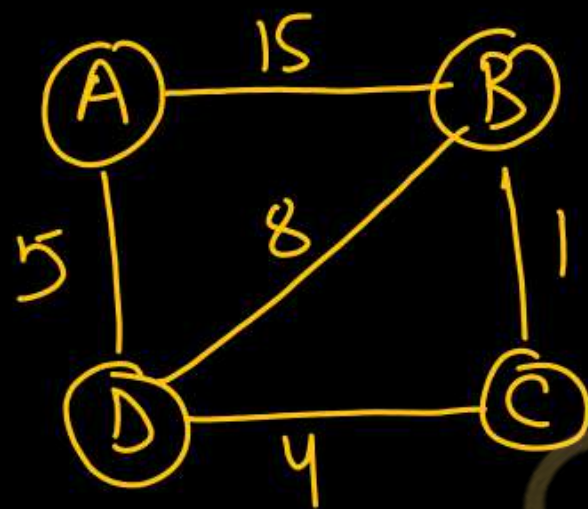
Step-3: Every node will create a topological global map of the network using these LSPs.

Example: At Node A



Similarly node B, C, D will acquire the global knowledge of the n/w.

Step-4: Now each router will apply dijkstra's algorithm to find the minimum cost route to every other router in the network and create its routing table.



Final Routing tables of all routers

Routing table of A

Node	Dist	Next hop
A	0	—
B	10	D
C	9	D
D	5	—

Routing table of C

Node	Dist	Next hop
A	9	B
B	1	—
C	0	—
D	4	—

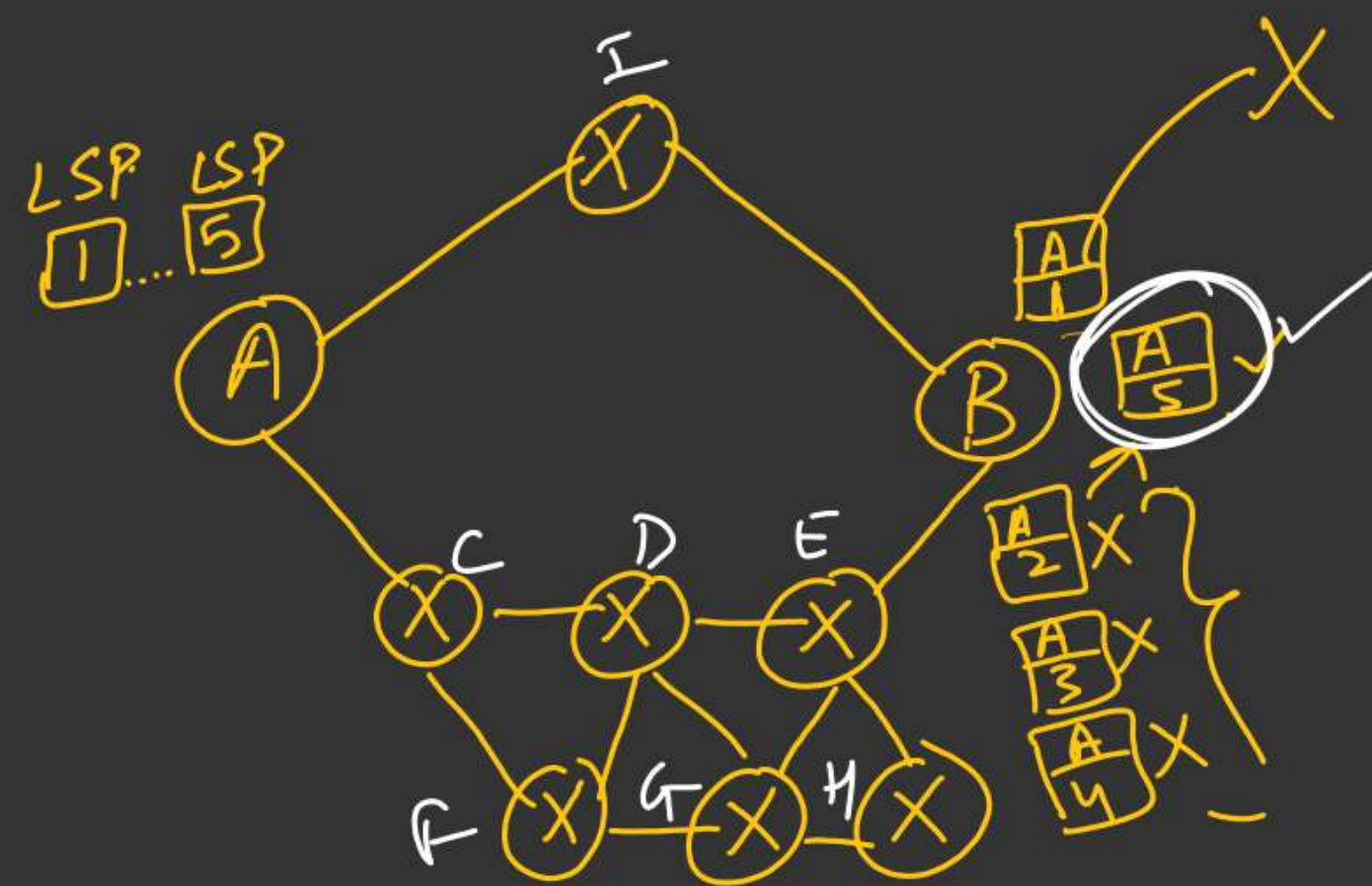
Routing Table of B

Node	Dis	Next hop
A	10	C
B	0	—
C	1	—
D	5	C

Routing table of D

Node	Dist	Next hop
A	5	—
B	5	C
C	4	—
D	0	—

Why Seq-no field in link state packet?



⇒ To keep the latest infoⁿ from each router.

⇒ What if this seq. no. field got corrupted.

Suppose seq. no. 10 is corrupted and changed to 100.

Seq. no. 11, 12, ..., 100 ⇒ All these LSPs are discarded
LSP with seq. no. '101' will be kept.

⇓
To avoid this situation, each router maintains a table with timer field.

Router	latest seq. No.	Timer
A	5	10
C	6	20
D	3	5
...		

Why to carry 'Age' field in every LSP?

To drop the packets if they are caught in infinite loop in the network and avoiding congestion/traffic.

Advantages of Link State protocols:

1. Fast network convergence.
2. Respond immediately for any networks changes by employing triggered flooded updates.
3. The complete and synchronized view about the network makes it hard for routing loops to exist .
4. Consume less bandwidth compared to Distance Vector protocols, as it does not engage periodic updates while maintaining the network.
5. bandwidth efficiency makes link state protocols more appropriate for large and complex networks.

Disadvantages of Link State protocols:

- Extensive databases such as neighbor table and Link State Table database, in addition to routing table require substantial amount of memory and CPU power.

Path Vector Routing:

- Useful for inter domain routing.
- The principle is similar to distance vector routing.
- **We assume that there is one node(speaker node) in each AS** that acts on behalf of the entire AS.
- Speaker node in an AS creates a routing table and advertises it to speaker nodes in neighboring ASs.
- Only speaker nodes in each AS can communicate with each other.
- A speaker node advertises the path in its autonomous system or other autonomous systems

1. Initialization:

Initial routing tables in path vector routing

Dest. Path

A1	AS1
A2	AS1
A3	AS1
A4	AS1
A5	AS1

A1 Table

AS 1

Dest. Path

C1	AS3
C2	AS3
C3	AS3

C1 Table

AS 3

Dest. Path

D1	AS4
D2	AS4
D3	AS4
D4	AS4

D1 Table

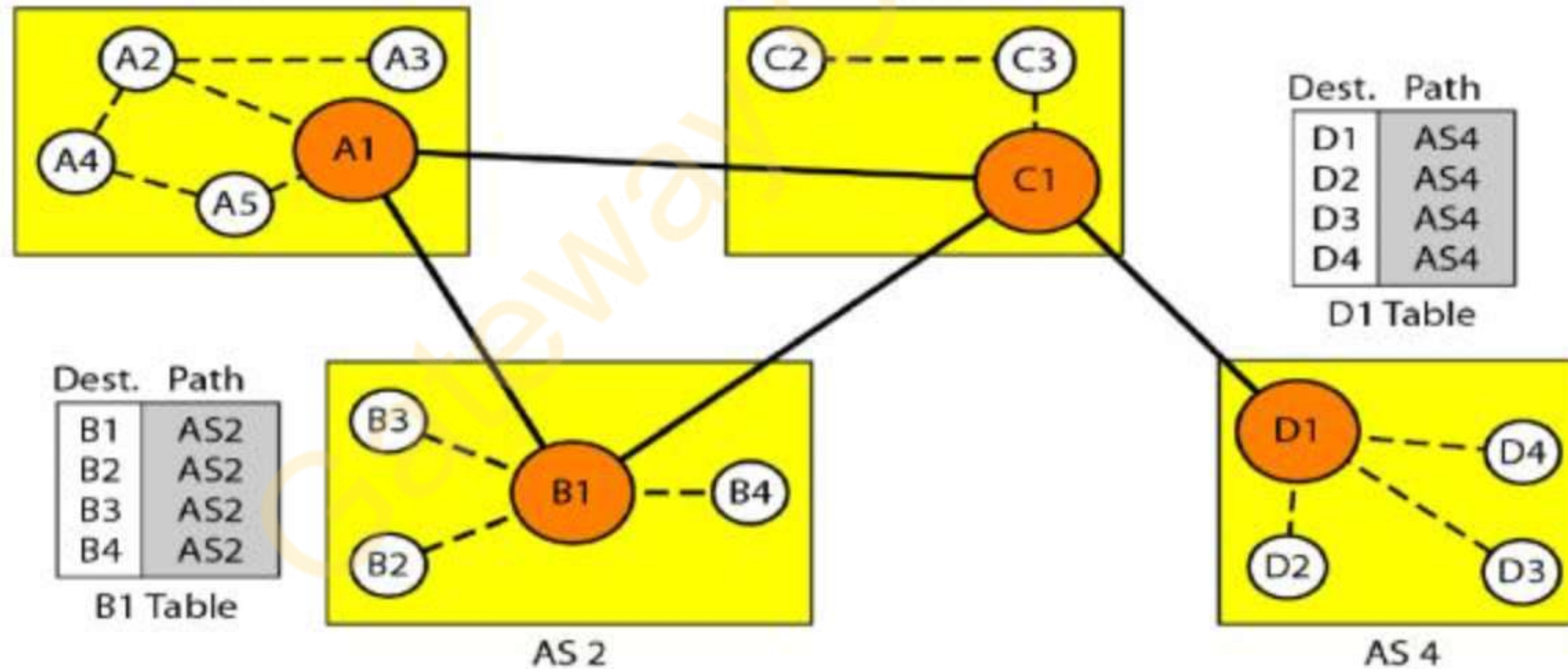
AS 4

Dest. Path

B1	AS2
B2	AS2
B3	AS2
B4	AS2

B1 Table

AS 2



2. Sharing :

- Just as in distance vector routing, a speaker in an AS shares its table with immediate neighbors.
- Node A1 shares its table with nodes B1 and C1. Node C1 shares its table with nodes D1, B1, and A1.
- A1. Node B1 shares its table with C1 and A1. Node D1 shares its table with C1.

Dest.	Path	Dest.	Path	Dest.	Path	Dest.	Path
A1	AS1	A1	AS2-AS1	A1	AS3-AS1	A1	AS4-AS3-AS1
...		...		...		...	
A5	AS1	A5	AS2-AS1	A5	AS3-AS1	A5	AS4-AS3-AS1
B1	<u>AS1-AS2</u>	B1	AS2	B1	AS3-AS2	B1	AS4-AS3-AS2
...		...		...		...	
B4	AS1-AS2	B4	AS2	B4	AS3-AS2	B4	AS4-AS3-AS2
C1	<u>AS1-AS3</u>	C1	AS2-AS3	C1	AS3	C1	AS4-AS3
...		...		...		...	
C3	AS1-AS3	C3	AS2-AS3	C3	AS3	C3	AS4-AS3
D1	<u>AS1-AS2-AS4</u>	D1	AS2-AS3-AS4	D1	AS3-AS4	D1	AS4
...		...		...		...	
D4	AS1-AS2-AS4	D4	AS2-AS3-AS4	D4	AS3-AS4	D4	AS4
A1 Table		B1 Table		C1 Table		D1 Table	

3. Updating:

When a speaker node receives a two-column table from a neighbor, it updates its own table by adding the nodes that are not in its routing table and adding its own AS and the AS that sent the table.

After a while each speaker has a table and knows how to reach each node in other ASs

a) Loop prevention: ✓

- When a router receives a message, it checks to see if its AS is in the path list to the destination.
- If it is, looping is involved and the message is ignored.

b) Policy routing: ✓

- When a router receives a message, it can check the path. If one of the AS in the path is against its policy, it can ignore that path and that destination.
- It does not update its routing table with this path, and it does not send this message to its neighbors.

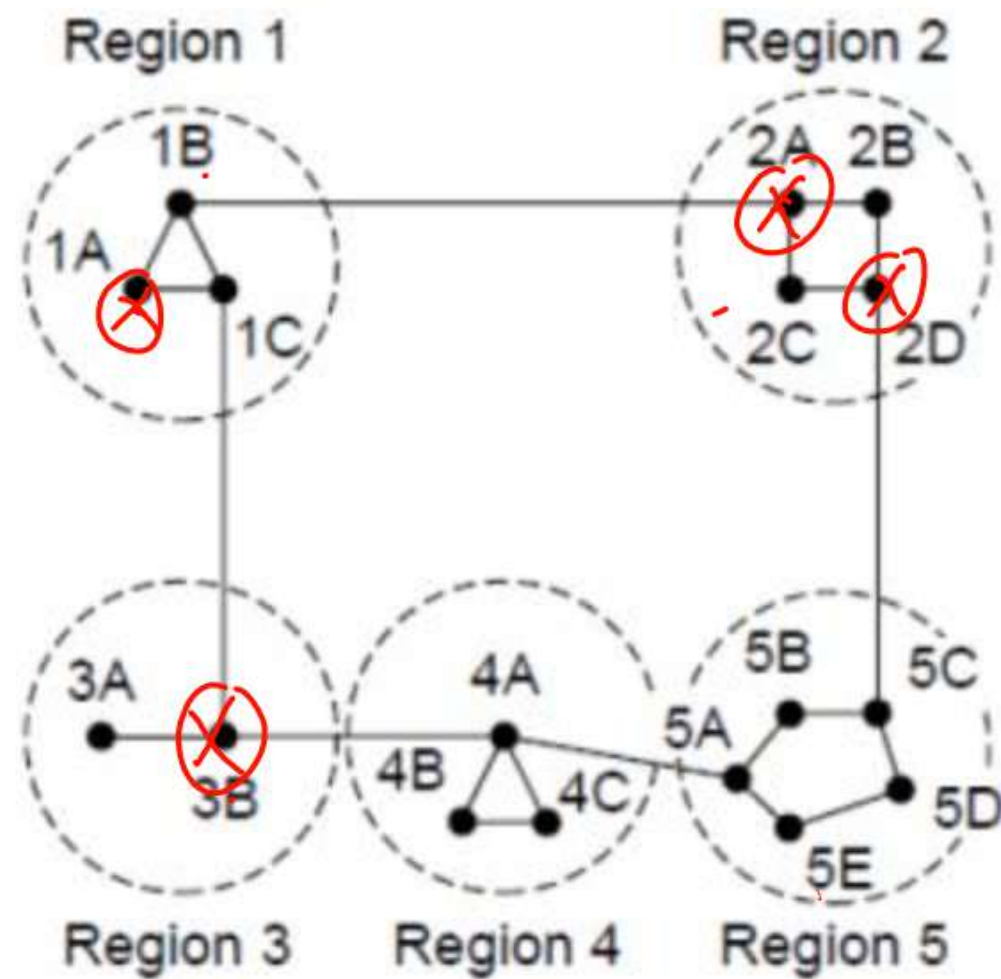
c) Optimum path:

- Each AS may have more than one path to a destination.
- For example, a path from AS4 to AS1 can be AS4-AS3-AS2-AS1, or it can be AS4-AS3-AS1.
- For the tables, we chose the one that had the smaller number of ASs (Just in case)

Gateway Classes

Hierarchical Routing:

- As networks grow in size, routing tables grow proportionally. More router memory consumed and more CPU time to scan them and more bandwidth to send status reports about them, is required.
- At a certain point, it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically, as it is in the telephone network.
- When hierarchical routing is used, the routers are divided into regions.
- Each router knows all the details about how to route packets to destinations within its own region but knows nothing about the internal structure of other regions.
- For huge networks, a two-level hierarchy may be insufficient; it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups, and so on..



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

AKTU PYQs

Q: What is unicast routing? Explain all unicast routing protocols in detail. (AKTU 2018-19)

↳ DVR + LSR + PVR

Gateway Classes



AKTU

B.Tech 6th Semester
CS IT & Allied

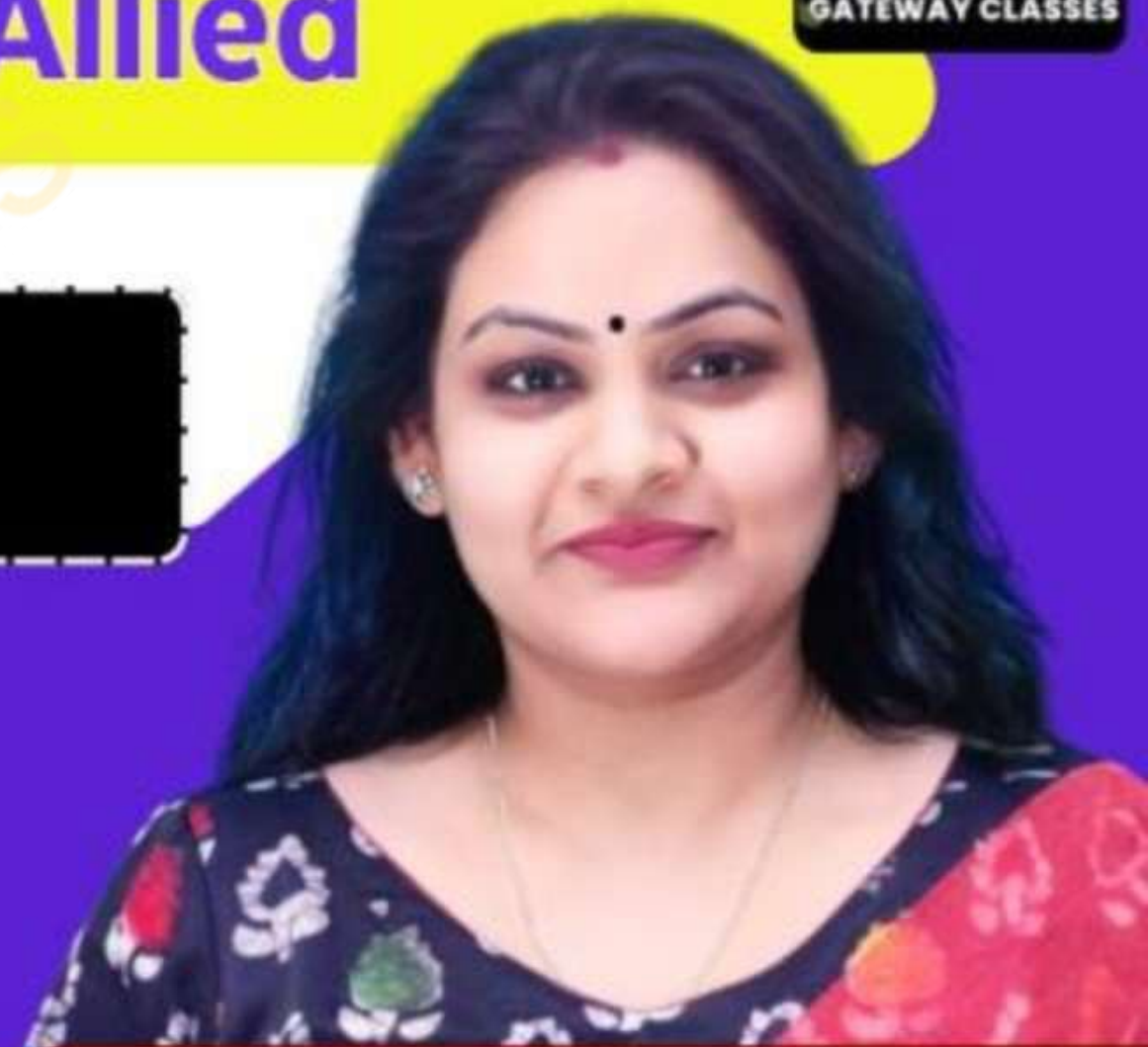


Computer Networks

Unit-3: Lecture 9

Today's Target

- Congestion Control ✓
- AKTU PYQs ✓



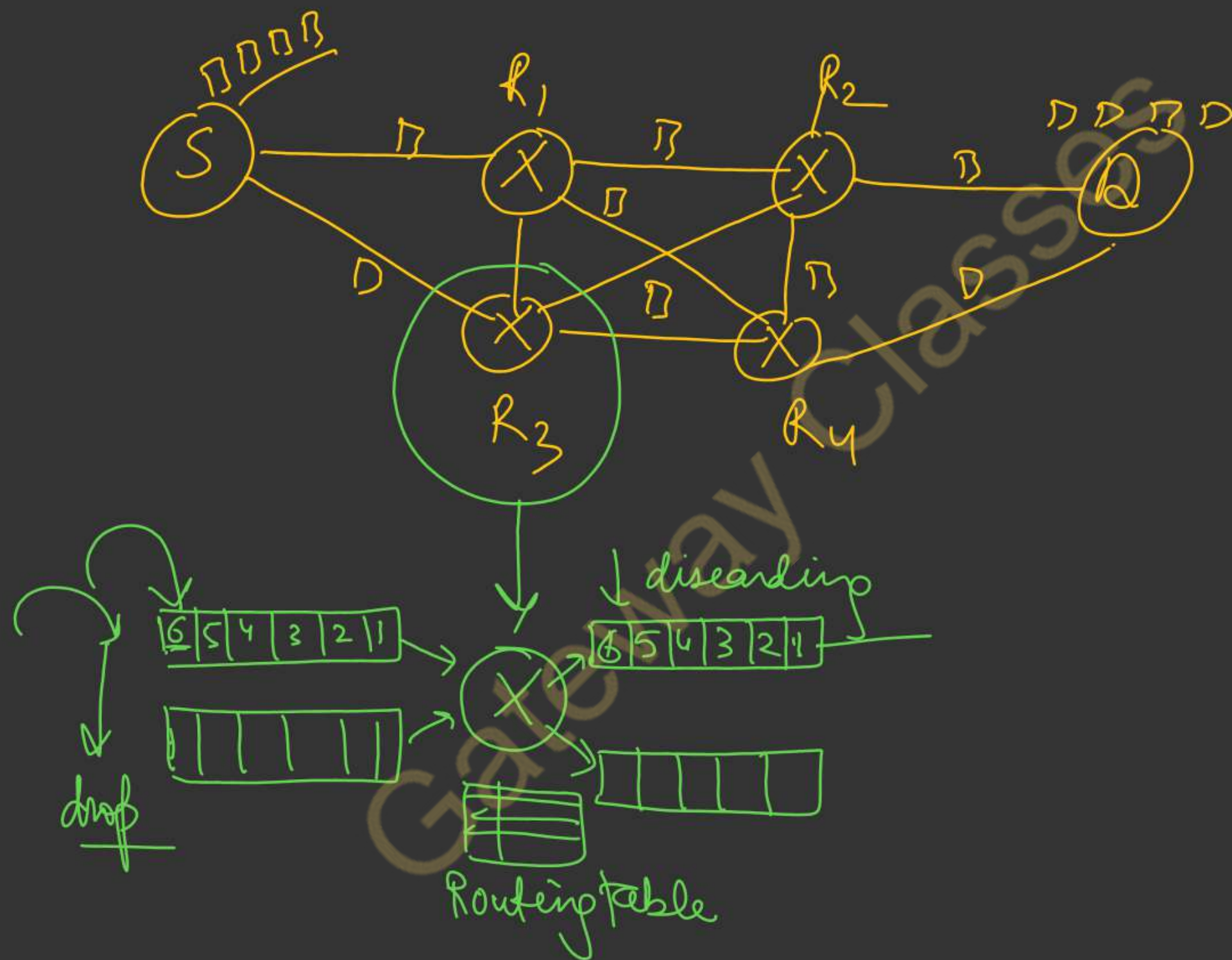
By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified

CONGESTION

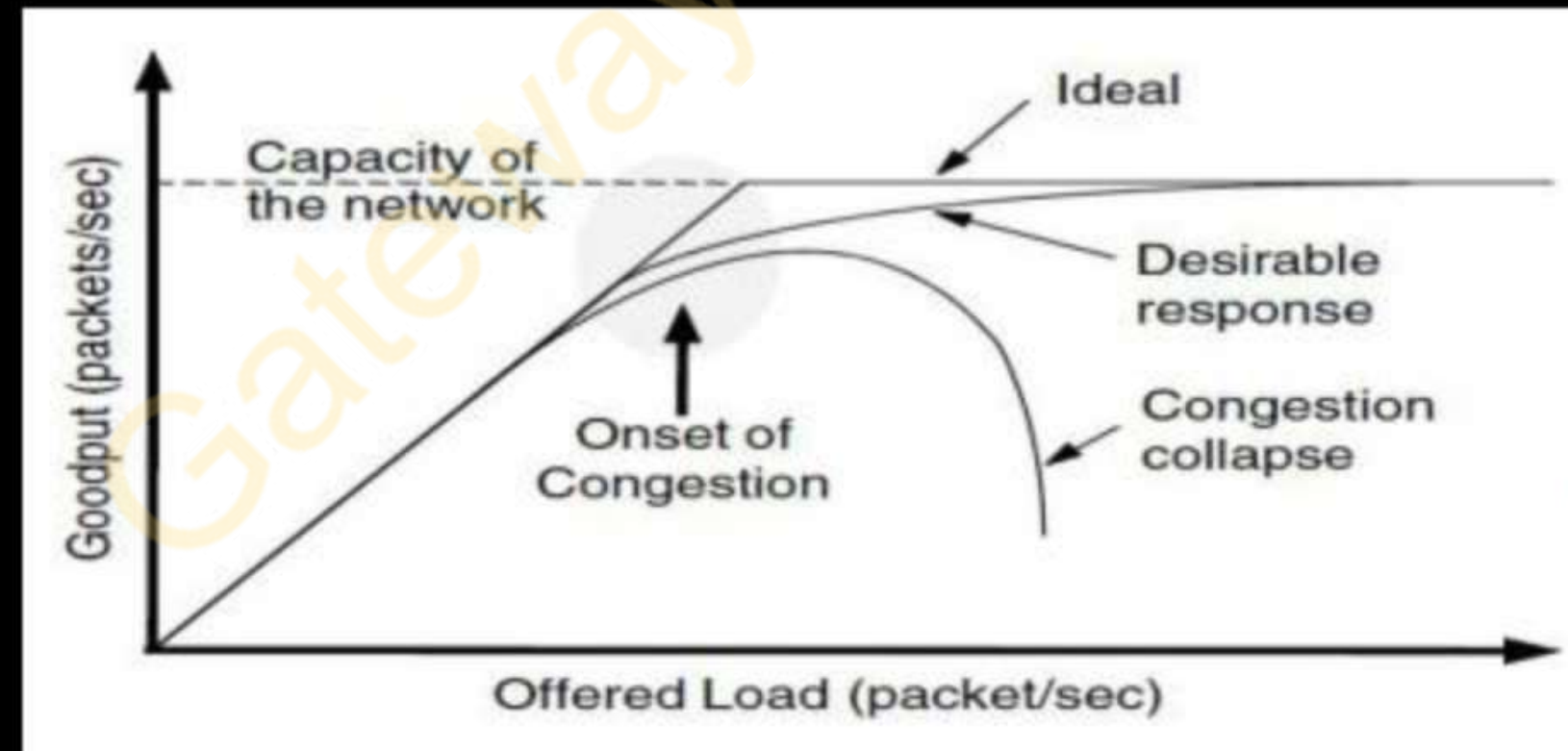
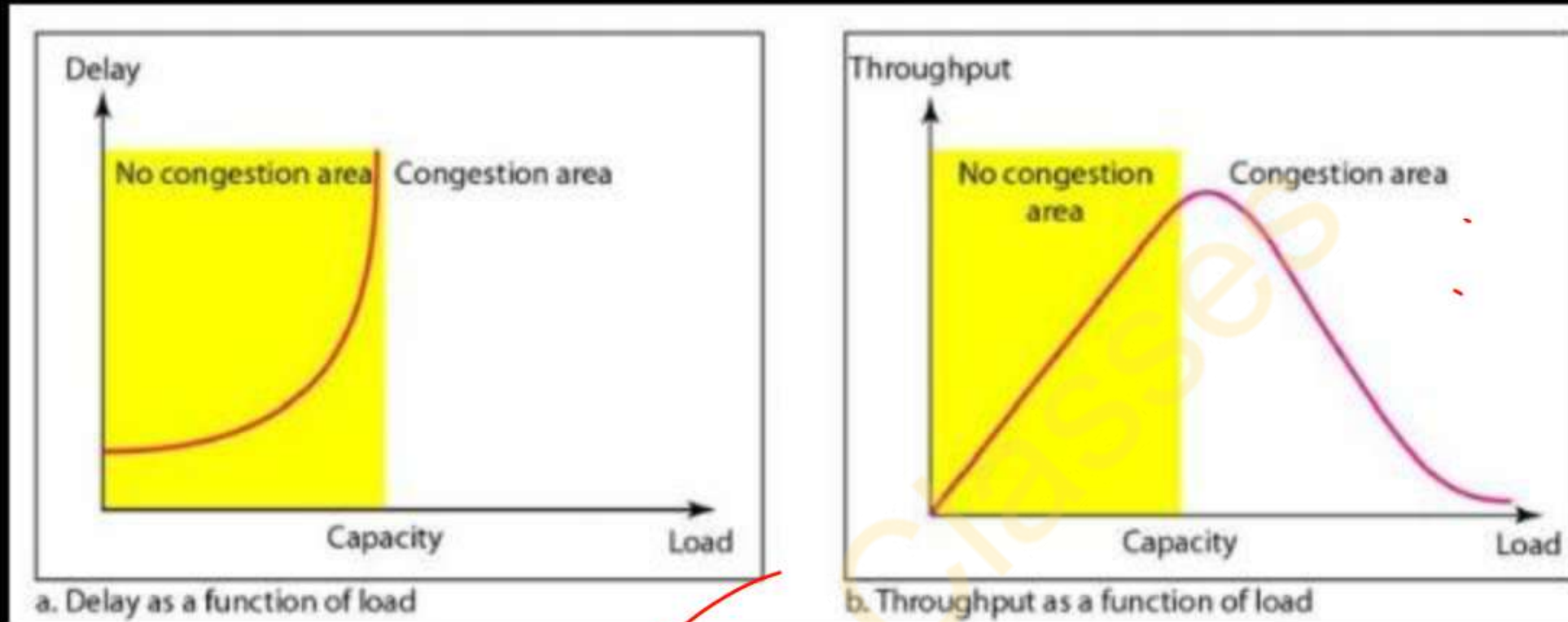
- Too many packets present in the network causes packet delay and loss that degrades performance.
This situation is called congestion.
- Congestion in a network may occur if load on the network is greater than its capacity.
- The network and transport layers share the responsibility for handling congestion.
- Network layer directly experiences it and determine what to do with the excess packets.
- The most effective way to control congestion is to reduce the load that transport layer is placing on the network.

Congestion control refers to the techniques to control the congestion and keep the load below the capacity.



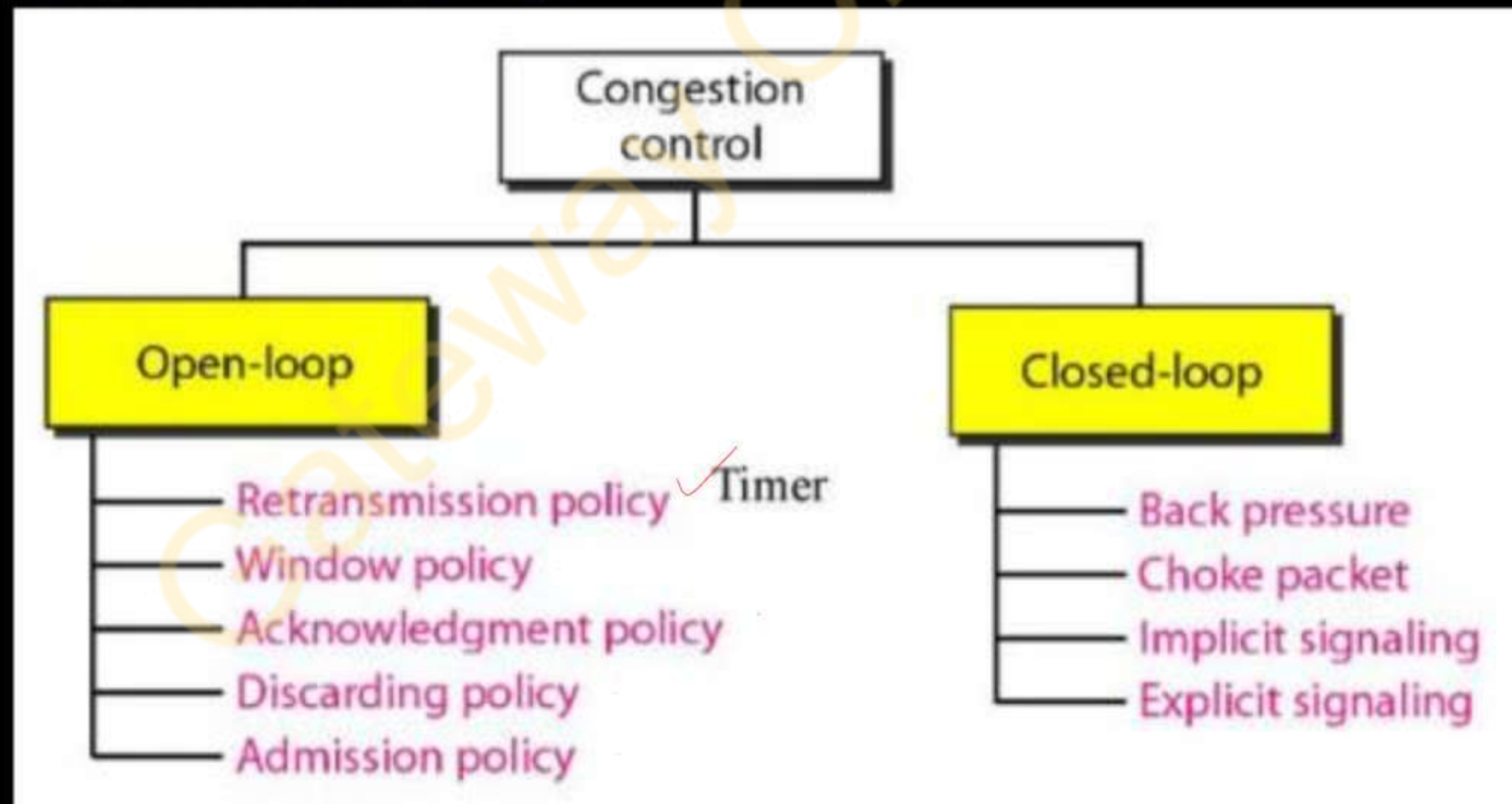
Network Performance: Congestion control involves two factors that measure performance of a network:

1. delay
2. throughput



Congestion Control

- Congestion control refers to techniques that can either prevent congestion, before it happens, or remove congestion, after it has happened.
- Congestion control mechanisms has two broad categories:
 1. open-loop congestion control (prevention)
 2. closed-loop congestion control (removal)



Open-Loop Congestion Control

- Policies are applied to prevent congestion before it happens.
- Congestion control is handled by either the source or the destination.

Retransmission Policy:

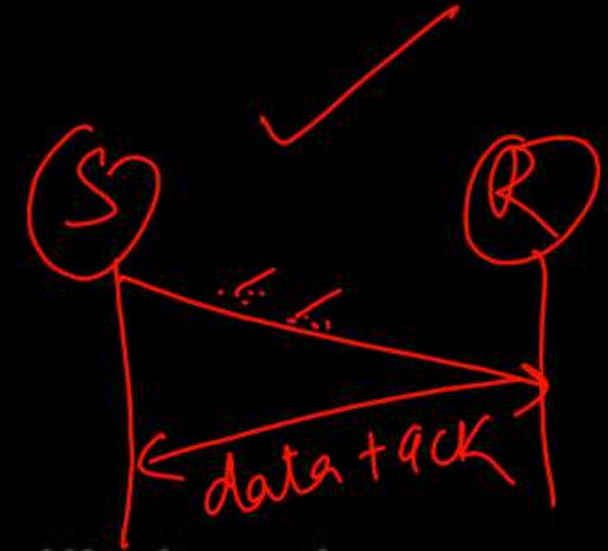
- If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted.
Retransmission may increase congestion in the network.
- The retransmission policy and the retransmission timers must be designed to optimize efficiency and time prevent congestion.

Window Policy:

- The type of window at the sender may also affect congestion.
- Selective Repeat window is better than Go-Back-N window for congestion control as Selective Repeat window tries to send the specific packets only that have been lost or corrupted.

Acknowledgment Policy: *use of piggybacking*

- If receiver doesn't ack every packet it receives; it may slow down sender and help prevent congestion.
- A receiver may send an ack only if it has a packet to be sent or a special timer expires.
- A receiver may decide to acknowledge only N packets at a time.
- Sending fewer acknowledgments reduce load on the network.



Discarding Policy:

- A good discarding policy by the routers may prevent congestion.
- In audio transmission, discard fewer sensitive packets when congestion is likely to happen, quality of sound is still preserved and congestion is prevented.

Admission Policy:

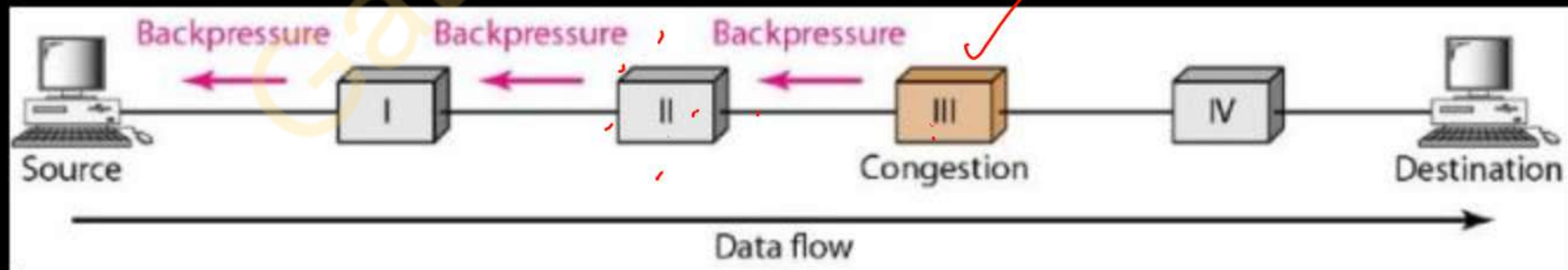
- A router can deny establishing a virtual circuit connection if there is congestion in the network or if there is a possibility of future congestion.

Closed-Loop Congestion Control:

Try to alleviate congestion after it happens.

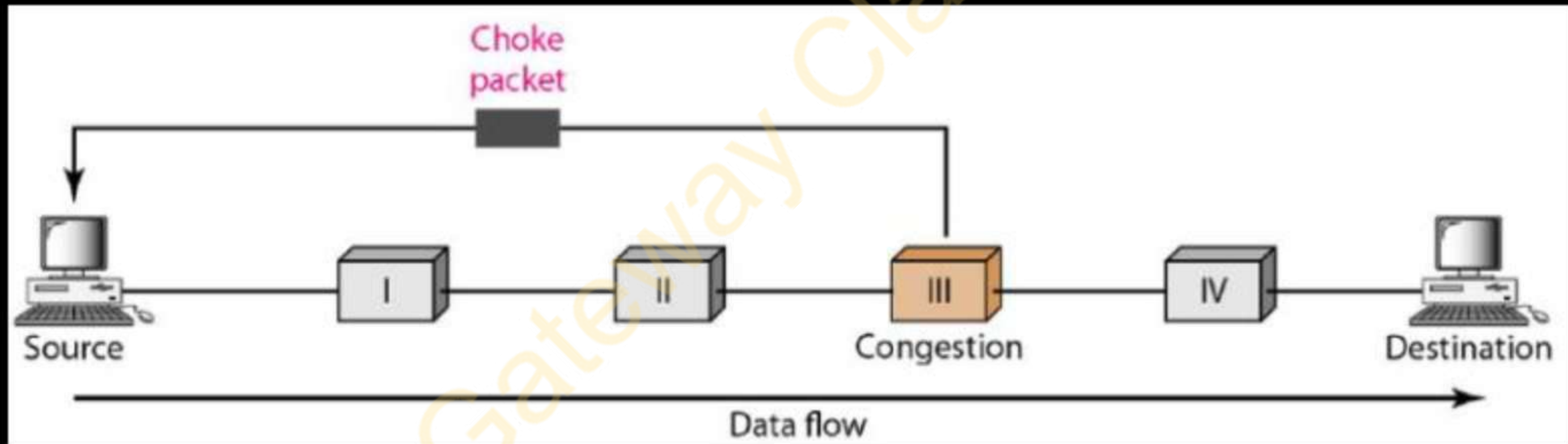
Backpressure:

- A congested node stops receiving data from the immediate upstream node/s.
- This may cause upstream node/s to become congested, and they also reject data from their upstream node/s and so on.
- Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source.
- Can be applied only to virtual circuit networks (each node knows the upstream node from which a flow of data is coming).



Choke Packet:

- A choke packet is a packet sent by a node to the source to inform it of congestion.
- Here the congested router warns the source station directly.
- The intermediate nodes through which the packet has traveled are not warned.
- Example: source quench ICMP message



Implicit Signaling

- No communication between the congested node/s and the source.
- Source guesses that there is a congestion somewhere in the network from other symptoms such as
- delay in receiving an ack is interpreted as congestion in the network; the source should slow down.
- Example: TCP congestion control

Explicit Signaling

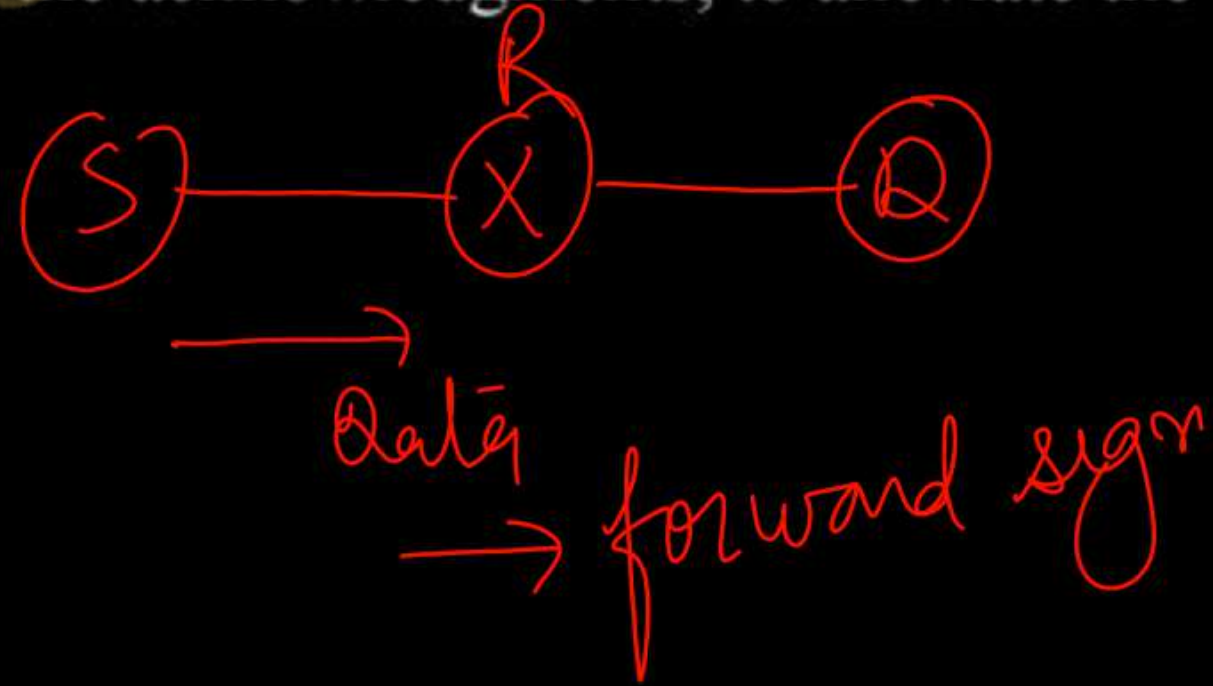
- The congested node can explicitly send a signal to source or destination.
- The signal is included in the packets that carry data.
- Can occur in either the **forward or the backward** direction.
- Example: **Frame Relay congestion**

Backward Signaling:

- A bit can be set in a packet moving in the direction opposite to the congestion to warn the source that there is congestion and that it needs to slow down to avoid the discarding of packets.

Forward Signaling:

- A bit can be set in a packet moving in the direction of the congestion to warn the destination that there is congestion.
- The receiver can use policies, such as slowing down the acknowledgments, to alleviate the congestion.



AKTU PYQs

Q: Describe few reasons of congestion in the network. (AKTU 2017-18)

Q: What is congestion? Briefly describe techniques that prevent congestion. (AKTU 2017-18)

↓
open-loop.

Gateway Classes

Thank
you